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CLINICAL STUDY PROTOCOL

An open-label, multicentre, integrated Phase 1 & 2 study to evaluate the safety, tolerability, radiation dosimetry and anti-tumour activity of Lutetium (^{177}Lu) rhPSMA-10.1 injection in men with metastatic castrate-resistant prostate cancer

BET-PSMA-121

Phase: 1 & 2

ClinicalTrials.gov ID: NCT05413850

WHO UT Number: U1111-1314-4775

EU CT Number: 2024-511537-35

IND Number (Lutetium (^{177}Lu) rhPSMA-10.1 Injection): 156938

IND Number (rhPSMA-7.3 (^{18}F)): 141561 (Phase 1 only)

Protocol Version 5.1, 24-Feb-2025

Revision Chronology:

Protocol Version 1, 15-Nov-2021

Protocol Version 2, 25-Jan-2022

Protocol Version 3, 11-Apr-2022

Protocol Version 3.1 (Germany), 26-Apr-2023

Protocol Version 4.0, 22-Sep-2023

Protocol Version 4.1 (EU), 22-Nov-2023

Protocol Version 5.0, 12-Dec-2024

Sponsor: Blue Earth Therapeutics Limited

Confidentiality Statement

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This document contains confidential information that must not be disclosed to anyone other than the Sponsor, the Investigator Team, Safety Review Committee, host organisation, and members of the Research Ethics Committee, unless authorised to do so.

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Blue Earth Therapeutics Limited

Clinical Study Protocol

An open-label, multicentre, integrated Phase 1 & 2 study to evaluate the safety, tolerability, radiation dosimetry and anti-tumour activity of Lutetium (^{177}Lu) rhPSMA-10.1 injection in men with metastatic castrate-resistant prostate cancer

Protocol Number:	BET-PSMA-121
Protocol Version / Date:	Protocol Version 5.1, 24-Feb-2025
Investigational Product:	For Phase 1 (completed): Lutetium (^{177}Lu) rhPSMA-10.1 Injection and POSLUMA® (flotufolastat F 18 injection; previously known during its development as rhPSMA-7.3 (^{18}F) injection) For Phase 2: Lutetium (^{177}Lu) rhPSMA-10.1 Injection
EU CT Number	2024-511537-35
IND Numbers:	156938 (Lutetium (^{177}Lu) rhPSMA-10.1 Injection); 141561 (rhPSMA-7.3 (^{18}F)) (Phase 1 only)
WHO UT Number	U1111-1314-4775
Development Phase:	1 & 2
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Chief Medical Officer

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Signing Time: 24-Feb-2025 | 4:25:31 PM CET
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Sponsor Signature (Name and Title) Date

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PROTOCOL AGREEMENT

I have read the protocol specified below. In my formal capacity as investigator, my duties include providing Blue Earth Therapeutics Limited with the information, as outlined in the protocol. It is understood that all information pertaining to the study will be held strictly confidential and that this confidentiality requirement applies to all study staff at this site. Furthermore, on behalf of the study staff and myself, I agree to maintain the procedures required to carry out the study in accordance with accepted Good Clinical Practice principles, applicable regulatory requirements, and to abide by the terms of this protocol.

Protocol Number: BET-PSMA-121

Protocol Title: An open-label, multicentre, integrated Phase 1 & 2 study to evaluate the safety, tolerability, radiation dosimetry and anti-tumour activity of Lutetium (^{177}Lu) rhPSMA-10.1 injection in men with metastatic castrate-resistant prostate cancer.

Protocol Version: 5.1 Protocol Date: 24-Feb-2025

_____ _____

Investigator Signature Date

_____

Print Investigator Name and Title

Site # _____

Site Name _____

Address _____

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Phone Number _____

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LIST OF ABBREVIATIONS

Abbreviation	Definition
^{18}F	Fluorine-18
^{177}Lu	Lutetium-177
^{99m}Tc	Technetium-99m
ADT	Androgen deprivation therapy
AE	Adverse event
ALP	Alkaline phosphatase
ALT	Alanine aminotransferase
AST	Aspartate aminotransferase
BOIN	Bayesian optimal interval
ceCT	Contrast enhanced computed tomography
CI	Confidence interval
CNS	Central nervous system
CR	Complete response
CRO	Clinical Research Organisation
CT	Computed tomography
CTCAE	Common Terminology Criteria for Adverse Events
DICOM	Digital Imaging and Communications in Medicine
DLT	Dose-limiting toxicity
DNA	Deoxyribonucleic acid
DPA	Data Protection Act
EAP	Efficacy Analysis Population
EBRT	External beam radiation therapy
ECG	Electrocardiogram
ECOG	Eastern cooperative oncology group
eCRF	Electronic case report form
EDC	Electronic data capture
EoT	End of Treatment
EU	European Union
FACT-P	Functional Assessment of Cancer Therapy-Prostate
FACT-RNT	Functional Assessment of Cancer Therapy-Radionuclide Therapy
FAS	Full Analysis Set
FDA	Food and Drug Administration
GBq	Gigabecquerel
GCP	Good Clinical Practice

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Abbreviation	Definition
GDPR	General Data Protection Regulation
GnRH	Gonadotropin-releasing hormone
Gy	Gray
HIPAA	Health Insurance Portability and Accountability Act
HR	Hazard ratio
IB	Investigator's Brochure
ICF	Informed consent form
ICH	International Council for Harmonisation
IEC	Independent Ethics Committee
IMP	Investigational medicinal product
IND	Investigational New Drug
IRB	Institutional Review Board
IRT	Interactive Response Technology
IV	Intravenous(ly)
MBq	Megabequerel
mCRPC	Metastatic castrate-resistant prostate cancer
mCi	Millicurie
MIRD	Medical Internal Radiation Dose
MRI	Magnetic resonance imaging
mSv	Millisievert
NAAD	Novel androgen axis drug
PCWG3	Prostate Cancer Clinical Trials Working Group 3
PET	Positron emission tomography
PFS	Progression-free survival
PK	Pharmacokinetic(s)
PKP	PK Population
PR	Partial response
PRO	Patient-reported outcomes
PSA	Prostate-specific antigen
PSA50	PSA decrease by at least 50% from baseline
PSMA	Prostate-specific membrane antigen
QC	Quality control
RCT	Randomised clinical trial
RECIST	Response Evaluation Criteria in Solid Tumours
rhPSMA	Radiohybrid prostate-specific membrane antigen

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Abbreviation	Definition
RLT	Radioligand therapy
RP2D	Recommended Phase 2 dose
rPFS	Radiographic progression free survival
RSI	Reference Safety Information
SAE	Serious adverse event
SAP	Statistical Analysis Plan
SiFA	Silicon-fluoride acceptor moiety
SPECT	Single-photon emission computed tomography
SRC	Safety Review Committee
SUSAR	Suspected unexpected serious adverse reaction
T3	Triiodothyronine
T4	Thyroxine
TEAE	Treatment-emergent adverse event
TSC	Trial Steering Committee
TSH	Thyroid-stimulating hormone
UK	United Kingdom
ULN	Upper limit of normal
UPCR	Urine protein creatinine ratio
US	United States
WBC	White blood cell

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PROTOCOL SYNOPSIS

Study Title	An open-label, multicentre, integrated Phase 1 & 2 study to evaluate the safety, tolerability, radiation dosimetry and anti-tumour activity of Lutetium (^{177}Lu) rhPSMA-10.1 injection in men with metastatic castrate-resistant prostate cancer
Protocol Number	BET-PSMA-121
Phase	1 & 2
Sponsor	Blue Earth Therapeutics Limited
Funding Organisation	Blue Earth Therapeutics Limited
Study Design	This is an interventional, open-label, integrated Phase 1 & 2 study to assess the safety, radiation dosing regimen and anti-tumour activity of Lutetium (^{177}Lu) radiohybrid prostate-specific membrane antigen (rhPSMA)-10.1 Injection (hereafter also referred to as the therapeutic investigational medicinal product [IMP], or IMP) in men with metastatic castrate-resistant prostate cancer (mCRPC). The study will consist of 2 parts: a non-randomised Phase 1 part, with safety, dose-finding, and dosimetry components, and a randomised Phase 2 part, with efficacy and safety assessments, and testing dosing regimens selected following analysis of the safety and dosimetry data in Phase 1. Both phases will include subjects with prostate-specific membrane antigen (PSMA)-positive mCRPC, which has progressed following prior therapy. Phase 1 will include a post-chemotherapy mCRPC cohort of subjects who have experienced disease progression on or after at least 1 novel androgen axis drug (NAAD) (e.g. abiraterone, enzalutamide) and at least 1 course (but no more than 2 courses) of taxane-based chemotherapy. Phase 2 will include subjects who have experienced disease progression on or after at least 1 NAAD (e.g. abiraterone, enzalutamide, apalutamide, darolutamide) but have not received previous taxane-based chemotherapy for the treatment of mCRPC. The treatment phase in Phase 1 of the study was completed as of Protocol Version 5.0; therefore, all sections specific to Phase 1 are marked as completed. Phase 1 (Completed): In the Phase 1 component, subjects will be enrolled into 1 of 2 dose levels, which will investigate the safety, tolerability, and radiation dosimetry of the therapeutic IMP. Safety and radiation dosimetry results of the Phase 1 component will be used to determine the recommended treatment dose and regimen (recommended Phase 2 dose [RP2D]) to be tested in the Phase 2 component. Eligible subjects enrolled in Phase 1 will sequentially enter into 1 of 2 dosing regimens testing 2 radioactivity levels of the therapeutic IMP: 5.55 Gigabecquerel [GBq] (150 millicurie [mCi]) or 7.40 GBq (200 mCi). Following the first dose, and depending on individual tolerability and safety, with particular attention to parameters relating to estimated absorbed radiation doses to normal organs, subjects will receive additional doses at the same radioactivity level separated by 6-week intervals. Both regimens will include a maximum of 3 cycles of therapy (subject to Safety Review Committee [SRC] confirmation for Cohort B) in order to assess the safety and radiation dosimetry of a “core” treatment regimen. In line with other PSMA-targeted pharmaceuticals, it is expected that the kidneys will be the radiation

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Study Design *(cont'd)*	dose-limiting organ. All subjects will undergo dosimetry assessments using blood and single-photon emission computed tomography /computed tomography (SPECT/CT) imaging at set intervals in the 7 days following each cycle of therapy in Phase 1. This radiation dosimetry assessment will permit organ and tumour absorbed dose estimates to be calculated for each subject and each cycle. In Phase 1, no subject will receive additional radioactivity if doing so would exceed any normal organ radiation dose limits defined in the protocol. In Phase 1, escalation between the 2 cohorts will be managed using a Bayesian optimal interval (BOIN; Liu and Yuan, 2015) design, with initially 3 subjects recruited to the low dose cohort (Cohort A). Each subject will participate in only one cohort, and subjects in Phase 1 will not participate in Phase 2 of the study. After completion of each cycle of the therapeutic IMP, the investigator will review all relevant data including radiation dosimetry and all available safety data for each subject. The decision to give the subject subsequent dose cycles will depend on individual safety and tolerability, and cumulative absorbed radiation doses to normal tissues and organs. Further cycles of the therapeutic IMP will not be given to subjects who did not tolerate previous administration or in whom the next dose would exceed the normal organ radiation absorbed dose limits. The occurrence of a dose-limiting toxicity (DLT) at any point from the first cycle of therapy to 6 weeks after the final cycle of therapy will lead to stopping dosing for that subject. Once the third subject in Cohort A has completed the DLT observation period (the period from the first administration of the therapeutic IMP until 6 weeks after the final cycle of therapy), a SRC will review all available data and decide on escalation to the next dose cohort (Cohort B) and the maximum number of cycles. In their decision, the SRC will incorporate the recommendation from the BOIN design (escalation may not always occur for example, in the event of safety concerns). Assuming 3 subjects have been treated in Cohort A and the decision to open Cohort B has been made, at least 9 further subjects will be enrolled in Cohort B. Once a combined total of at least 12 subjects have completed dosing and the DLT observation period (i.e. from first administration of the therapeutic IMP to 6 weeks after the final cycle of therapy with the IMP received by the subject), an SRC meeting will be convened to review the safety and dosimetry data acquired thus far and to select an RP2D for testing in the Phase 2 component of the study. Data from Phase 1 were supportive of continuing development to Phase 2. Phase 2: The Phase 2 component of the study will involve 2-3 dosing cohorts to investigate the safety, efficacy, radiation dosimetry and optimal dose of the IMP. In Phase 2, subjects will be initially randomised to 1 of 2 dosing regimens, with approximately 30 subjects randomised to each cohort as follows: • Cohort 2A: 2 doses at 10.00 GBq (270 mCi) followed by up to 5 additional doses at 7.40 GBq (200 mCi), all doses administered at 6-weekly intervals. • Cohort 2B: Up to 8 doses at 7.40 GBq (200 mCi). The first 3 doses will be administered at 3-weekly intervals, with the remaining doses being administered at 6-weekly intervals.

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Study Design *(cont'd)*	A decision will be made on whether or not to open a third cohort (Cohort 2C) based on ongoing review of safety and preliminary efficacy data in Cohorts 2A and 2B. The decision on whether or not to open Cohort 2C will be made by the Trial Steering Committee (TSC) after reviewing all available safety and preliminary efficacy data and when approximately 35 subjects have been randomised and dosed in Cohorts 2A and 2B. Further information is provided in the TSC Charter. • Cohort 2C, if opened, will include approximately 10 subjects treated with 2 doses at 14.80 GBq (400 mCi) followed by up to 4 additional doses at 7.40 GBq (200 mCi), all doses administered at 6-weekly intervals. If Cohort 2C is not opened, additional subjects will be randomised to Cohort 2A and Cohort 2B, respectively, resulting in a total of approximately 35 subjects in each cohort. If unacceptable rates of toxicity occur in Cohorts 2A or 2B, enrolment into the trial will be put on hold and a protocol amendment may be submitted to test a reduced radioactivity cohort. Each cohort will enrol subjects to assess whether the IMP achieves anti-tumour responses using a decrease in prostate specific antigen (PSA) by at least 50% from baseline (PSA50 response) as a surrogate biomarker endpoint. Following safety review, Cohort 2A or 2B may be closed for futility (i.e. estimated PSA50 response rate ≤20% within a cohort). This decision will be based on the recommendation of the TSC. In this case, additional subjects may be randomised to the remaining cohorts to complete study enrolment. During the conduct of Phase 2, the Sponsor may perform additional interim reviews of the emerging data to assess preliminary efficacy and safety. Full details of such reviews will be described in the Statistical Analysis Plan (SAP). Image-based dosimetry will be performed to determine normal organ radiation absorbed doses for each individual subject. These absorbed doses will be used to determine the maximum number of cycles an individual subject can receive without exceeding agreed normal organ dose limits. In certain subjects in Phase 2 who demonstrate no evidence of progression (defined as radiological progression or clinical progression as determined by the investigator), and an absence of clinically significant normal organ toxicity in the judgement of the investigator, it may be appropriate for the treating physician to continue treatment with additional cycles (up to the maximum number of cycles for that dosing cohort), where this may result in the kidney or bone marrow radiation dose limits being cautiously exceeded according to the rules provided in the protocol.
Study Rationale	Prostate cancer is the second most frequent cancer diagnosis made in men, and the fifth leading cause of death worldwide (Rawla 2019). Prostate cancer is most commonly diagnosed in men aged >65 years (Daniyal 2014), and is largely asymptomatic in the early stages, with tumours detected by increased concentrations of PSA in peripheral blood and/or an abnormal digital rectal examination (Daniyal 2014; Kohaar 2019). Diagnosis is confirmed by a prostate biopsy (Kohaar 2019). Prostate-specific antigen is used as a tumour biomarker (Kohaar 2019) and serum levels of PSA have been shown to positively correlate with the risk of metastatic disease or subsequent disease recurrence or progression (Kamaleshwaran 2012; Esfahani 2015; Ecke 2016; Nishimura 2018). Primary treatment of prostate cancer can be divided into radical and non-radical treatment, the former applicable predominantly to patients with curable disease and the latter for those whose disease has spread, and

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Study Rationale (*cont'd)*	treatment is aimed at prolonging life. However, treatment options remain limited and survival following progression after receiving NAAD is less than 2 years on average (Schaeffer 2021). Radiohybrid PSMA-ligands are a new class of radiopharmaceuticals currently under evaluation in prostate cancer diagnosis and treatment. The only PSMA-targeting therapeutic radiopharmaceutical is Lutetium (^{177}Lu) PSMA-617 (Pluvicto™) which was approved for the treatment of men who have progressed following NAAD and taxane chemotherapy in mCRPC. It was approved based on data from a Phase 3 randomised, controlled trial, which demonstrated increased overall survival in patients with mCRPC who had previously received chemotherapy in the castrate-resistant phase of their disease (VISION Phase 3 trial [NCT03511664]; Sartor 2021). In the more recent PSMAfore trial, lutetium (^{177}Lu) vipivotide tetraxetan was shown to prolong radiographic progression-free survival (rPFS) compared to androgen receptor pathway inhibitor change in taxane-naïve patients with PSMA-positive mCRPC (Morris 2024). Patient selection for such therapies is based on the use of a variety of PSMA-targeted positron emission tomography (PET) tracers to determine tumour expression of PSMA. Furthermore, the dosing regimens have often not been informed by formal dose-ranging studies, posing the question as to whether the therapeutic index has yet to be optimised for patients with mCRPC. Non-clinical dosimetry data for ^{177}Lu rhPSMA-10.1 indicates preserved tumour uptake with improved tumour to normal organ ratios including the kidney, which is the dose-limiting organ with this class of compounds. Therefore, this integrated Phase 1 & 2 study aims to establish the safety, recommended treatment regimen and preliminary efficacy of Lutetium (^{177}Lu) rhPSMA-10.1 Injection in men with mCRPC.	
Phase 1 (Completed)		
Primary Objective and Endpoints	Objective: <u>Phase 1</u> To establish the RP2D regimen by evaluation of the safety and tolerability of intravenous (IV) administration of Lutetium (^{177}Lu) rhPSMA-10.1 Injection in subjects with mCRPC.	Primary Endpoints: <u>Phase 1</u> • Incidence of DLTs during the DLT observation period. • Frequency and nature of treatment-emergent adverse events (TEAEs).
Secondary Objectives and Endpoints	Objectives: <u>Phase 1</u> 1. To determine the whole-body radiation dosimetry and pharmacokinetics of Lutetium (^{177}Lu) rhPSMA-10.1 Injection. 2. To determine the organ-specific radiation dosimetry of Lutetium (^{177}Lu) rhPSMA-10.1 Injection (organ exposure to radiation) after each administration.	Endpoints: <u>Phase 1</u> • Terminal half-life of activity concentrations in blood. • Specific whole-body absorbed dose (Gray [Gy]/GBq), specific absorbed dose to the tumour lesions (Gy/GBq), specific absorbed dose per organ (Gy/GBq) and cumulative absorbed organ and whole-body absorbed doses (Gy).

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Secondary Objectives and Endpoints *(cont'd)*	Objectives: Phase 1 *(cont'd)* 3. To describe observed anti-tumour activity of Lutetium (^{177}Lu) rhPSMA-10.1 Injection.	Endpoints Phase 1 *(cont'd)* • Number of subjects with ≥50% reduction in PSA level from baseline at 12 weeks after first therapeutic IMP administration. • Number of subjects with best response in PSA level ≥50% from baseline to the end of the dosing period. • PSA Progression-free survival (PFS): Time interval from first cycle of therapeutic IMP administration to PSA progression as defined by the Prostate Cancer Clinical Trials Working Group 3 (PCWG3) criteria, or death, whichever comes first. • Number of subjects who remain PSA progression-free at fixed 6-month intervals throughout the study. • Duration of response as defined by time interval from achieving a ≥50% reduction in PSA compared to baseline to the PSA returning to baseline or evidence of radiological progression, as defined by PCWG3, or death. • rPFS: Time interval from first therapeutic IMP administration to the date when the first site of disease is found to progress radiographically, or death, whichever occurs first, as defined by PCWG3. • Number of subjects who remain radiographic progression-free at fixed 6-month intervals throughout the study. • Number of subjects with confirmed complete response (CR) or partial response (PR) based on PCWG3-recommended application of the Response Evaluation Criteria in Solid Tumours v1.1 (RECIST v1.1) criteria.

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Secondary Objectives and Endpoints *(cont'd)*	Objectives: Phase 1 *(cont'd)*	Endpoints Phase 1 *(cont'd)* • Evaluation of PSMA PET/CT defined disease response after 2 treatment cycles. • Overall survival at fixed 6-month intervals throughout the study. • Time to PSA Progression: Time interval from first cycle of therapeutic IMP administration to PSA progression as defined by the PCWG3 criteria. • Time to Radiographic Progression: Time interval from first therapeutic IMP administration to the date when the first site of disease is found to progress radiographically as defined by PCWG3.
Phase 2		
Primary Objective and Endpoints	Objectives: Phase 2 1. To evaluate the efficacy of Lutetium (^{177}Lu) rhPSMA-10.1 Injection.	Endpoints: Phase 2 • Number of subjects with an anti-tumour response defined as a ≥50% reduction in PSA level from baseline to the end of treatment (EoT).
Secondary Objectives and Endpoints	Objectives: Phase 2 1. To evaluate the efficacy of Lutetium (^{177}Lu) rhPSMA-10.1 Injection.	Endpoints: Phase 2 • Number of subjects with ≥50% reduction in PSA level from baseline at 12 weeks after first IMP administration. • PSA PFS: Time interval from the date of randomisation to PSA progression as defined by PCWG3 criteria, or death. • Number of subjects who remain PSA progression-free at fixed 6-month intervals throughout the study. • Duration of response as defined by time interval from achieving a ≥50% reduction in PSA compared to baseline to the PSA returning to baseline or evidence of radiological progression, as defined by PCWG3.

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Secondary Objectives and Endpoints *(cont'd)*	Objectives: Phase 2 *(cont'd)*	Endpoints: Phase 2 *(cont'd)* • rPFS: Time interval from the date of randomisation to the date when the first site of disease is found to progress radiographically, or death, whichever occurs first, as defined by PCWG3. • Number of subjects who remain radiographic progression-free at fixed 6-month intervals throughout the study. • Number of subjects with confirmed CR or PR based on PCWG3-recommended application of RECIST v1.1. • Overall survival, defined as the time interval from the date of randomisation to death. • Overall survival at fixed 6-month intervals throughout the study. • Time to PSA progression: Time interval from the date of randomisation to PSA progression as defined by PCWG3 criteria. • Time to radiographic progression: Time interval from the date of randomisation to the date when the first site of disease is found to progress radiographically as defined by PCWG3.
	2. To further evaluate the safety profile of Lutetium (^{177}Lu) rhPSMA-10.1 Injection using the dosing regimen (RP2D) recommended by the Phase 1 results.	• Frequency and nature of TEAEs.
	3. To further characterise the whole-body distribution and dosimetry of Lutetium (^{177}Lu) rhPSMA-10.1 Injection.	• Specific absorbed dose to the tumour lesions (Gy/GBq), specific absorbed dose per organ (Gy/GBq).

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Secondary Objectives and Endpoints *(cont'd)*	Objectives: Phase 2 *(cont'd)* 4. To describe the influence of Lutetium (^{177}Lu) rhPSMA-10.1 Injection on the health-related quality of life (HRQoL) of treated subjects using Functional Assessment of Cancer Therapy – Prostate (FACT-P).	Endpoints: Phase 2 *(cont'd)* • Changes in patient-reported outcomes (PROs) assessed by FACT-P. PROs will be measured from baseline up to the EoT: o Change from baseline in FACT-P total score. o Proportion of subjects with improvement or worsening in the FACT-P total score and subscales. o Time to worsening in the FACT-P total score and subscales, defined as the time from randomisation to worsening, clinical progression, or death, whichever occurs first.
Exploratory Objective and Endpoint	Objective: Phase 2 1. To describe the influence of Lutetium (^{177}Lu) rhPSMA-10.1 Injection on the HRQoL of treated subjects using the Functional Assessment of Cancer Therapy – Radionuclide Therapy (FACT-RNT).	Endpoint: Phase 2: • Changes in PROs assessed by the FACT-RNT. The PROs will be measured from baseline up to the EoT.
Study Sites	Phase 1 will be conducted in up to 5 centres in the United States (US) and European Union (EU) (Completed). Phase 2 will be conducted in up to approximately 25 centres in the US, the EU, and the United Kingdom.	
Investigational Medicinal Product(s), Including Control Products	Phase 1 (Completed): Therapeutic IMP: Lutetium (^{177}Lu) rhPSMA-10.1 Injection. Diagnostic IMP: POSLUMA® (flotufolastat F 18 injection; previously known during its development as rhPSMA-7.3 (^{18}F) injection). Phase 1: Dose escalation schema commencing at 5.55 GBq (150 mCi) per cycle, to a maximum of 7.40 GBq (200 mCi) per cycle. Phase 2: IMP: Lutetium (^{177}Lu) rhPSMA-10.1 Injection. Phase 2: Subjects will be enrolled into 2–3 dosing cohorts: • Cohort 2A: 2 doses at 10.00 GBq (270 mCi) followed by up to 5 additional doses at 7.40 GBq (200 mCi); all doses administered at 6-weekly intervals. • Cohort 2B: Up to 8 doses at 7.40 GBq (200 mCi). The first 3 doses will be administered at 3-weekly intervals, with the remaining doses being administered at 6-weekly intervals. • Cohort 2C (optional): 2 doses at 14.80 GBq (400 mCi) followed by up to 4 additional doses at 7.40 GBq (200 mCi) administered at 6-weekly intervals.	

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Study and Subject Duration	Overall study duration: Approximately 6 years from Q1 2022. • Phase 1: approximately 3 years from Q1 2022 (Completed) • Phase 2: approximately 3 years from Q1 2025 Phase 1 Subject Duration (Completed): Dosing with the therapeutic IMP will continue depending on the number of 6-week cycles administered. All subjects will participate in efficacy follow-up until centrally-determined radiographic progression, death, or loss to follow-up, whichever occurs first. After centrally-determined radiographic progression, subjects will enter a passive long-term safety follow-up until death or loss to follow-up. Phase 2 Subject Duration: Dosing with the IMP will continue depending on the number of cycles administered. All subjects will participate in efficacy follow-up until centrally-determined radiographic progression, death, or loss to follow-up, whichever occurs first. After centrally-determined radiographic progression, subjects will enter a passive long-term safety follow-up until death or loss to follow-up.
Planned Interim Analyses	During the conduct of Phase 2, a decision will be made on whether or not to open a third cohort (Cohort 2C) based on an ongoing safety and preliminary efficacy data review in Cohorts 2A & 2B by the TSC. Following safety review, Cohort 2A or 2B may be closed for futility (i.e. estimated PSA50 response rate ≤20% within a cohort). This decision will be based on the recommendation of the TSC. In Phase 2, the Sponsor may perform additional interim reviews of the emerging data to assess preliminary efficacy and safety. Full details of such reviews will be described in the SAP. Due to the exploratory nature of this study, no adjustment for multiplicity (i.e. alpha) will be made.
Statistics: Primary Analysis Plan and Rationale for Number of Subjects	Phase 1 (Completed): In Phase 1, efficacy endpoints will be summarised descriptively using point estimates and measures of uncertainty. For example, the number of subjects with a PSA reduction ≥50%, with 95% confidence intervals (CIs); median duration of response estimated using Kaplan-Meier methodology, together with 95% CIs. At least 12 subjects (Cohort A: at least 3 subjects; Cohort B: at least 9 subjects) are proposed for the Phase 1 dose escalation part of the study, which is consistent with other early phase studies and enables collation of high-quality radiation dosimetry data to inform later clinical development as well as first assessment of safety. Phase 2: In Phase 2, point estimates for the number of subjects with PSA50 responses will be presented together with 95% CIs for each cohort. To compare the differences between the cohorts, the difference in PSA50 responses will be calculated, along with 95% CIs. This summary will be for descriptive purposes only. Median duration of response will be estimated using Kaplan-Meier methodology and presented together with 95% CIs. Methods will also be used to compare the rates in each cohort to a historical control. Bayesian methods may also be used to estimate probability distributions and subsequent inferences, for example, to estimate the probability that the PSA response rate exceeds a historical rate by a certain amount. Full details will be described in the SAP. Secondary endpoints will be summarised descriptively, with 95% CIs, where applicable.

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Statistics: Primary Analysis Plan and Rationale for Number of Subjects *(cont'd)*	For Phase 2, the pooled cohort size of approximately 70 subjects was selected for enrolment and treatment based on simulations under a range of scenarios for PSA50 response rate when performing interim analyses after 20 evaluable subjects, then subsequently every 10 evaluable subjects using Bayesian posterior probabilities (Lee 2020). In particular, under a range of potential scenarios, the simulations showed that the average sample size will be between 51 and 61 (hence the approximately 70 subjects) to enable reliable estimation of PSA50 response when compared to relevant historical controls (Morris 2024). The individual cohort sample size of approximately 30 subjects per cohort was selected to provide sufficient exploratory data on dose optimisation without exposing an excessive number of subjects to the IMP at this stage of development. The Cohort 2C sample size (approximately 10 subjects) is based on assessing gross differences in efficacy or safety. If data from this cohort are compelling, a protocol amendment may be submitted to increase its sample size.

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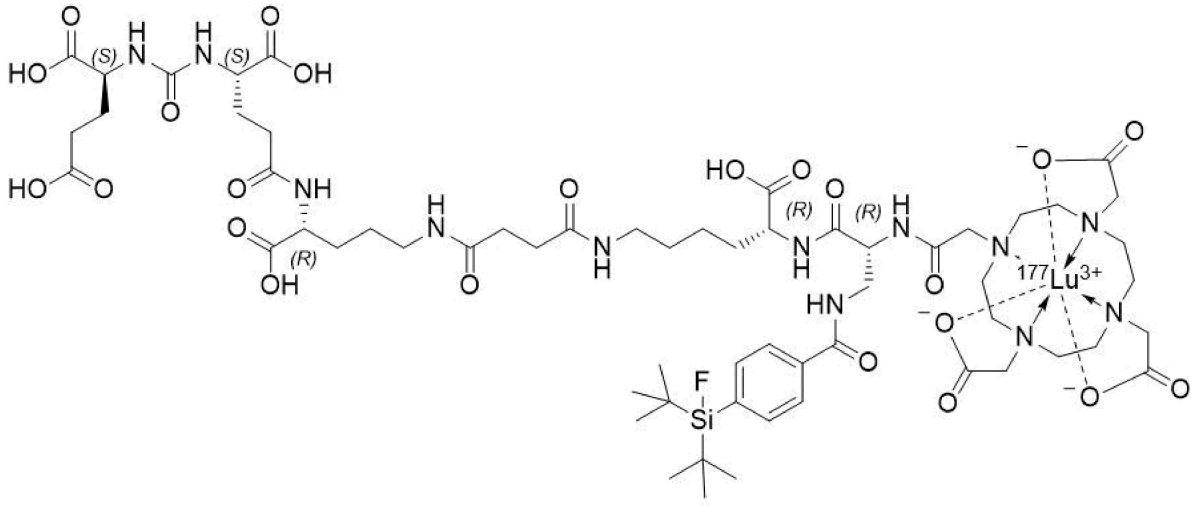
1. BACKGROUND

The therapeutic investigational medicinal product (IMP) is Lutetium (^{177}Lu) radiohybrid prostate-specific membrane antigen (rhPSMA)-10.1 injection (hereafter also referred to as the therapeutic IMP or IMP), an investigational radiopharmaceutical solution containing rhPSMA-10.1 and ^{177}Lu rhPSMA-10.1. rhPSMA-10.1 and ^{177}Lu rhPSMA-10.1 are respectively the chemical precursor / non-radioactive carrier and radiopharmaceutical drug substance, in which the beta emitting radionuclide ^{177}Lu with a radioactive half-life of 6.7 days provides a means for systemic radiation therapy.

^{177}Lu-rhPSMA-10.1 is designed to target prostate-specific membrane antigen (PSMA), also known as glutamate carboxypeptidase II or folate hydrolase 1, which is a 750-amino acid type II transmembrane glycoprotein with significantly increased expression (100- to 1000-fold) in prostate cancer cells when compared to other PSMA-expressing tissues such as the kidney, proximal small intestine or salivary glands (Smith-Jones 2000, Ghosh 2004).

The molecular structure of the drug substance comprises a PSMA binding motif (Glutamate-urea), conjugated via a peptide-like spacer to a silicon-fluoride acceptor moiety (SiFA) and lutetium-chelator complex (^{177}Lu-DO3A) (Figure 1).

Figure 1: Structure of ^{177}Lu-rhPSMA-10.1



1.1 Overview of Non-clinical Studies

A number of non-clinical dosimetry studies with ^{177}Lu-rhPSMA-10.1 have been performed in different mouse models in order to establish biodistribution and dosimetry, with an additional biodistribution study in minipigs.

Data from the biodistribution and dosimetry studies showed a typical uptake pattern known from other PSMA ligands with rapid blood clearance and majority of the radioactivity excreted through the kidneys and urinary bladder. Brain values were low for all time periods, consistent with the molecule not crossing the blood brain barrier.

Overall, results from the non-clinical studies support the development of ^{177}Lu-rhPSMA-10.1 to be taken forward into clinical development.

Further details of the non-clinical studies performed with ^{177}Lu-rhPSMA-10.1, including Good Laboratory Practice toxicology studies are provided in the Lutetium (^{177}Lu) rhPSMA-10.1 Injection Investigator's Brochure (IB).

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1.2 Overview of Clinical Studies

The Sponsor (Blue Earth Therapeutics Limited) is beginning the clinical development programme for Lutetium (^{177}Lu) rhPSMA-10.1 Injection with an open-label, multicentre, integrated Phase 1 & 2 study to evaluate the safety, tolerability, radiation dosimetry and anti-tumour activity of the therapeutic IMP in men with metastatic castrate-resistant prostate cancer (mCRPC). This study will consist of 2 parts: a non-randomised Phase 1 part, with safety, dose-finding, and dosimetry components, and a randomised Phase 2 part with efficacy and safety assessments and testing dosing regimens selected following analysis of the safety and dosimetry data in Phase 1. In the Phase 1 component, subjects will be enrolled into 1 of 2 dose levels, which will investigate the safety, tolerability, and radiation dosimetry of the therapeutic IMP. The results of the Phase 1 component will be used to determine the recommended treatment regimen (recommended Phase 2 doses [RP2D]) to be tested in the Phase 2 component.

As of Protocol Version 5.0, the treatment phase in the Phase 1 part of the study is completed, and all sections related to Phase 1 in the protocol are marked as completed. A data summary from the Phase 1 part of the study is presented in Section 2.1.2. The Phase 2 component of this study will enrol approximately 70 subjects who will be randomised into 2-3 dosing cohorts to investigate the safety, efficacy, radiation dosimetry, and dose optimisation of the IMP.

2 STUDY RATIONALE

2.1 Background

Prostate cancer is the second most frequent cancer diagnosis made in men, and the fifth leading cause of death worldwide (Rawla 2019). Prostate cancer is most commonly diagnosed in men aged >65 years (Daniyal 2014). It is largely asymptomatic in the early stages, with tumours often detected by increased concentrations of prostate-specific antigen (PSA) in peripheral blood and/or an abnormal digital rectal examination (Daniyal 2014; Kohaar 2019). Diagnosis is confirmed by a prostate tissue biopsy (Kohaar 2019). Prostate-specific antigen is used as a tumour biomarker (Kohaar 2019) and serum levels of PSA have been shown to positively correlate with the risk of metastatic disease or subsequent disease recurrence or progression (Kamaleshwaran 2012; Esfahani 2015; Ecke 2016; Nishimura 2018).

Primary treatment of prostate cancer can be divided into radical and non-radical treatment, the former applicable predominantly to patients with curable disease and the latter for those whose disease has spread, and treatment is aimed at prolonging life. Radical primary treatment options include radical prostatectomy, radical external beam radiation therapy (EBRT) and brachytherapy. In patients without curable disease, the current mainstay of treatment is based on androgen deprivation therapy (ADT). Standard ADT (e.g. gonadotropin-releasing hormone [GnRH] agonists/antagonists), which alters pituitary sex hormone signalling to switch off the production of androgens, and the newer novel androgen axis drugs (NAAD), which can be given together in combination with GnRH agonists/antagonists, can stabilise metastatic prostate cancer for an average of 12 to 24 months after which point the disease becomes castrate-resistant. When this happens, eligible patients can commence NAAD (e.g. abiraterone, enzalutamide) if they have not already received them during the hormone-sensitive phase. NAAD have been shown to extend survival in the castrate-resistant setting (Schaeffer 2021) however, patients will continue to progress as their disease no longer responds to the suppression of androgen signalling. Patients with metastatic castrate-resistant disease progression following treatment with NAAD have limited treatment options with taxane-based

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chemotherapy the mainstay. Recently, inhibitors of poly (adenosine diphosphate-ribose) polymerases (e.g. olaparib) have shown efficacy in a subset of men who carry germline mutations in deoxyribonucleic acid (DNA) repair enzymes. However, treatment options remain limited and survival following progression on chemotherapy is limited to months. Increasingly, systemic radiation therapy through the use of radiopharmaceuticals targeting PSMA is emerging as a promising investigational class of compound in men who have progressed following ADT/NAAD and are castrate-resistant (Yadav 2019). Such agents may permit exploitation of the inherent radiation sensitivity of prostate cancer by delivering ionising radiation to disseminated lesions in a way that EBRT cannot.

The therapeutic effectiveness of systemic radiopharmaceuticals is determined by the specificity of targeting and the innate characteristics of the particle emitted, including potential range, effective travel distance in tissue, and the magnitude of energy emitted. Radionuclides are classified by their emission of particles, including alpha and beta particles (De Vincentis 2019). Beta particles induce their anti-cancer effects by causing DNA damage. The ability to induce DNA damage drives efficacy but also underlies toxicity experienced due to normal tissue damage (e.g. irradiation of haematological progenitor cells in the bone marrow during circulation).

It is reported that only 10% of primary prostate tumours or metastatic lesions are PSMA-negative (Maurer 2016). Prostate-specific membrane antigen's transmembrane location in cancer, with a large extracellular domain and its enzyme activity, allow for the development of specific inhibitors which bind to the enzymatic site and their subsequent internalisation after ligand binding (Kiess 2015). Although 5 to 10% of men with prostate cancer do not express PSMA, and PSMA may not be expressed uniformly within the tumour, high levels of PSMA expression are observed across the spectrum of aggressive and poorly differentiated prostate cancer, making PSMA a promising therapeutic target (De Vincentis 2019). In addition, the increasing availability of PSMA-targeted positron emission tomography (PET)/computed tomography (CT) imaging agents permits the selection of patients with PSMA expression in advance of therapeutic dosing. Those patients with significant PSMA-negative prostate cancer who are less likely to benefit from a PSMA-targeted agent can be identified in advance.

Investigational therapeutic agents targeting PSMA can be radiolabelled with different radionuclides, including alpha (e.g. actinium-225 [^{225}Ac] and thorium-227 [^{227}Th]) and beta emitters (e.g. Lutetium-177 [^{177}Lu] and iodine-131 [^{131}I]). These agents can be targeted to PSMA via different vehicles, such as antibodies (e.g. ^{227}Th-radiolabelled anti-PSMA Immunoglobulin G1) and small-molecule ligands (e.g. ^{225}Ac-PSMA-617) (De Vincentis 2019). Early clinical studies reporting surrogate markers, such as PSA response rates, supported the assertion that PSMA-targeted systemic radiopharmaceuticals are a promising class of investigational compounds (Kratochwil 2019). In 2022, the therapeutic radioligand lutetium (^{177}Lu) vipivotide tetraxetan (^{177}Lu PSMA-617; Pluvicto™) was approved for the treatment of men who have progressed following NAAD and taxane chemotherapy in mCRPC. Efficacy and safety data supporting approval were from two prospective, open-label randomised controlled trials in subjects with mCRPC (TheraP Phase 2 trial [NCT03392428; Hofman et al, 2021; Hofman et al, 2022] and VISION Phase 3 trial [NCT03511664; Sartor et al 2021; Morris et al, 2021]). In the more recent PSMAfore trial, lutetium (^{177}Lu) vipivotide tetraxetan was shown to prolong radiographic progression-free survival (rPFS) compared to androgen receptor pathway inhibitor change in taxane-naïve patients with PSMA-positive mCRPC (Morris et al 2024).

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2.1.1 Rationale for Study

Available data for radioligand therapy support a strong correlation between the amount of radiation delivered to the tumour and the response to treatment. Whole-body mean standardised uptake values (SUV_{mean}) have been reported to correlate with absorbed dose to tumour lesions and PSA response (Violet 2019), and a higher whole-body SUV_{mean} is associated with improved survival (Kuo 2022). Based on the available data and precedent in other therapeutic areas, enhancing the absorbed radiation dose to the tumour with PSMA-targeted radioligands may achieve improved clinical outcomes in patients with PSMA-positive prostate cancer.

The existing dosing regimens for PSMA-targeted radioligands have often not been informed by formal dose-ranging studies, posing the question as to whether these dosing regimens have been optimised to maximise therapeutic efficacy in this class of compounds.

Pre-therapeutic dosimetry with 1 compound from the class of rhPSMA compounds (^{177}Lu-rhPSMA-7.3C) has been recently reported (Feuerecker 2022). Encouragingly, it suggested that the binding and kinetic characteristics of ^{177}Lu-rhPSMA-7.3C could lead to similar improved absorbed radiation doses to tumour deposits, yet at lower injected radioactivity levels compared with other ^{177}Lu-radiolabelled PSMA-targeted compounds in more advanced stages of development. In this study of 6 patients, tumour uptake was approximately 2.5 × higher in an intra-patient comparison with ^{177}Lu-PSMA-I&T with similarly elevated key normal organ uptake. Building on this observation, adjustments have been made to the structure of ^{177}Lu-rhPSMA-7.3C to optimise its pharmacological and pharmacokinetic (PK) properties for PSMA-targeted radioligand therapy, in particular to reduce normal organ uptake, which has yielded the selected compound ^{177}Lu rhPSMA-10.1. Non-clinical dosimetry data for ^{177}Lu rhPSMA-10.1 indicate preserved or enhanced tumour uptake with improved tumour to normal organ ratios including the kidney, which is the dose-limiting organ with this class of compounds.

The IMP belongs to a class of compounds known as 'radiohybrids' with many versions of the class tested pre-clinically for properties that might improve the therapeutic index in humans. Fourteen subjects to date have received the actual IMP (Lutetium (^{177}Lu) rhPSMA-10.1 Injection) at the Department of Nuclear Medicine, University Hospital Augsburg, Germany under their physician's personal responsibility (exempt from a manufacturing authorisation as per Section 13, Subsection 2b of the German Medicinal Products Act). Four of these patients received a low dose of 1 GBq of Lutetium (^{177}Lu) rhPSMA-10.1 Injection and 1 GBq of ^{177}Lu-PSMA-I&T separated by 1 week, for the purpose of pre-therapeutic dosimetry, and then therapeutic cycles of 7.40 to 7.70 GBq of Lutetium (^{177}Lu) rhPSMA-10.1 Injection for up to 6 cycles, with each cycle separated by 6 weeks. Dosimetry data for these 4 patients confirmed a distribution of radiation in keeping with the literature describing other ^{177}Lu-radiolabelled PSMA-targeted radiopharmaceuticals and tumour to kidney absorbed dose ratios were improved, when compared with ^{177}Lu-PSMA-I&T on an intra-patient basis (Rinscheid 2023). Therapeutically, all 4 patients showed a decrease of PSA, and 3 of the 4 patients experienced a decline in PSA of >80%, with radiological responses to treatment. Of these 3 patients, 1 had a complete biochemical and radiological response to treatment. Progression-free survival (PFS) was not reached for 2 patients at 24 months and 18 months of follow-up, respectively, and was 12 months and 15 months in the other 2 patients. As of July 2023, all patients were alive with one patient showing an ongoing complete radiologic and biochemical response more than 2 years after starting radioligand therapy (RLT) (Dierks 2024).

In summary, the selected IMP utilises the same PSMA-targeting moiety and the same beta emitting radioisotope (^{177}Lu) as ^{177}Lu-rhPSMA-617, which demonstrated improved overall survival in a Phase 3 randomised clinical trial (RCT) in men with mCRPC alongside other

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encouraging trial data. In addition, a very similar compound to the IMP, and the IMP itself, have been utilised in Germany in patients with mCRPC and demonstrated preliminary evidence of efficacy at doses similar to those being tested in this protocol. However, neither of these German experiences took place within formal clinical trials. Furthermore, non-clinical data are supportive of the potential for properties, which may enhance the therapeutic index in humans compared to other similar agents in development. Taking these data into account, there is sufficient evidence to suggest the development of this therapeutic IMP has the potential to benefit patients with mCRPC and warrants evaluation in clinical trials.

2.1.2 Preliminary Data from Phase 1

Preliminary Phase 1 dosimetry data were available for 13 subjects at Cycle 1, 11 subjects at Cycle 2, and 10 subjects at Cycle 3. Absorbed doses per unit activity among at-risk organs were highest in the lacrimal glands, kidneys, and salivary glands; in keeping with data from other ^{177}Lu-PSMA RLTs. A summary of absorbed doses for key at-risk organs are shown in Table 1. Tumour dosimetry in the Phase 1 part of the study was in a sufficient range to plausibly offer therapeutic benefit to patients with mCRPC.

Overall, preliminary data from the Phase 1 part of the study demonstrated tumour to normal organ ratios that compare favourably to the class of compounds and were supportive of the continued development of the IMP. The data from Phase 1 were supportive of continuing development to Phase 2.

A detailed summary of the currently available data from the Phase 1 component of the study is presented in the Lutetium (^{177}Lu) rhPSMA-10.1 Injection IB.

Table 1: Phase 1 Normal Organ Dosimetry Summary

At-risk Organ	Mean Absorbed Dose ± SD, Gy/GBq
Lacrimal Glands	0.620 ± 0.33
Kidneys	0.299 ± 0.099
Salivary Glands	0.134 ± 0.078
Bone Marrow (Image-based)	0.011 ± 0.020
Bladder Wall	0.018 ± 0.01
Brain	0.003 ± 0.004
Heart	0.014 ± 0.003
Intestine	0.002 ± 0.001
Liver	0.035 ± 0.019
Lungs	0.050 ± 0.056
Stomach Wall	0.004 ± 0.003

Abbreviations: GBq=Gigabecquerel; Gy=gray; SD=standard deviation.

2.2 Known and Potential Benefits and Risks to Participating Subjects

2.2.1 Potential Benefits to Participating Subjects

Prostate-specific membrane antigen-targeted small molecules radiolabelled with Lutetium (^{177}Lu) have been shown to improve overall survival for patients with mCRPC that has progressed following NAAD and chemotherapy (Sartor 2021). The Phase 3 VISION RCT (NCT03511664; Endocyte) investigated ^{177}Lu-PSMA-617 given at 7.40 GBq with 6-weekly

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intervals (median of 5 cycles) compared with best supportive care in subjects with mCRPC that has progressed following NAAD treatment and chemotherapy. Overall survival was extended in the intervention arm by 4 months (Hazard Ratio [HR] 0.62; 95% Confidence Interval [CI] 0.52-0.74; p<0.001) when compared with the control arm. In addition, rPFS was extended by 5.3 months (HR 0.40; 99% CI 0.29-0.57; p<0.001). The Phase 2 TheraP study tested ^{177}Lu-PSMA-617 given at 8.5 GBq (reducing by 0.5 GBq per cycle) at 6-week intervals in men with mCRPC with progression following NAAD treatment, where cabazitaxel was considered the next appropriate standard treatment (Hofman 2021). This RCT compared the ^{177}Lu-PSMA-617 group with cabazitaxel in the control group. The proportion of men with a PSA reduction of ≥50% was 66% in the intervention group vs 44% in the control group. The proportion of subjects who had not progressed in the intervention group at 12 months follow-up was 19% (95% CI 12 to 27%) compared with 3% (95% CI 1 to 9%) in the control group. In the more recent PSMAfore trial, lutetium (^{177}Lu) vipivotide tetraxetan was shown to prolong rPFS compared to androgen receptor pathway inhibitor change in taxane-naïve patients with PSMA-positive mCRPC (Morris 2024). These 3 prospective RCTs demonstrate that PSMA-targeted small molecules that are radiolabelled with ^{177}Lu offer benefit to patients with mCRPC.

The purpose of this integrated Phase 1 & 2 study is to evaluate whether the IMP, which is also a ^{177}Lu-radiolabelled small molecule with PSMA as a target, has the potential to offer benefit to subjects with mCRPC.

2.2.2 General Risks of Participating in Early Phase Clinical Trials

2.2.2.1 Phase 1 (Completed)

The therapeutic IMP being investigated in this integrated Phase 1 & 2 study has not been formally evaluated in humans. The Phase 1 component of the study design involves establishing the optimal dose, so subjects may receive a dose which is insufficient to exert any therapeutic effect, or they may receive a dose that causes toxicity.

2.2.2.2 Phase 2

In the Phase 2 component, the study design aims to evaluate the efficacy and further characterise the safety of the IMP. There may be no efficacy observed with the IMP, or a reduced efficacy compared to other treatment options available to subjects taking part. In addition, further safety information may emerge at any point in clinical development as the number of subjects exposed to the IMP increases and different administered radioactivity and treatment schedules are tested. Based on the data available, the benefit-risk assessment is in favour of the continuing development of the IMP at the start of this study. However, at any stage during clinical development, it may be necessary to stop further development as the benefit-risk assessment no longer supports continuing.

2.2.3 Investigational Medicinal Product Specific Risks

2.2.3.1 Phase 1 (completed)

It is anticipated that the risks of participating in this study relate predominantly to exposure to radiation. The administered activity of Lutetium (^{177}Lu) rhPSMA-10.1 Injection is between 5.55 GBq and 7.40 GBq, with a minimum 6-week interval between therapeutic doses. As the product is administered by a medical professional in a hospital setting, the risk of administering higher amounts of activity is reduced. The injected radioactivity is measured prior to

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administration, and by measuring residual activity in the injection equipment after administration. A tolerance of ±10% injected radioactivity is generally permitted in nuclear medicine and would usually not be deemed an overdose or underdose, respectively. An overdose of >10% of the therapeutic IMP or >20% of the diagnostic IMP is considered an AE and should be reported as such. The short/acute risks and potential longer-term risks are outlined below.

2.2.3.2 Phase 2

The Phase 2 component of this study is investigating which dosing regimen will provide the most appropriate balance between risk and benefit. In Cohort 2A and 2C, subjects will receive 10.00 GBq and 14.80 GBq activity for the first 2 cycles, respectively, followed by multiple cycles of 7.40 GBq activity. In Cohort 2B, subjects will receive more frequent dosing of 7.40 GBq activity, with initially shorter dosing intervals. Both of these approaches may lead to a higher frequency or severity of toxicity compared to currently available therapies in the class of PSMA RLT, without improving efficacy.

In addition, subjects participating in Phase 2 have not received chemotherapy for mCRPC. For those subjects who are eligible to receive chemotherapy, they may experience greater benefit by not participating in this trial and instead choosing standard of care chemotherapy, which has been shown to be a safe and efficacious treatment in this setting.

2.2.3.3 Short-term/Acute Risks of Radiation

2.2.3.3.1 Phase 1 (Completed)

The Phase 3 VISION trial provides valuable data concerning the potential toxicity that a subject may experience acutely whilst taking part in a PSMA-targeting therapeutic study and should be reviewed by investigators participating in this study (Sartor 2021). It is expected that the safety data observed with the therapeutic IMP will be similar to that seen in the VISION trial. As well as targeting PSMA-expressing tumour deposits, a systemically injected PSMA-targeted radiopharmaceutical also delivers radiation to normal organs and tissues. The radiation dose delivered to key normal organs and tissues defines the potential risks for subjects. In the VISION trial, the most commonly experienced toxicity was haematological toxicity. Forty-seven percent of subjects experienced some bone marrow suppression, of which 23% were Grade 3 or 4. Anaemia (12.9%; Grade 3-4) and thrombocytopenia (7.9%; Grade 3-4) were the most commonly affected cell lineages. The haematological toxicity experienced in the VISION trial is most likely caused by radiation dose delivered to progenitor cells in the bone marrow whilst the radioactive compound is circulating in plasma. Whilst in most cases myelosuppression would be expected to resolve with time, there is the potential for harm to occur to subjects whilst their bone marrow is suppressed e.g. cardiovascular events secondary to anaemia, bleeding events secondary to thrombocytopenia and infections during periods of low white cell counts. In addition, prolonged myelosuppression lasting weeks or months may occur. Dry mouth and potentially dry eyes may be caused by uptake of the compound in the salivary and lacrimal glands. In the VISION trial, 39% of subjects experienced a dry mouth (no Grade 3-4 events). Based on the experience reported with other ^{177}Lu-PSMA agents, toxicity experienced by subjects is expected to resolve over time. Additional potential risks exist from the radiation exposure received by the kidney, which is the organ of excretion and has tissues that express the PSMA receptor. In the short-term, this radiation exposure to the kidney may cause an acute deterioration in renal function. In addition to haematological and renal toxicity, subjects in the VISION trial frequently reported fatigue (49% all grades, 7% Grade 3-4), and nausea and vomiting (39.3% all grades, 1.5% Grade 3-4). There may be

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other short-term/acute toxicities from receiving the therapeutic IMP that are not yet known and were not observed with other PSMA-targeted small molecule radiopharmaceutical compounds.

2.2.3.3.2 Phase 2

The short-term/acute effects described in the Phase 1 section remain ongoing risks in Phase 2. Haematological toxicity was limited in Phase 1, with 1 subject each experiencing Grade 2 and Grade 3 anaemia, and another subject experiencing Grade 2 thrombocytopenia. However, in Phase 2, the radioactivity under evaluation is significantly higher in terms of both single injections, and cumulatively injected radioactivity over all cycles. Therefore, the risk remains that observed haematological toxicity may be more frequent or more severe than has been reported in other trials of the same class of PSMA RLT. Additionally in Phase 1, 8 out of 13 subjects experienced Grade 1 dry mouth. It is plausible that in Phase 2, this toxicity may increase in frequency and severity. Similarly, whilst no acute deterioration in renal function was observed in Phase 1, the higher cumulative radioactivity under evaluation in Phase 2 means that this risk remains and will continue to require careful monitoring.

A detailed summary of the currently available safety data from the Phase 1 component of the study is presented in the Lutetium (^{177}Lu) rhPSMA-10.1 Injection IB.

2.2.3.4 Potential Long-term Risks of Radiation (Phase 1 and Phase 2)

A delayed onset radiation nephropathy is a plausible risk based on the literature derived from EBRT and there have been some cases observed with PSMA-directed RLT at very high cumulative injected activities, where patients developed Grade 3 to 4 chronic kidney disease with biopsy-confirmed severe acute tubular injury and subacute glomerular thrombotic microangiopathy (Schäfer 2023). The cells of the kidney are slowly repairing and therefore, radiation toxicity can present as a delayed reaction. Preliminary data presented in the literature report long-term declines in renal function in a proportion of patients receiving PSMA RLT (Steinhelfer 2024). The relationship between dose or dosing schedules and the clinical presentation is not well understood, but renal dysfunction can present anywhere from 6 months to years after radiation exposure (Breitz 2004). It is not yet established whether there is a generalisable dose limit that should be applied to the kidney in order to reduce the risk of radiation nephropathy. The shortened life expectancy of the subject population in this study may mean observing the onset of a delayed renal impairment is not always possible. As a result of this plausible risk, subjects will be followed until death or loss to follow-up in order to capture any delayed renal toxicity that might be attributed to the therapeutic IMP and radiation exposure. A small risk (approximately 2 to 3% during remaining lifetime) of developing a secondary haematological malignancy due to the radiation exposure to the bone marrow during treatment has also been described based on experience with other ^{177}Lu-radiolabelled small molecules, which target another cancer via the somatostatin receptor (Lutathera® Product Information). Acute myeloid leukaemia and a myelodysplastic syndrome have both been observed with these ^{177}Lu-radiolabelled systemically delivered radiopharmaceuticals, occurring approximately 55 months and 28 months after the End of Treatment (EoT), respectively (Lutathera® Product Information). The short life expectancy of the subjects enrolled in this study means it may not be possible to observe the onset of these conditions. Long-term follow-up until death or loss to follow-up will be performed to ensure any such events in subjects are captured. There may be other longer-term risks from receiving this IMP which are not yet known and were not observed with other PSMA-targeted small molecule radiopharmaceutical compounds.

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To assess the short-term and potential long-term risks described above, quantitative radiation dosimetry will be performed for each subject receiving therapeutic doses of Lutetium (^{177}Lu) rhPSMA-10.1 Injection and the data provided to the treating investigator. In Phase 1, multi-timepoint dosimetry will be performed in the days after dosing for every cycle of therapy. In Phase 2, a reduced dosimetry protocol will be performed with multi-time point dosimetry being acquired at Cycle 1, and then single time point dosimetry being performed from Cycle 2 onwards. The rationale for this decision is directly evidenced by the dosimetry data acquired in Phase 1 of this study and will allow the subject imaging burden to be reduced whilst maintain sufficient dosimetric accuracy. These data will be used by the investigator to ensure key organ defined dose limits are not exceeded, and that the exposure of normal organs and tissues to radiation is quantified.

Subject selection in this study will include pre-therapeutic PSMA PET/CT imaging. This is to identify subjects with PSMA-positive prostate cancer and to exclude IMP administration to those with no/minimal PSMA expression, such that they are not exposed to the risks of therapeutic radiation when the likelihood of benefit is small. In addition, the inclusion criteria have been selected to ensure subjects have acceptable bone marrow reserve, and that baseline renal function is >60 mL/min.

In addition to dosimetry assessments, to assess the risks of radiation exposure, subjects will return after dosing for haematology, coagulation, and biochemistry laboratory testing as well as adverse event (AE) monitoring. Subjects in Phase 1 will also have thyroid functional panel laboratory testing. The decision to give the subject subsequent dose cycles will depend on individual safety and tolerability, and cumulative absorbed radiation doses to normal tissues and organs. Further cycles of Lutetium (^{177}Lu) rhPSMA-10.1 Injection will not be given to subjects who did not tolerate previous administration or in whom the next dose would exceed the protocol-defined normal organ radiation absorbed dose limits (see Table 3). For Phase 1, the occurrence of a dose-limiting toxicity (DLT) at any point from the first cycle of therapy to 6 weeks after the final cycle of therapy will lead to stopping dosing for that subject (see Section 4.1.1.4). For Phase 2, rules regarding dose delay and discontinuation will be applied per protocol (see Section 4.1.2.2).

2.2.4 Specific Risks Relating to Diagnostic Scans and Dosimetry Assessments

The risks from the diagnostic scans mainly relate to the intravenous (IV) injection, the relatively low radiation emitted by the radiopharmaceutical (i.e. PSMA PET/CT and technetium-99m [^{99m}Tc] bone scan) and the radiation emitted by the CT (i.e. PET or single-photon emission computed tomography / computed tomography [SPECT/CT] transmission scan, or contrast enhanced computed tomography [ceCT] scan).

Intravenous injection carries a small risk of injection site reactions, infection, and haematoma. There is a potential risk of a hypersensitivity reaction requiring the availability of resuscitative equipment and personnel.

For Phase 1, the mean effective dose per unit administered activity of rhPSMA-7.3 (^{18}F) injection is 14.1 microsievert (µSv)/Megabecquerel (MBq) (Tolvanen 2021). An administered activity of 8 millicurie (mCi) (296 MBq) (recommended dose) will result in an effective dose of 4.2 millisievert (mSv).

For Phase 2, the effective dose from the administration of a PSMA PET/CT tracer injection will vary depending on the tracer used, as well as the site-specific protocol that is employed. As a guide, the effective dose is expected to be between 4 mSv and 7 mSv.

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The effective dose due to the CT scans (CT transmission scan associated with PET or SPECT, and ceCT scan) should be in accordance with the ALARA (as low as reasonably achievable) principle. The maximum effective dose due to a CT transmission scan will vary from site to site, but as a guide, a dose ranging from 0.5 mSv to 7 mSv could be expected (McCollough 2015).

Examples of effective doses from the ceCT scan for anatomical cross-sectional imaging are as follows: pelvic CT 6 mSv; abdominal CT 8 mSv; chest CT 7 mSv. The effective dose from a bone scan would be 4 mSv (McCollough 2015).

The cumulative radiation exposure received from diagnostic imaging in this protocol is relatively low in comparison to the radiation exposure received from the administration of doses of the IMP and is of less concern in subjects with metastatic cancer and severely shortened life expectancy from their disease.

3 STUDY OBJECTIVES

3.1 Phase 1 (Completed)

3.1.1 Primary Objective

The primary objective in Phase 1 of the study is to establish the RP2D regimen by evaluation of the safety and tolerability of IV administration of Lutetium (^{177}Lu) rhPSMA-10.1 Injection in subjects with mCRPC.

3.1.2 Secondary Objectives

The secondary objectives in Phase 1 of the study are:

1. To determine the whole-body radiation dosimetry and PK of Lutetium (^{177}Lu) rhPSMA-10.1 Injection.
2. To determine the organ-specific radiation dosimetry of Lutetium (^{177}Lu) rhPSMA-10.1 Injection (organ exposure to radiation) after each administration.
3. To describe observed anti-tumour activity of Lutetium (^{177}Lu) rhPSMA-10.1 Injection.

3.2 Phase 2

3.2.1 Primary Objective

The primary objective in Phase 2 of the study is to evaluate the efficacy of Lutetium (^{177}Lu) rhPSMA-10.1 Injection.

3.2.2 Secondary Objectives

The secondary objectives in Phase 2 of the study are:

1. To evaluate the efficacy of Lutetium (^{177}Lu) rhPSMA-10.1 Injection.
2. To further evaluate the safety profile of Lutetium (^{177}Lu) rhPSMA-10.1 Injection using the dosing regimen (RP2D) recommended by the Phase 1 results.
3. To further characterise the whole-body distribution and dosimetry of Lutetium (^{177}Lu) rhPSMA-10.1 Injection.

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4. To describe the influence of Lutetium (^{177}Lu) rhPSMA-10.1 Injection on the health-related quality of life (HRQoL) of treated subjects using Functional Assessment of Cancer Therapy – Prostate (FACT-P).

3.2.3 Exploratory Objective

The exploratory objective in Phase 2 of the study is to describe the influence of Lutetium (^{177}Lu) rhPSMA-10.1 Injection on the HRQoL of treated subjects using the Functional Assessment of Cancer Therapy – Radionuclide Therapy (FACT-RNT).

4 STUDY DESIGN

4.1 Study Overview

This is an interventional, open-label, integrated Phase 1 & 2 study to assess the safety, radiation dosing regimen and anti-tumour activity of Lutetium (^{177}Lu) rhPSMA-10.1 Injection in men with PSMA-positive mCRPC, which has progressed following prior therapy. The study will consist of 2 parts: Phase 1 with safety, dose-finding and dosimetry components, and Phase 2, with an assessment of efficacy and safety utilising the dose selected from Phase 1. In the non-randomised Phase 1 component, subjects will be enrolled into 1 of 2 dose levels, which will investigate the safety, tolerability, and radiation dosimetry of the therapeutic IMP. Safety and radiation dosimetry results of the Phase 1 component will be used to determine the recommended treatment dose and regimen (RP2D) for testing in the Phase 2 component.

In the randomised Phase 2 component of this study, subjects will be initially randomised to one of 2 dosing regimens (Cohort 2A and Cohort 2B), with the option to open a third cohort (Cohort 2C) based on ongoing safety and preliminary efficacy review. The decision on whether or not to open Cohort 2C will be made by the Trial Steering Committee (TSC) after reviewing all safety and preliminary efficacy data and when approximately 35 subjects have been randomised and dosed in Cohorts 2A and 2B.

Approximately 82 subjects will be enrolled into the study, with at least 12 subjects (Cohort A: at least 3 subjects; Cohort B: at least 9 subjects) in Phase 1 and approximately 70 subjects in Phase 2 (Cohorts 2A and 2B: approximately 30-35 subjects each; Cohort 2C [if opened]: approximately 10 subjects). The initial dose-finding component of the study (Phase 1) will be conducted in up to 5 centres in the United States (US) and the European Union (EU); the treatment phase is completed. Phase 2 will be conducted in up to approximately 25 centres in the US, the EU and the United Kingdom (UK).

The planned total study duration is approximately 6 years from Q1 2022, with the Phase 1 part of the study planned for approximately 3 years from Q1 2022, and the Phase 2 part of the study planned for 3 years from Q1 2025.

4.1.1 Phase 1 (Completed)

Following consent and a preliminary eligibility check, subjects will have a PET/CT scan to ascertain whether the subject has PSMA-positive disease. The tracer used for this PET/CT scan will be POSLUMA® (flotufolastat F 18 injection; previously known during its development as rhPSMA-7.3 (^{18}F) injection) (hereafter also referred to as the diagnostic IMP) at an administered activity of 296 MBq (8 mCi). The presence of focal avid lesions on the scan image will be interpreted as PSMA-positive disease using an assessment tool based on visual interpretation of the image by a local reader (criteria described in the inclusion criteria and Image Acquisition Guidelines). To be eligible for treatment, subjects must have

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PSMA-positive disease on the PET/CT with no significant PSMA-negative disease detected on baseline cross-sectional imaging (ceCT or magnetic resonance imaging [MRI]). The criteria for 'significant' PSMA-negative disease are described in the exclusion criteria and Image Acquisition Guidelines. Subjects without any PSMA-positive disease or with significant PSMA-negative disease will be considered screen failures.

Eligible subjects with PSMA-positive disease will sequentially enter into one of the two cohorts testing two dosing regimens of the therapeutic IMP:

- Cohort A: 5.55 GBq (150 mCi) (maximum of 3 cycles)
- Cohort B: 7.40 GBq (200 mCi) (maximum of 3 cycles [subject to the recommendation of a Safety Review Committee {SRC}])

Following the first dose, and depending on individual tolerability and safety, with particular attention to parameters relating to estimated absorbed radiation doses to normal organs, subjects will receive additional doses at the same radioactivity level separated by 6-week intervals. Both regimens will include a maximum of 3 cycles of therapy (subject to the recommendation of an SRC for Cohort B) in order to assess the safety and radiation dosimetry of a "core" treatment regimen (Table 2 and Figure 2). The maximum number of cycles a subject can receive is pre-specified based on absorbed dose estimates acquired from human dosimetry data from an earlier compound from the same class of 'radiohybrid' molecules as the therapeutic IMP (^{177}Lu-rhPSMA-7.3C) and preliminary human dosimetry from a single patient who has received the therapeutic IMP; however, subjects may not be eligible for additional cycles dependent on cumulative absorbed radiation doses to normal organs and the presence of toxicity. Individual subjects will be closely monitored throughout all treatment cycles for observation of DLT (DLT is defined in Section 4.1.1.4). Further subjects may be added to a cohort in line with recommendations outlined in Section 4.1.1.2. All subjects will undergo dosimetry assessments using blood and SPECT/CT imaging at set intervals in the 7 days following each cycle of therapy in Phase 1. This radiation dosimetry assessment will permit organ and tumour absorbed dose estimates to be calculated for each subject and each cycle. No subject will receive additional radioactivity if doing so would exceed any normal organ radiation dose limits defined in the protocol.

Escalation between the two cohorts will be managed using a Bayesian optimal interval (BOIN; Liu and Yuan 2015) design, with initially 3 subjects recruited to the low dose cohort (Cohort A). The number of subjects recruited to Cohort B will depend on the observed safety data from prior cohort; however, it is planned to recruit at least 12 subjects (Cohort A: at least 3 subjects; Cohort B: at least 9 subjects) between the 2 cohorts that are opened for enrolment. A cohort will be considered as 'evaluable' for the purposes of dose-escalation once enrolled subjects have completed the DLT observation period. The DLT observation period is defined as the period from first administration of the therapeutic IMP to 6 weeks after the final cycle of therapy with the therapeutic IMP received by each subject in a cohort. If any subject in a cohort fails to receive at least 2 cycles of therapy for reasons other than the presence of a DLT or because a second cycle would exceed normal organ dose limits (e.g. intercurrent illness meaning the subject withdraws from the study), then an additional subject will be recruited to replace them.

In this study, a treatment cycle is defined as 6 weeks following the last administration of the therapeutic IMP (Figure 2). The 6-week dosing interval is based on the known time to nadir for toxicity observed with other PSMA-targeted radiopharmaceuticals, which is generally thought to be 3 to 5 weeks post-dose. Additionally, a 6-week dosing interval has been tested in a Phase 3 study with ^{177}Lu-PSMA-617 and been shown to be efficacious (Sartor 2021).

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However, in the case of logistical or administrative issues, or non-DLT which requires dose delay, the cycle can be extended by up to 28 days (giving a maximum dosing interval of 10 weeks). The decision to open Cohort B will be taken by a SRC comprising the Principal Investigators from the Phase 1 sites, an independent medical oncologist, and a nuclear medicine physician. The Sponsor's statistics/data management representative, medical representative and drug safety representative will also attend (see Section 15). Safety and radiation dosimetry data will be reviewed from every cycle of the completed prior cohort to make the escalation decision.

Table 2: Phase 1 Radioactivity Levels

Cohort	Radiation Level per Cycle		Max Cycles	Dosing Interval	Planned Number of Subjects
	GBq	mCi			
A	5.55	150	3	6 weeks	3
B	7.40	200	3[a]	6 weeks	9

Abbreviations: GBq=Gigabecquerel; Max=maximum; mCi=millicurie; SRC=safety review committee.

[a] Subject to the recommendation of an SRC.

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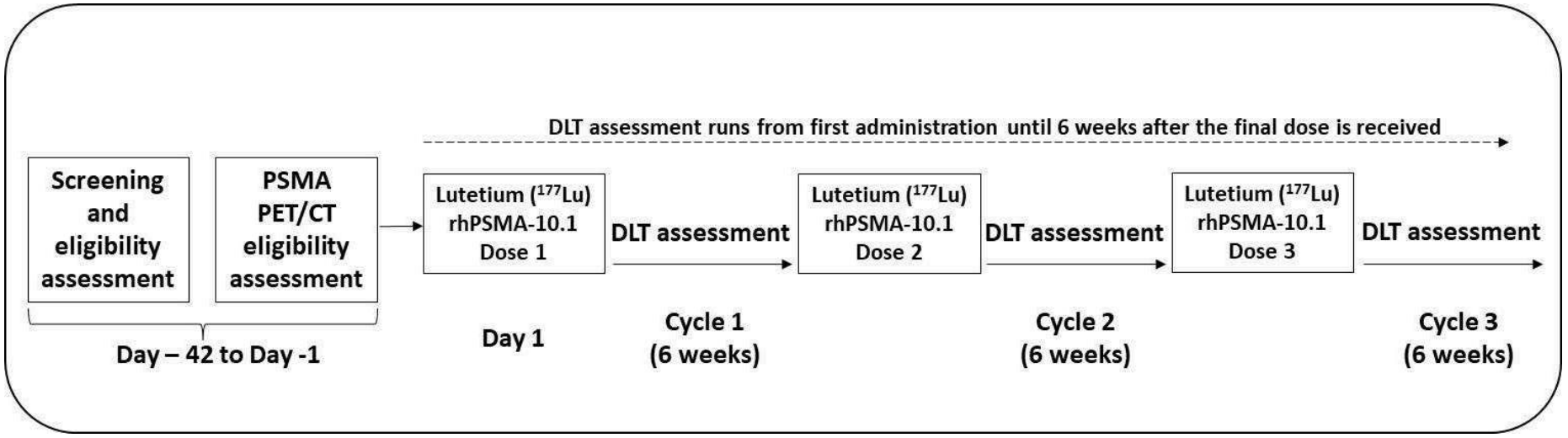
Figure 2: Study Schema: Phase 1 Component (Completed)



Abbreviations: CT=computed tomography; DLT=dose-limiting toxicity; PET=positron emission tomography; PSMA=prostate-specific membrane antigen; rhPSMA=radiohybrid prostate-specific membrane antigen.

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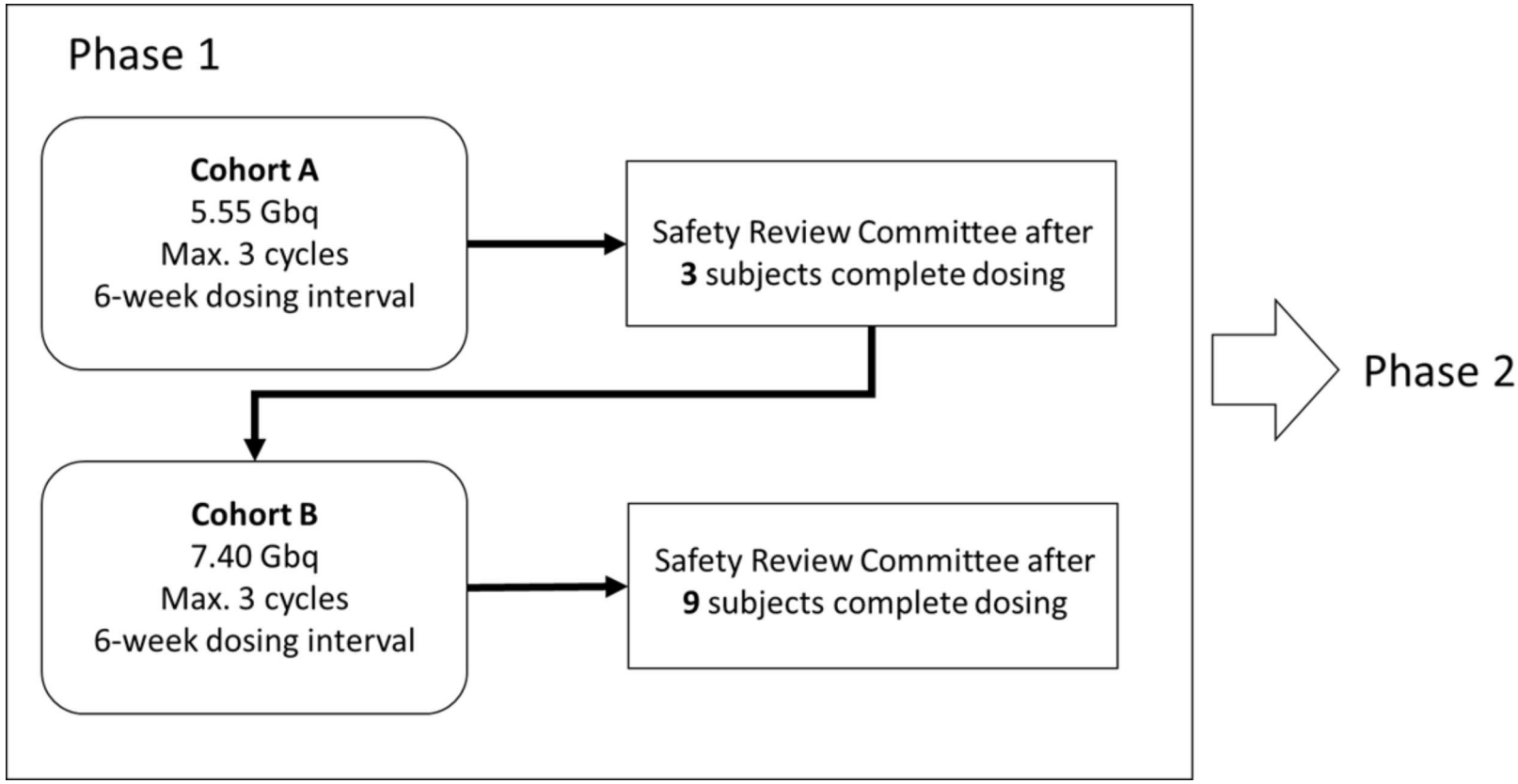
Figure 3: Radioactivity Dose-ranging Schema: Phase 1 Component (Completed)



Abbreviations: GBq=Gigabecquerel; RP2D=recommended Phase 2 dose.
Note: Safety Review Committee review all available data to assign RP2D. Dosing in the next cohort will not start until all subsequent cycles from the previous cohort are complete.

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Each subject will participate in only one cohort, and subjects in Phase 1 will not participate in Phase 2 of the study. After completion of each cycle of the therapeutic IMP, the investigator will review all relevant data including radiation dosimetry and all available safety data for each subject.

Administration of subsequent treatment cycles will depend on the following:

- Subjects exhibiting DLTs at any point during the DLT observation period will be discontinued from further treatment.
- A dose delay of up to 28 days to allow time for any toxicity to improve or resolve is acceptable where the investigator assesses in their clinical judgement, that this is appropriate. Subjects with worsening or persisting toxicity, which in the clinical judgement of the investigator would make them ineligible for further dosing will be discontinued, even if the experienced toxicity does not meet the definition of a DLT.
- Cumulative key organ defined radiation doses derived from dosimetry analysis will not exceed the conventionally accepted dose limits (the 2 organ dose limits, which are most relevant to patient safety, will be 2 Gray [Gy] in the bone marrow and 23 Gy in the kidney). If further treatment would lead to any limits listed in Table 3 being exceeded, then a subject will not receive additional treatment cycles.

Key organ defined dose limits are taken from several sources and are presented below in Table 3. It is currently unclear if these estimates, derived from published historical experience with EBRT, are the most appropriate thresholds when delivering ionising radiation with low dose rate beta emitting radiopharmaceuticals. However, they serve as an important safety threshold for early phase clinical trials in the absence of long-term clinical safety data with the therapeutic IMP. Dose to other normal organs will be determined using the Medical Internal Radiation Dose (MIRD) Committee S-value methodology, and dose estimates will be presented after each cycle of therapy. Organs considered to be at risk are based on published experience with other PSMA-targeted agents and initial compassionate use experience with ^{177}Lu-rhPSMA-7.3C and the therapeutic IMP itself. The kidney and bone marrow are highlighted due to the potential seriousness of toxicity experienced with these organs and the known experience with other PSMA-targeted radiopharmaceuticals. As described in Section 4.1.1, it is expected that the kidney will be the radiation dose-limiting organ with this compound. Estimated absorbed radiation dose limits are provided in Table 3. Of note, these are set at 23 Gy for the kidneys and 2 Gy for the bone marrow.

Table 3: Phase 1 Conventional Organ Estimated Dose Limits

Target Organs[a]	Estimated Dose Limit (Gy)	Ref
Bladder Wall	65	Marks 2010
Brain	60	Marks 2010
Heart	26	Marks 2010
Intestine	45	Marks 2010
Kidneys	23	Dawson 2010
Liver	30	Marks 2010
Lungs	20	Marks 2010

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Target Organs[a]	Estimated Dose Limit (Gy)	Ref
Bone Marrow	2	DeNardo 2002; ICRP 2012
Salivary Glands	24	Eisbruch 1999
Stomach Wall	45	Marks 2010

Abbreviations: Gy=Gray; Ref=references.

[a] The estimated absorbed dose to a sphere to provide an estimate of the dose to the lacrimal glands will also be calculated during radiation dosimetry assessments. However, given the methodological challenges of accurately measuring absorbed dose in such a small organ, a dose limit will not be applied for the purpose of discontinuing dosing. Instead, absorbed dose estimates will be correlated with any reported ocular symptoms.

4.1.1.1 Phase 1 Scientific Justification for Choice of Cohort Radioactivity Level

The choice of injected radioactivity in each cohort is based on initial human dosimetry and therapeutic data available with an earlier compound from the same class of 'radiohybrid' molecules (i.e. ^{177}Lu-rhPSMA-7.3C) and from a single patient who has received the therapeutic IMP. ^{177}Lu-rhPSMA-7.3C has been evaluated in humans at TUM in Germany under the physician's personal responsibility (exempt from a manufacturing authorisation as per Section 13, Subsection 2b of the German Medicinal Products Act). ^{177}Lu-rhPSMA-7.3C has an additional carboxylic acid group (on the linker between the ^{177}Lu-chelator complex and the glutamate-urea PSMA binding site) but is otherwise structurally identical to the therapeutic IMP. Lutetium (^{177}Lu) rhPSMA-10.1 Injection has been selected as the study drug based on this minor structural optimisation, which results in improved biodistribution in head-to-head non-clinical comparison studies with ^{177}Lu-rhPSMA-7.3C. The in-human experience from 6 subjects with mCRPC who received ^{177}Lu-rhPSMA-7.3C (Table 4) has been extrapolated to inform the dose-ranging schema chosen for Phase 1 of this study. Of note, 2 of the 6 subjects received therapeutic doses of ^{177}Lu-rhPSMA-7.3C. Subject 1 received 3.3 GBq followed by a second cycle of 3.3 GBq, and Subject 2 received 3.9 GBq followed by a second cycle of 3.9 GBq. Both subjects experienced a reduction in PSA levels, with Subject 2 experiencing a >50% decline in PSA and an observed radiological response. No safety data were available.

Table 4: ^{177}Lu-rhPSMA-7.3C Activities: Technical University of Munich

Activity (GBq)	Number of Cycles	Number of Subjects	Reason	Notable Findings
Low dose (approx. 1)	1	6	Initial radiation dosimetry estimates	--
3.3	2	1	Therapeutic dose	Decline in PSA
3.9	2	1	Therapeutic dose	>50% decline in PSA and a radiological response

Abbreviations: GBq=Gigabecquerel; PSA=prostate-specific antigen.

From this experience in Germany, pre-therapeutic planar dosimetry with ^{177}Lu-rhPSMA-7.3C has given the below values (Table 5) for the estimated absorbed radiation doses (Feuerecker 2022). In addition, planar dosimetry was performed after the injection of a therapeutic radioactivity dose in 2 subjects.

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Table 5: ^{177}Lu-rhPSMA-7.3C Organ Absorbed Radiation Doses

Region of Interest	Mean Estimated Absorbed Radiation Dose (Gy/GBq) Using Planar Dosimetry – pre-therapeutic (n = 6)	Mean Estimated Absorbed Radiation Dose (Gy/GBq) Using Planar Dosimetry – post-therapeutic (n = 2)
Kidney	1.65	1.59
Liver	0.19	0.20
Parotid Gland	2.35	1.59
Submandibular Gland	2.10	1.97
Lacrimal Glands	5.92	3.82
Bone Marrow	0.12[a]	0.14[b]

Abbreviations: GBq=Gigabecquerel; Gy=Gray.

[a] Low confidence estimate. Only 2 of 4 evaluable subjects were able to be analysed due to the presence of metastatic lesions in the region of interest selected in the lumbar spine.

[b] Low confidence estimate. Only 1 of 2 evaluable subjects were able to be analysed due to the presence of metastatic lesions in the region of interest selected in the lumbar spine.

Planar dosimetry is understood to overestimate absorbed radiation doses to some organs including the kidney (Lehnert 2018). Similarly, it is known that measuring salivary gland and lacrimal gland absorbed doses with planar imaging is challenging due to the very small size of these organs and lack of 3-dimensional anatomical correlation.

In addition to the 6 subjects who received ^{177}Lu-rhPSMA-7.3C, as of 15 November 2021, a single subject at the Department of Nuclear Medicine, University Hospital Augsburg, received the actual therapeutic IMP under the physician's personal responsibility (exempt from a manufacturing authorisation as per Section 13, Subsection 2b of the German Medicinal Products Act). The subject received a pre-therapeutic dose of 1 GBq and then 3 therapeutic cycles of 7.40 GBq of the therapeutic IMP Dosimetry is available for the pre-therapeutic dose and the first two therapeutic cycles. Of note, the dosimetry protocol combined SPECT/CT and planar acquisitions as presented in Table 6. Of note, the subject experienced a >50% decline in PSA and a radiological response following 2 cycles of 7.40 GBq Lutetium (^{177}Lu) rhPSMA-10.1 Injection. No safety data were available.

Table 6: Dosimetry for Pre-therapeutic Dose and First Two Therapeutic Cycles for a Single Subject Treated with Lutetium (^{177}Lu) rhPSMA-10.1 Injection

Region of Interest	Mean Estimated Absorbed Radiation Dose (Gy/GBq) Using hybrid SPECT and Planar Dosimetry – pre-therapeutic (n = 1)	Mean Estimated Absorbed Radiation Dose (Gy/GBq) Using hybrid SPECT and Planar Dosimetry – post-therapeutic (n = 1) Cycle 1	Mean Estimated Absorbed Radiation Dose (Gy/GBq) Using hybrid SPECT and Planar Dosimetry – post-therapeutic (n = 1) Cycle 2
Kidney	1.11	0.66	0.67
Liver	0.18	0.15	0.08
Salivary Glands (planar)	0.35	0.23	0.14
Adrenals	0.01	0.02	0.02
Brain	0.01	0.02	0.02
Bone Marrow	0.06	0.06	0.05
Breast	0.01	0.02	0.02

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Region of Interest	Mean Estimated Absorbed Radiation Dose (Gy/GBq) Using hybrid SPECT and Planar Dosimetry – pre-therapeutic (n = 1)	Mean Estimated Absorbed Radiation Dose (Gy/GBq) Using hybrid SPECT and Planar Dosimetry – post-therapeutic (n = 1) Cycle 1	Mean Estimated Absorbed Radiation Dose (Gy/GBq) Using hybrid SPECT and Planar Dosimetry – post-therapeutic (n = 1) Cycle 2
Gallbladder Wall	0.01	0.02	0.02
Lower Large Intestine Wall	0.01	0.02	0.02
Small Intestine	0.01	0.02	0.02
Stomach Wall	0.01	0.02	0.02
Upper Large Intestine Wall	0.01	0.02	0.02
Heart Wall	0.01	0.03	0.02
Lungs	0.01	0.03	0.02
Muscle	0.01	0.02	0.02
Pancreas	0.01	0.02	0.02
Osteogenic Cells	0.05	0.08	0.08
Skin	0.01	0.02	0.02
Spleen	0.08	0.10	0.05
Testes	0.01	0.02	0.02
Thymus	0.01	0.02	0.02
Thyroid	0.01	0.02	0.02
Urinary Bladder Wall	0.01	0.02	0.02
Total Body	0.02	0.02	0.03
Effective Dose [mSv/MBq]	0.04	0.04	0.03

Abbreviations: GBq=Gigabecquerel; Gy=Gray; MBq=Megabecquerel; mSv=millisievert; SPECT=single-photon emission computed tomography.

Based on the non-clinical data available before study initiation, the clinical experience with pre-therapeutic doses of ^{177}Lu-rhPSMA-7.3C in 6 subjects, therapeutic doses of ^{177}Lu-rhPSMA-7.3C in 2 subjects, and both pre-therapeutic and therapeutic doses of the therapeutic IMP (Lutetium (^{177}Lu)-rhPSMA-10.1 Injection) in 1 patient, it is anticipated that the kidneys are likely to be the radiation dose-limiting organ. The cohort dosing regimens in Phase 1 using the therapeutic IMP have been designed to allow a maximum cumulative injected activity of 16.65 GBq (450 mCi) in Cohort A and 14.80 GBq (400 mCi) in Cohort B, where a higher per cycle radioactivity level will be tested. Based on the data acquired with ^{177}Lu-rhPSMA-7.3C, this would give a theoretical maximum absorbed radiation dose to the kidneys of 26.64 Gy in Cohort A, and 23.68 Gy in Cohort B when applying the value of 1.6 Gy/GBq for kidney absorbed radiation dose obtained with ^{177}Lu-rhPSMA-7.3C. If using the single patient experience with Lutetium (^{177}Lu) rhPSMA-10.1 Injection, the theoretical maximum absorbed radiation dose to the kidneys for Cohort A would be 11.0 to 18.5 Gy, depending on whether the pre-therapeutic or therapeutic dosimetry estimates are used. For Cohort B, these estimates are 9.8 to 16.4 Gy. It is recognised that these assumptions need to be tested in a formal Phase 1 clinical trial setting with high-quality prospective dosimetry.

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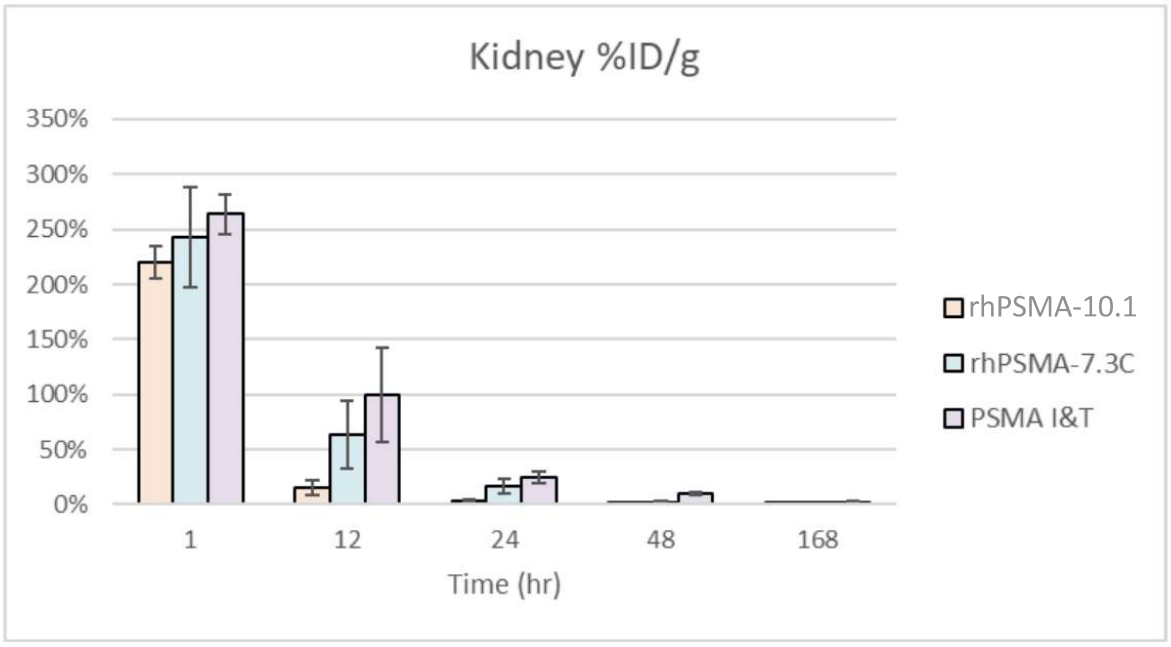
Subjects shall not receive additional cycles of radioactivity if the absorbed dose estimates predict that any of the normal tissue dose limits listed in Table 3 would be exceeded. Importantly in Phase 1, all subjects will have radiation dosimetry calculated after each cycle so the maximum cumulative injected activity for the therapeutic IMP will be defined prospectively by the clinical study and the above assumptions tested.

Thus, the data described above derived using ^{177}Lu-rhPSMA-7.3C and the single patient who received the therapeutic IMP act as a bridge to this study. Non-clinical comparative biodistribution studies in mice and minipigs have been completed to explore similarities and differences in biodistribution between ^{177}Lu-rhPSMA-7.3C and ^{177}Lu rhPSMA-10.1 (see ^{177}Lu rhPSMA-10.1 IB). These data showed that renal uptake values with ^{177}Lu rhPSMA-10.1 was significantly lower than ^{177}Lu-rhPSMA-7.3C due to more rapid renal clearance (see Figure 4 and Table 7), and fits with the limited clinical experience reported above.

Figure 4: Renal Uptake Values in Mouse Model (^{177}Lu-rhPSMA-10.1, ^{177}Lu-rhPSMA-7.3C, and ^{177}Lu-PSMA-I&T)



Abbreviations: ^{177}Lu=Lutetium-177; h r=hour; ID/g=injected dose per gram; PSMA=prostate-specific membrane antigen; rh=radiohybrid.
Note: Error bars indicate one standard deviation.

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Table 7: Renal Uptake Values in Mouse Model (^{177}Lu rhPSMA-10.1, ^{177}Lu-rhPSMA-7.3C, and ^{177}Lu-PSMA-I&T)

Kidney %ID/g					
Time (hr)	**1**	**12**	**24**	**48**	**168**
^{177}Lu-rhPSMA-10.1	220.1% ± 14.9%	15.2% ± 7.0%	3.3% ± 1.2%	0.8% ± 0.1%	0.3% ±0.0%
^{177}Lu-rhPSMA-7.3C	243.3% ± 45.52%	63.8% ± 31.09%	16.7% ± 6.99%	1.8% ± 0.84%	0.4% ± 0.05%
^{177}Lu-PSMA-I&T	263.9% ± 18.04%	99.3% ± 43.10%	24.5% ± 5.56%	9.7% ± 0.91%	2.4% ± 0.22%
Kidney ID/g					
^{177}Lu-rhPSMA-10.1	72.6% ± 5.32%	4.5% ± 2.14%	0.9% ± 0.34%	0.3% ± 0.04%	0.1% ± 0.01%
^{177}Lu-rhPSMA-7.3C	77.7% ± 4.43%	18.6% ± 6.53%	4.5% ± 1.39%	0.6% ± 0.23%	0.1% ± 0.0%
^{177}Lu-PSMA-I&T	78.8% ± 2.63%	26.0% ± 7.54%	8.2% ± 1.76%	3.2% ± 0.29%	0.8% ± 0.06%

Abbreviations: ^{177}Lu=Lutetium-177; hr=hour; ID=injected dose; PSMA=prostate-specific membrane antigen; rh=radiohybrid.

The radiation dosimetry performed during Phase 1 of the study will guide the decision to administer subsequent cycles of therapy to subjects and whether opening Cohort B is appropriate. Importantly, at all points during the study, any subject in whom the delivery of additional radiation is expected to exceed organ absorbed dose limits defined in Table 3 will be discontinued from further dosing.

4.1.1.1.1 Protocol Version 4.0 Dose Justification Update – September 2023

Radiation dosimetry data from the first dose cohort (A) in Phase 1 is supportive of the administration of additional cycles of therapy without breaching pre-defined organ dose limits (Table 3). A third cycle has been added at the 7.40 GBq dose level to permit a more extensive assessment of safety and is likely to be a closer match to the chosen recommended Phase 2 dose which will be evaluated in the next phase of development. Additionally, variation in absorbed dose estimates to tumour and normal organs are expected as cycles progress within the same patient, and a third cycle allows better characterisation of these changes. Furthermore, the single patient dosimetry data acquired from a clinical experience at the Department of Nuclear Medicine, University Hospital Augsburg (Table 6) has now expanded to include data from 4 patients and this has been presented academically (Rinscheid, 2023). These data are further supportive of permitting 3 cycles of therapy at the 7.40 GBq level from a radiation dosimetry perspective. The decision to escalate to the next dosing cohort remains the decision of the SRC after review of all relevant safety and radiation dosimetry data from Cohort A. With the additional cycle, the maximum cumulative injected activity will be 22.20 GBq (600 mCi) in Cohort B.

4.1.1.2 Phase 1 Dose Escalation Rules

Once the third subject in Cohort A has completed the DLT observation period following therapeutic IMP administration (the period from the first administration of therapeutic IMP, to 6 weeks after the final cycle of therapy with the IMP received by the subject), an SRC will review all available data and decide on escalation to Cohort B and the maximum number of cycles (see Section 15).

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In their decision, the SRC will incorporate (but is not obligated to follow, for example, in the event of safety concerns) the recommendation from the BOIN design. For the BOIN design, the target DLT rate is set at 0.3 (i.e. 30%), with a safety boundary to dose escalate of ±40% as recommended by Liu and Yuan (2015). As such, if the observed DLT rate at the current dose is ≤0.236, the BOIN recommendation is to dose escalate. If the observed DLT rate is between 0.236 and 0.359, the BOIN recommendation is to add more subjects to the current cohort. If the observed DLT rate is ≥0.359, the BOIN recommendation is to de-escalate; however, for the special case of Cohort A, the SRC may recommend that an additional lower dose cohort may be opened. An application for a protocol amendment will be submitted for approval if this occurs.

Assuming 3 subjects have been treated in Cohort A and the decision to open Cohort B has been made, at least 9 subjects will be enrolled in Cohort B, although the final number will be determined by the SRC and guided by safety data. Once all enrolled subjects have received their cycles of the therapeutic IMP and completed the DLT observation period, the SRC will review all available data and make a recommendation regarding the RP2D to be tested in the Phase 2 component of the study.

Dose escalation/de-escalation will continue until at least 12 subjects in total have completed dosing and the DLT observation period. The BOIN design permits addition of 1 or more subjects to an open cohort, and the estimated DLT rate can be calculated using the total number of evaluable subjects in the cohort in the denominator, and then this DLT rate can be compared with the boundary limits of 0.236 and 0.359 to deduce the BOIN recommendation to escalate, de-escalate, stay, or close the cohort. Due to the simplicity of this algorithm, the exact numbers of DLTs leading to each decision can be tabulated for the maximum cohort size of 12 and appears in Table 8.

Table 8: Dose Escalation/De-escalation Rule for the BOIN Design

	Number of Evaluable Subjects at the Current Dose											
Actions[a]	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Escalate if # of DLT ≤	0	0	0	0	1	1	1	1	2	2	2	2
De-escalate if # of DLT ≥	1	1	2	2	2	3	3	3	4	4	4	5
Eliminate if # of DLT ≥	NA	NA	3	3	4	4	5	5	5	6	6	7

Abbreviations: BOIN=Bayesian optimal interval; DLT=dose-limiting toxicity; NA=not applicable.

[a] When none of the actions (i.e. escalate, de-escalate or eliminate) is triggered, stay at the current dose for treating the next cohort of subjects. Note that "# of DLT" is the number of subjects with at least 1 DLT, and "NA" means that a dose cannot be eliminated before treating 3 evaluable subjects.

When 12 subjects have completed dosing or the study has been stopped for any reason, an SRC meeting will be convened to review the data acquired and where appropriate, to select an RP2D for Phase 2 of the study (see Section 4.1.1.5).

4.1.1.3 Phase 1 Dose Modification, Stopping Rules, and Discontinuation Criteria

The decision to administer the therapeutic IMP will always be based on assessment by the subject's treating physician/investigator.

If a DLT is experienced by a subject during the DLT observation period (i.e. from first administration of the therapeutic IMP to 6 weeks after the final cycle of therapy with the IMP received by the subject), which is deemed non-related to the therapeutic IMP, that subject will be withdrawn from treatment and another subject will be recruited to replace them. However, the subject will remain in the study for safety assessments.

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Adverse events reported outside of the DLT observation period (i.e. occurring >6 weeks after the final administered cycle of therapy) will not be classified as DLTs for the purpose of evaluating whether the higher dose level cohort can be opened, but their grade, frequency, seriousness, relation to the therapeutic IMP and expectedness will be evaluated for the purposes of safety reporting, and for recommending the Phase 2 dosing regimen to take forward into further clinical trials.

If the next scheduled cycle of radioactivity for a subject is predicted to exceed the absorbed radioactivity dose limit to normal organs, the investigator will have the option to assign the subject the radioactivity dose from a lower dose cohort in order to remain below the normal organ dose limits. This is only applicable for subjects in Cohort B who can be given 5.55 GBq instead of 7.40 GBq.

Where a subject requires surgery during the treatment phase of the study, the therapeutic IMP may be temporarily discontinued providing that the surgery is not with therapeutic intent for prostate cancer. Following surgery, the investigator should assess the subject's suitability for restarting the therapeutic IMP and discuss this with the subject. A maximum dose delay of 8 weeks is acceptable for surgical treatment (i.e. a maximum of 14 weeks from the last administration of therapeutic IMP). If a subject requires a dose delay of >8 weeks, the investigator should withdraw the subject from the clinical study.

Radiotherapy during the study is not permitted due to the critical importance of radiation dosimetry and the need to correlate dose estimates from the therapeutic IMP administrations with any observed responses and/or toxicity. If a subject requires radiotherapy, then the investigator should withdraw them from the clinical study. Subjects who have received the therapeutic IMP will continue to be followed for safety.

Based on available data with other ^{177}Lu-labelled PSMA-targeting small molecules, it is anticipated that bone marrow suppression may be experienced by subjects receiving multiple cycles of the therapeutic IMP. Where a subject experiences toxicity, and the investigator judges that it is in the subject's best interests to receive additional cycles of therapy, then a dose delay of up to 28 days is acceptable to allow this to improve or resolve. The management of any bone marrow suppression should proceed as per usual medical practice at the institution and the medical judgement of the investigator. However, if subjects require bone marrow stimulating agents (e.g. sargramostim, filgrastim, epoetin alfa, eltrombopag), then they should not receive further radioactivity. Subjects can receive blood products (e.g. transfusion of packed red blood cells, transfusion of platelets) and receive further therapeutic cycles of radiation provided the investigator judges that this is in their best interests (unless this toxicity represents a DLT in which case they are not permitted further cycles of therapy).

Specifically, if any of the following occur, no further treatment will be administered:

- Subject withdraws their consent to further treatment.
- Subject requires any other form of radiotherapy treatment (e.g. external beam).
- Subject requires surgery with therapeutic intent for prostate cancer.
- Subject requires surgery that is not with therapeutic intent for prostate cancer but results in a dose delay of >8 weeks.
- The cumulative absorbed dose to normal organs exceeds any of the dose limits listed in Table 3 and/or the administration of the next cycle would lead to those limits being exceeded.

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- Occurrence of any DLT at any point from administration of the first dose until 6 weeks following the final dose.
- When considering additional cycles of therapy, subjects with worsening or persisting toxicity, which in the clinical judgement of the investigator would make them ineligible for further dosing, even if the experienced toxicity does not meet the definition of a DLT (a dose delay of no more than 28 days to give time for any toxicity to improve or resolve is acceptable where the investigator assesses this is in the subject's best interests).

4.1.1.4 Phase 1 Dose-limiting Toxicity Definitions

Dose-limiting toxicities are defined as any of the following treatment-emergent therapeutic IMP related or possibly related AEs according to National Cancer Institute – Common Terminology Criteria for Adverse Events (CTCAE) scale version 5.0, that occur during the defined DLT observation period (i.e. from first administration of the therapeutic IMP to 6 weeks after the final cycle of therapy with the IMP received by the subject):

- Grade 4 febrile neutropenia
- Grade 4 neutropenia >5 days duration
- Grade 4 haematologic toxicity lasting ≥7 days
- Any Grade 3 or higher non-haematological toxicity but specifically *excluding* the following:
  - Grade 3 fatigue <7 days
  - Grade 3 hypertension in the absence of maximal medical therapy
  - Grade 3 or 4 electrolyte abnormality that lasts <72 hours, is not clinically complicated, and resolves spontaneously or responds to conventional medical intervention
  - Grade 3 nausea, vomiting or diarrhoea which resolves within 72 hours in the absence of maximal medical therapy
  - Grade 3 rash that resolves to ≤Grade 1 within 3 weeks
  - Grade 3 or 4 elevation in serum amylase and/or lipase that are not associated with clinical or radiographic evidence of pancreatitis
- Grade 3 or higher thrombocytopenia with bleeding
- Grade 3 or higher acute kidney injury (defined as creatinine of >3× baseline or absolute value of >4.0 mg/dL)
- Grade 3 or higher abnormalities in aspartate amino transferase/alanine amino transferase (AST/ALT) and/or bilirubin
- Grade 3 or higher electrolyte abnormalities where the subject experiences related clinical symptoms (regardless of duration)
- Liver enzyme elevation meeting criteria for Hy's Law (concomitant ALT or AST >3× upper limit of normal (ULN) AND total bilirubin >2× ULN AND alkaline phosphatase [ALP] <2× ULN AND no other reason for liver injury)

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- Any AE requiring a dose delay of >28 days, unless related to the underlying disease or extraneous causes
- Death not clearly related to the underlying disease or an extraneous cause

4.1.1.5 Declaration of the Recommended Phase 2 Dose

The RP2D is defined as the radioactivity dose and number of cycles that, in the judgement of the SRC, permits the optimum therapeutic index, which balances tumour absorbed radiation dose against the absorbed radiation dose to normal organs and any observed clinical toxicity related to the study drug. It will also consider the average number of cycles tolerated by the subjects at that radioactivity level and the accumulation of toxicity, if any. It will also consider normal organ radiation dose limits. The RP2D may not be the highest radioactivity level given as a single injection, as the cumulative radioactivity administered will also be factored into the decision-making. Preliminary data on anti-tumour response (Phase 1 secondary endpoints) may also be factored into the selection of the RP2D. The assignment of the RP2D will be based on the scientific judgement of a convened SRC, which will comprise the Principal Investigators taking part in Phase 1, plus an independent medical oncologist and a nuclear medicine physician. In addition, the Sponsor's statistics/data management representative, medical representative and drug safety representative will attend. If both Phase 1 dosing regimens are safe and well tolerated with no clear advantage for the subject of one over the other, then due consideration will also be given to the known radiobiology of beta emitting radiopharmaceuticals, the practicalities of dosing at clinical sites, and the challenges of the radioactive supply chain in the selection of the regimen to take into Phase 2. The RP2D may be a greater number of cycles than tested in the Phase 1 part of the study if radiation dosimetry estimates permit this without exceeding normal organ dose limits. For example, in the event that three 7.40 GBq treatment cycles are well tolerated without DLT by subjects in Cohort B and dosimetry estimates predict that up to 5 cycles can be given in Phase 2 without exceeding normal organ dose limits. In this case, the RP2D would be 5 cycles of 7.40 GBq, separated by 6-week intervals.

In order to have sufficient information to assign the RP2D, a meeting of the SRC will be convened when subjects in the final radioactivity dose-ranging cohort (i.e. Cohort B or the lower cohort if Cohort B was not opened or could not be completed due to DLTs) have completed the DLT observation period. The SRC will review the data for the dose-finding cohorts and the acquired safety and dosimetry data for each subject and decide whether to proceed with the Phase 2 dosing regimen.

4.1.2 Phase 2

Following consent, subjects will have a PSMA PET/CT scan to ascertain whether the subject has PSMA-positive disease and to rule out PSMA-negative disease. The PSMA PET/CT tracer selected by the investigator should be approved under marketing authorisation where available, or already be in use at the study site via an applicable local authorisation or permission for use at the study site. The PSMA PET/CT scan will be sent for central review to determine eligibility. Baseline cross-sectional imaging must also be completed and submitted to the Central Imaging vendor for all subjects.

In order to reduce unnecessary interventions and radiation exposure, where PSMA PET/CT imaging or cross-sectional imaging has been recently completed as part of standard of care and the images are available for submission to the central reader, these images may be used to determine eligibility, provided that the scan was acquired no more than 28 days prior to informed consent.

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Eligible subjects with PSMA-positive mCRPC will be initially randomised to one of two dosing regimens, with approximately 30 subjects randomised to each cohort as follows:

- Cohort 2A: 2 doses at 10.00 GBq (270 mCi) followed by up to 5 additional doses at 7.40 GBq (200 mCi), all doses administered at 6-weekly intervals.
- Cohort 2B: Up to 8 doses at 7.40 GBq (200 mCi). The first 3 doses will be administered at 3-weekly intervals, with the remaining doses being administered at 6-weekly intervals.

A decision will be made on whether or not to open a third cohort (Cohort 2C) based on ongoing review of safety and preliminary efficacy data in Cohorts 2A and 2B. The decision to open Cohort 2C will be discussed and taken by the TSC after reviewing all available safety and preliminary efficacy data and when approximately 35 subjects have been randomised and dosed in Cohorts 2A and 2B. As part of the data review, the TSC will specifically assess the safety profile from the 2 initial 10.00 GBq (270 mCi) doses administered in Cohort 2A. For example, a high frequency of severe cytopenia from the 2 initial 10.00 GBq (270 mCi) doses in Cohort 2A would preclude the opening of Cohort 2C given the higher radioactivity in the first 2 cycles in that cohort. Further information is provided in the TSC Charter.

- Cohort 2C, if opened, will include approximately 10 subjects treated with 2 doses at 14.80 GBq (400 mCi), followed by up to 4 additional doses at 7.40 GBq (200 mCi), all doses administered at 6-weekly intervals.

If Cohort 2C is not opened, additional subjects will be randomised to Cohort 2A and Cohort 2B respectively, resulting in a total of approximately 35 subjects in each cohort (Figure 5). If unacceptable rates of toxicity occur in both Cohorts 2A or 2B, enrolment into the trial will be put on hold and a protocol amendment may be submitted to test a reduced radioactivity cohort.

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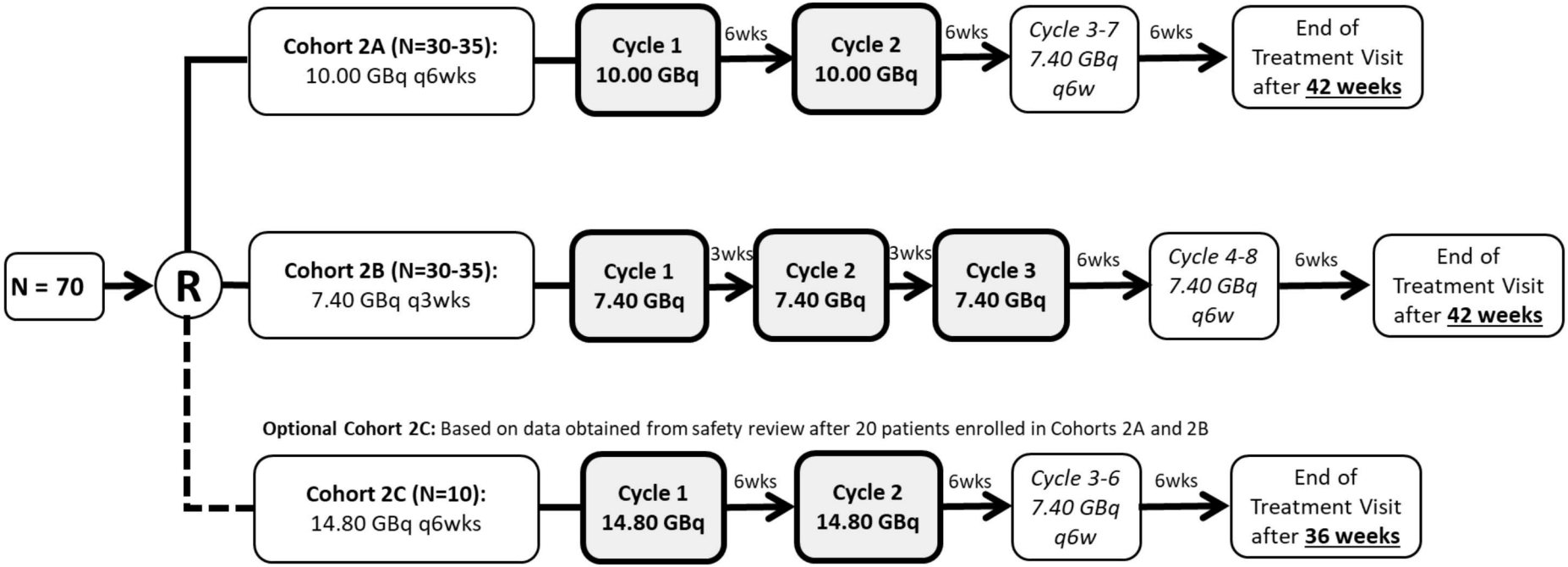
Figure 5 Study Schema: Phase 2 Component



Abbreviations: GBq= Gigabecquerel; q3w=every 3 weeks; q6w=every 6 weeks; R=randomisation.

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Each cohort will enrol subjects to assess whether the IMP achieves anti-tumour responses using a decrease in PSA by at least 50% from baseline (PSA50 response) as a surrogate biomarker endpoint. Following the safety review, Cohort 2A or 2B may be closed for futility (i.e. estimated PSA50 response rate ≤20% within a cohort). This decision will be based on the recommendation of the TSC (see Section 15). In this case, additional subjects may be randomised to the remaining cohorts to complete study enrolment. During the conduct of Phase 2, the Sponsor may perform additional interim reviews of the emerging data to assess preliminary efficacy and safety. Full details of such reviews will be described in the Statistical Analysis Plan (SAP).

Image-based dosimetry will be performed to determine normal organ radiation absorbed doses for each individual subject. These absorbed doses will be used to determine the maximum number of cycles an individual subject can receive without exceeding agreed normal organ dose limits.

4.1.2.1 Phase 2 Scientific Justification for Choice of Cohort Radioactivity Levels

Dosimetry data from Phase 1 indicate a consistent trend across all subjects that the absorbed tumour dose and therefore therapeutic index, reduces with cycle number. The highest tumour dose, per unit of administered activity, for all subjects was achieved at Cycle 1. An average decrease in the mean specific absorbed tumour dose of 32% and 49% was observed from Cycle 1 to Cycle 2, and Cycle 1 to Cycle 3, respectively.

This observed dosimetry data formed the basis and justification for the choice of dosing in Cohorts 2A and 2C in Phase 2, which involves front-loading the activity through administering higher activities at earlier cycles. This approach will allow a higher cumulative tumour absorbed dose to be achieved compared to a fixed dosing regimen with an equivalent cumulative administered activity, with no corresponding increase in normal organ dose. Comparing the efficacy and safety of this dosing approach with one where the time interval between the first 3 cycles is shortened (Cohort 2B), will allow the dosing regimen to be optimised for Phase 3.

4.1.2.2 Phase 2 Dose Modification, Stopping Rules, and Discontinuation Criteria

The investigator is responsible for the ongoing safety monitoring of subjects taking part in the Phase 2 trial. Importantly, at any stage where an investigator feels that continuing dosing is not in the subject's best interests, then they should discontinue dosing.

Additionally, the dose delay and discontinuation rules presented in Table 9 should be applied during Phase 2 as a minimum standard.

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Table 9: Phase 2 Dose Delay and Discontinuation Rules

Safety Event	Action
Myelosuppression (anaemia, thrombocytopenia, leukopenia, neutropenia)	• Grade 3 or above: withhold dosing until Grade 1 or baseline is achieved • Recurrent Grade 3 or above myelosuppression: permanently discontinue dosing
Acute renal toxicity (defined as confirmed serum creatinine increase of 40% or greater from baseline during dosing)	• Withhold dosing until improvement[a] or return to baseline is achieved
Electrolyte or metabolic abnormalities	• Grade 2 or above: withhold dosing until Grade 1 or baseline is achieved
AST or ALT elevation	• AST or ALT >5 times ULN in the absence of liver metastases: discontinue dosing
Other non-haematologic toxicity	• Any serious adverse reaction requiring a treatment delay of >4 weeks: discontinue dosing • Any recurrent Grade 3 or Grade 4 toxicity, or persistent or intolerable Grade 2 toxicity: discontinue dosing • Any unacceptable toxicity: discontinue dosing

Abbreviations: ALT=alanine aminotransferase; AST=aspartate aminotransferase; ULN=upper limit of normal.

[a] Improvement: serum creatinine has improved sufficiently that the investigator assesses that the next cycle may be administered even though a return to baseline has not been achieved.

Where a subject requires surgery during the treatment phase of the study, the IMP may be temporarily discontinued providing that the surgery is not with therapeutic intent for prostate cancer. Following surgery, the investigator should assess the subject's suitability for restarting the IMP and discuss this with the subject. A maximum dose delay of 8 weeks is acceptable for surgical treatment (i.e. a maximum of 14 weeks from the last IMP administration for subjects on a 6-week dosing schedule, or 11 weeks for subjects on a 3-week dosing schedule). If a subject requires a dose delay of >8 weeks, the investigator should withdraw the subject from the clinical study.

In the case of logistical or administrative issues, which require dose delay, the cycle can be extended by up to 28 days (giving a maximum dosing interval of 10 weeks for subjects on a 6-week dosing schedule, or 7 weeks for subjects on a 3-week dosing schedule).

Radiotherapy during the study is not permitted due to the critical importance of radiation dosimetry and the need to correlate dose estimates from the IMP administrations with any observed responses and/or toxicity. If a subject requires radiotherapy, then the investigator should withdraw them from the clinical study. Subjects who have received the IMP will continue to be followed for safety.

4.1.2.3 At-risk Organ Radiation Dose Limits in Phase 2

The widely referenced upper absorbed dose limits for at-risk organs in RLT clinical trials, are based on tolerance values reported in the literature for EBRT. These values do not consider the inherent differences between EBRT and RLT and have been applied in the absence of sufficient prospective data assessing the dose-toxicity relationship in RLT (Wahl 2021). There is concern that implementation of EBRT dose limits in RLT clinical trials is impairing the process of treatment optimisation and is potentially resulting in the undertreatment of patients (Kiess 2024). In addition, there is evidence that these tolerances can often be safely exceeded

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in clinical practice in patients receiving multiple courses or extended cycles of RLT (Mader 2023, Derlin 2022). However, it is important to note that this evidence is generally not derived from prospective clinical trials, and it is likely that many patients have insufficiently rigorous long-term follow-up.

Based on the average radiation absorbed dose estimates to normal organs obtained in Phase 1, and the maximum cumulative activity of 60 GBq (1620 mCi) in Phase 2, the planned dosing regimens are not designed to exceed the organ radiation dose limits that were applied in Phase 1 (see Table 10). However, given the inter-patient variation in absorbed doses among at-risk organs, it is likely that some subjects will reach the kidney or bone marrow dose limits before completing all intended cycles of treatment. Image-based bone marrow dosimetry will enable absorbed dose values to be measured for key at-risk organs for every subject throughout the study treatment period.

In certain subjects, it may be appropriate for the treating physician to continue with additional cycles (up to the maximum number of cycles for that dosing cohort), where this may result in the kidney or bone marrow radiation dose limits being cautiously exceeded. This would only be permitted if the subject meets the following continuation criteria:

- The subject demonstrates no evidence of progression (defined as radiological progression or clinical progression as determined by the investigator),

  and

- The subject demonstrates an absence of clinically significant normal organ toxicity in the judgement of the investigator

Note that PSA progression alone does not mean that subjects should be prevented from receiving further cycles of therapy, where the treating physician judges this to still be in the subject's best interest. In the event that a subject has a complete biochemical response (i.e. undetectable PSA) and has no visible disease on imaging, the investigator may wish to discontinue dosing when the dose limits have been exceeded.

Subjects must be re-checked against these criteria after every treatment cycle, where normal organ dose limits (as defined in Table 3) have been exceeded. In addition, the subject must be re-consented if the next dosing cycle is predicted to exceed the normal organ dose limits. The subject must be re-consented to ensure that they are re-informed of the potential risks of acute, delayed, and cumulative toxicities that may result from the IMP, and the uncertainty regarding appropriate radiation dose limits.

Additionally, an absolute upper dose limit will also be applied to critical organs for subjects who meet these continuation criteria; these are shown in Table 11. These dose limits have been set based on statistical modelling from the Phase 1 dosimetry data, with due consideration of the existing evidence available for dose-toxicity relationships within RLT. For all other organs not listed in Table 11, the same dose limits that were used in Phase 1 (see Table 3) will be applied.

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Table 10: Modelled Cumulative Normal Organ and Tumour Absorbed Doses for the Three Dosing Cohorts in Phase 2

Dosing Cohort	Cumulative Administered Activity, GBq	Cumulative Normal Organ Dose (Gy)			
		Kidney	Salivary	Bone Marrow	
				Image-based	Blood-based
A	57	18.1	8.2	0.75	1.76
B	59	18.8	8.5	0.78	1.82
C	60	18.9	8.6	0.79	1.85

Abbreviations: GBq=Gigabecquerel; Gy=Gray.

Table 11: Dose Limits and Absolute Upper Dose Limits for the Kidneys and Bone Marrow for Phase 2

Organ at Risk	Dose Limit[a], Gy	Absolute Upper Dose Limit, Gy
Kidney	23	35
Bone marrow	2	2.5

Abbreviations: Gy=Gray.
[a] Dose limit triggering subject re-consent.

4.1.3 Follow-up Period (Phase 1 and Phase 2)

For both Phase 1 and Phase 2, following the treatment phase, subjects will enter into an efficacy and safety follow-up phase until centrally-determined radiographic progression, death, or loss to follow-up, whichever occurs first. Subjects who experience radiographic progression and/or initiate treatment with another anti-cancer therapy will enter into a separate long-term safety follow-up period assessing safety information at 6-monthly intervals using chart review only.

5 CRITERIA FOR EVALUATION

5.1 Phase 1 (Completed)

5.1.1 Primary Endpoints

The primary endpoints in Phase 1 of the study are:

1. Incidence of DLTs during the DLT observation period. The definition of DLTs in this study is described in Section 4.1.1.4.
2. Frequency and nature of treatment-emergent adverse events (TEAEs).

5.1.2 Secondary Endpoints

The secondary endpoints in Phase 1 of this study are:

1. Terminal half-life of activity concentrations in blood.
2. Specific whole-body absorbed dose (Gy/GBq), specific absorbed dose to the tumour lesions (Gy/GBq), specific absorbed dose per organ (Gy/GBq) and cumulative absorbed organ and whole-body absorbed doses (Gy).
3. Number of subjects with ≥50% reduction in PSA level from baseline at 12 weeks after first therapeutic IMP administration.

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4. Number of subjects with best response in PSA level ≥50% from baseline to the end of the dosing period.
5. PSA PFS: Time interval from first cycle of therapeutic IMP administration to PSA progression as defined by the Prostate Cancer Clinical Trials Working Group 3 (PCWG3) criteria, or death, whichever comes first.
6. Number of subjects who remain PSA progression-free at fixed 6-month intervals throughout the study.
7. Duration of response as defined by time interval from achieving a ≥50% reduction in PSA compared to baseline to the PSA returning to baseline or evidence of radiological progression, as defined by PCWG3, or death.
8. Radiographic PFS: Time interval from first therapeutic IMP administration to the date when the first site of disease is found to progress radiographically, or death, whichever occurs first, as defined by PCWG3.
9. Number of subjects who remain radiographic progression-free at fixed 6-month intervals throughout the study.
10. Number of subjects with confirmed CR or partial response (PR) based on PCWG3-recommended application of the Response Evaluation Criteria in Solid Tumours v1.1 (RECIST v1.1) criteria.
11. Evaluation of PSMA PET/CT defined disease response after 2 treatment cycles.
12. Overall survival at fixed 6-month intervals throughout the study.
13. Time to PSA progression: Time interval from first cycle of therapeutic IMP administration to PSA progression as defined by the PCWG3 criteria.
14. Time to radiographic progression: Time interval from first therapeutic IMP administration to the date when the first site of disease is found to progress radiographically as defined by PCWG3.

5.2 Phase 2

5.2.1 Primary Endpoint

The primary endpoint in Phase 2 of the study is the number of subjects with an anti-tumour response defined as a ≥50% reduction in PSA level from baseline to the EoT.

5.2.2 Secondary Endpoints

The secondary endpoints in Phase 2 of the study are the:

1. Number of subjects with ≥50% reduction in PSA level from baseline at 12 weeks after first IMP administration.
2. PSA PFS: Time interval from the date of randomisation to PSA progression as defined by PCWG3 criteria, or death.
3. Number of subjects who remain PSA progression-free at fixed 6-month intervals throughout the study.
4. Duration of response as defined by time interval from achieving a ≥50% reduction in PSA compared to baseline to the PSA returning to baseline or evidence of radiological progression, as defined by PCWG3.

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5. Radiographic PFS: Time interval from the date of randomisation to the date when the first site of disease is found to progress radiographically, or death, whichever occurs first, as defined by PCWG3.
6. Number of subjects who remain radiographic progression-free at fixed 6-month intervals throughout the study.
7. Number of subjects with confirmed CR or PR based on PCWG3-recommended application of RECIST v1.1.
8. Overall survival, defined as the time interval from date of randomisation to death.
9. Overall survival at fixed 6-month intervals throughout the study.
10. Time to PSA progression: Time interval from the date of randomisation to PSA progression as defined by PCWG3 criteria.
11. Time to radiographic progression: Time interval from the date of randomisation to the date when the first site of disease is found to progress radiographically as defined by PCWG3.
12. Frequency and nature of TEAEs.
13. Specific absorbed dose to the tumour lesions (Gy/GBq), specific absorbed dose per organ (Gy/GBq).
14. Changes in patient-reported outcomes (PROs) assessed by the FACT-P. The PROs will be measured from baseline up to the EoT:
    - Change from baseline in FACT-P total score.
    - Proportion of subjects with improvement or worsening in the FACT-P total score and subscales.
    - Time to worsening in the FACT-P total score and subscales, defined as the time from randomisation to worsening, clinical progression, or death, whichever occurs first.

5.2.3 Exploratory Endpoint

The exploratory endpoint in Phase 2 of the study is the change in PRO assessed by the FACT-RNT. PROs will be measured from baseline up to the EoT.

6 SUBJECT SELECTION

6.1 Study Population

Men with PSMA-positive mCRPC who meet all the inclusion and none of the exclusion criteria will be eligible for participation in this study.

6.2 Inclusion Criteria

1. Male subjects, 18 years of age or older.
2. Willing to provide signed and dated written informed consent form (ICF) prior to any study-specific procedures.
3. Willing to comply with required lifestyle restrictions following administration of the IMP per local regulations.
4. Histologically confirmed adenocarcinoma of the prostate.

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5. Serum testosterone levels <50 ng/dL (1.73 nmol/L) after surgical or continued chemical castration.

   Note: If a sample has been taken within the last 3 months before the screening date and there has been no change to medication it can be used to confirm eligibility.

6. Presence of disease target or non-target lesions (per RECIST v1.1) on CT/MRI

   and/or

   Presence of disease on full body ^{99m}Tc bone scan.

   Note: Imaging scans obtained no more than 28 days prior to informed consent may be used to determine imaging eligibility, if available for submission to the central reader.

7. Positive disease expression of PSMA as confirmed on PSMA PET/CT scan.

   Note:

   For Phase 1 only: Positive disease is defined as having at least 1 PSMA-positive lesion of any size with higher uptake than normal liver using visual assessment. The lesion can be bone, lymph node or viscera. At least one PSMA-positive lesion should be visible on the CT/MRI and ≥1.5 cm in the short axis. Further details regarding the interpretation of the diagnostic images can be found in the Image Acquisition Guidelines.

   For Phase 2 only: Positive disease is defined as:

   - Having at least 1 PSMA-positive lesion of any size with a higher uptake than normal parotid using visual assessment. The lesion can be bone, lymph node or viscera, and
   - Having the majority of PSMA-expressing disease with equivalent or higher uptake than normal liver using visual assessment.

   PSMA PET/CT scans obtained no more than 28 days prior to informed consent may be used to determine PSMA eligibility, if available for submission to the central reader. Further details regarding this inclusion criterion can be found in the Image Review Charter.

8. At least 28 days or 5 half-lives (whichever is longer) elapsed between last anti-cancer treatment administration and the initiation of study treatment (except for Luteinising Hormone-releasing Hormone or GnRH).

9. Resolution of all previous treatment-related toxicities to CTCAE Version 5.0 grade of ≤1 (except for chemotherapy-induced alopecia and Grade 2 peripheral neuropathy or Grade 2 urinary frequency which are allowed).

10. Prior major surgery must be at least 12 weeks prior to study entry.

11. Eastern Cooperative Oncology Group (ECOG) performance status of 0-2 with a life expectancy ≥6 months.

12. Adequate bone marrow reserve and organ function as demonstrated by blood count, and serum biochemistry at baseline:

    For Phase 1 only:

    - Platelet count ≥150 × 10^9/L
    - White blood cell (WBC) count ≥3.0 × 10^9/L
    - Neutrophil count of ≥1.5 × 10^9/L
    - Haemoglobin ≥10 g/dL

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- Estimated glomerular filtration rate (using Chronic Kidney Disease Epidemiology Collaboration Creatinine Equation [2009]) (Levey 2009) ≥60 mL/min
- Total bilirubin <1.5 × ULN (except if confirmed history of Gilbert's disease)
- Serum albumin ≥30 g/L
- AST <2 × the ULN
- ALT <2 × the ULN

For Phase 2 only:

Bone marrow reserve:

- Platelet count ≥150 × 10^9/L
- WBC count ≥3.0 × 10^9/L
- Neutrophil count of ≥1.5 × 10^9/L
- Haemoglobin ≥9 g/dL

Renal:

- Estimated glomerular filtration rate (using Chronic Kidney Disease Epidemiology Collaboration Creatinine Equation [2009]) (Levey 2009) ≥60 mL/min

Hepatic:

- Total bilirubin <2 × the institutional ULN. For subjects with known Gilbert's Syndrome, ≤3 × ULN is permitted
- AST <3 × the ULN (<5 × ULN in the presence of confirmed liver metastasis)
- ALT <3 × the ULN (<5 × the ULN in the presence of confirmed liver metastasis)
- Serum albumin ≥30 g/L

13. Male subjects must not father children or donate sperm during the study and for at least 6 months after the last study treatment. In addition, they must agree to use highly effective contraception for this same period to protect partners from any exposure to the IMP. For fertile males with partners who are women of childbearing potential, highly effective contraception should be used during the study and for at least 6 months after the last study treatment. Highly effective methods of contraception include:
    - combined (oestrogen and progestogen containing) hormonal contraception associated with inhibition of ovulation (oral, intravaginal, transdermal)
    - progestogen-only hormonal contraception associated with inhibition of ovulation (oral, injectable, implantable)
    - intrauterine device
    - intrauterine hormone-releasing system
    - bilateral tubal occlusion
    - vasectomised partner
    - sexual abstinence, only where this is the preferred and usual lifestyle of the subject

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A woman is considered of childbearing potential i.e. fertile, following menarche and until becoming post-menopausal unless permanently sterile. Permanent sterilisation methods include hysterectomy, bilateral salpingectomy, and bilateral oophorectomy. A postmenopausal state is defined as no menses for 12 months without an alternative medical cause.

A man is only considered to be infertile if he has had bilateral orchidectomy or successful vasectomy with laboratory confirmed aspermia.

14. Study phase-specific inclusion criteria:

a) For Phase 1 mCRPC only: Subjects who have experienced disease progression on or after at least 1 NAAD (e.g. abiraterone, enzalutamide) and at least 1 course (but no more than 2 courses) of taxane-based chemotherapy.

b) For Phase 2 mCRPC only: Subjects who have experienced disease progression on or after at least 1 NAAD (e.g. abiraterone, enzalutamide, apalutamide, darolutamide), but have not received previous taxane-based chemotherapy for the treatment of mCRPC.

- Prior NAAD can be given as part of doublet or triplet therapy during treatment for metastatic hormone sensitive prostate cancer, or during the castrate resistant phase of disease.
- Prior chemotherapy as part of the treatment of metastatic hormone sensitive disease is permitted.
- Prior first-generation androgen receptor inhibitors (i.e. bicalutamide) are permitted but do not count as prior NAAD for eligibility in this trial.

Note: Subjects must have progressive mCRPC. Disease progression is defined by the "criteria for progression at trial entry" (Table 3 in PCWG3 guidelines; Scher 2016; Appendix 2).

6.3 Exclusion Criteria

1. Known hypersensitivity to the therapeutic IMP, PSMA PET/CT tracer, or diagnostic IMP (for Phase 1 only) or any of its constituents.
2. Presence of PSMA-negative disease:

   Note:

   For Phase 1 only: defined as any large PSMA-negative lymph node >1 cm in the short axis and/or a PSMA-negative bone metastasis which has a significant soft tissue component suggesting ongoing disease activity and/or a PSMA-negative solid organ metastasis >1 cm in the long axis. In addition, subjects with significant low PSMA-expressing disease should be excluded. Further details are provided in the Image Acquisition Guidelines.

   For Phase 2 only: defined as any large PSMA-negative lymph node >2.5 cm in the short axis and/or a PSMA-negative bone metastasis which has a significant soft tissue component suggesting ongoing disease activity and/or a PSMA-negative solid organ metastasis >1 cm in the long axis.

   Further details are provided in the Image Review Charter.

3. Diffuse marrow infiltration of disease ('superscan' appearance on full body ^{99m}Tc bone scan). A superscan is defined as bone scintigraphy in which there is excessive skeletal

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radioisotope uptake in relation to soft tissues along with absent or faint activity in the genitourinary tract and soft tissues due to diffuse bone/bone marrow metastases.

4. Symptomatic spinal cord compression, or clinical or radiological findings that are indicative of impending spinal cord compression.
5. Known history of haematological malignancy.
6. Known history of central nervous system (CNS) metastases.

   Exception: Subjects with a history of CNS metastases who have received and completed therapy (e.g. surgery, radiotherapy), and are neurologically stable, asymptomatic and do not require corticosteroids or anti-convulsants to control neurological symptoms will be eligible. Discrete dural metastases are permitted but diffuse leptomeningeal disease is not. For subjects with a history of CNS metastases, baseline imaging and subsequent radiological imaging for assessing treatment response must include MRI or ceCT evaluation of the brain.
7. Histological findings consistent with neuroendocrine phenotype of prostate cancer.
8. Known history of other solid malignancy that may reduce life expectancy and/or may interfere with disease assessment.

   Exception: Subjects with histopathologically confirmed prior malignancy that has been treated, and who have been disease-free for >3 years. Subjects with treated non-melanoma skin cancer and non-muscle invasive bladder cancer will be eligible.
9. Unresolved urinary tract obstruction defined as radiographic evidence of hydronephrosis with or without ureteric stent/nephrostomy. Where the clinical team judges that the subject's hydronephrosis is not obstructing, and renal function meets the inclusion criteria, the subject may undergo ^{99m}Tc mercaptoacetyltriglycerine scanning during the screening period and if the result is non-obstructed, the subject can be eligible for the study.
10. Any uncontrolled significant medical, psychiatric, or surgical condition or laboratory finding that would pose a risk to subject safety or interfere with study participation or interpretation of individual subject results. For cardiac conditions, this includes, but is not limited to, New York Heart Association class III or IV congestive heart failure, history of congenital prolonged QT syndrome, and myocardial infarction diagnosed within 6 months prior to enrolment.
11. Ongoing treatment with bisphosphonates for bone-targeted therapy.

    Exception: Subjects will be eligible if they have received a stable dose of bisphosphonates for at least 6 weeks prior to enrolment. These subjects must have had renal function monitored since initiation of bisphosphonate therapy, with a stable pattern observed, and must have an eGFR $\geq$ 60 mL/min.
12. Severe urinary incontinence that would preclude safe disposal of radioactive urine.
13. Single kidney or renal transplant or any concomitant nephrotoxic therapy that might put the subject at high risk of renal toxicity during the study in the judgement of the investigator.
14. Clinically significant abnormalities on a single 12-lead electrocardiogram (ECG) at screening.
15. Previously received external beam irradiation to a field that includes more than 30% of the bone marrow or kidneys.

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16. Sponsor employees or investigator site personnel directly affiliated with this study, and their immediate families. Immediate family is defined as a spouse, parent, child, or sibling, whether biological or legally adopted.
17. Previous treatment with any of the following: PSMA-targeted radionuclide therapy, Strontium-89, Samarium-153, Rhenium-186, Rhenium-188, Radium-223, hemi-body irradiation.
18. Subjects with bilateral hip replacements or any significant metallic implants or objects, which may in the opinion of the investigator, affect image quality and/or dosimetry calculations (for Phase 1 only).
19. Transfusion of blood products for the sole purpose of meeting the eligibility criteria for this clinical study.
20. Participation in other studies involving IMP(s) within 28 days or 5 half-lives (whichever is longer) prior to study entry and/or during study participation.

7 PRIOR AND CONCURRENT MEDICATIONS

7.1 Prior Medications

All prior medications received for the treatment of prostate cancer will be documented at screening. The dose, route, unit, frequency of administration, indication for administration and dates of administration will be captured.

7.2 Concomitant Medications

All subjects should continue their usual medications (including analgesia and ADT) during the study unless the investigator recommends that these should be altered in order to take part, or the subject's medical condition changes justifying alterations per conventional medical practice. Specifically, background ADT should continue during the study.

Special consideration should be given to providing subjects with timely analgesia in the event of breakthrough pain during the study, especially during or in preparation for lying on the scanner bed for SPECT imaging.

All concomitant medication and concurrent therapies will be documented at screening and throughout the study; any changes in concomitant medication during the study will be recorded on the electronic case report form (eCRF). The dose, route, unit, frequency of administration, and indication for administration and dates of medication will be captured.

7.3 Prohibited Medications

Other than Lutetium (^{177}Lu) rhPSMA-10.1 Injection and background ADT, no other prostate cancer treatments should be administered whilst receiving the IMP during the study. If it is standard institutional practice, ongoing NAADs may be continued with background ADT. Where the investigator judges that other prostate cancer treatments are necessary, the subject should be withdrawn from the study and enter the passive safety follow-up. If subjects require bone marrow stimulating agents to manage haematological toxicity deemed to be related to the therapeutic IMP (e.g. sargramostim, filgrastim, epoetin alfa, eltrombopag), then they should be discontinued from the study.

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There are no other prohibited medications in this study (other than those listed in the exclusion criteria), and treatment should proceed as per usual medical practice at the institution and the medical judgement of the investigator.

8 STUDY TREATMENTS

8.1 Treatments Administered

The IMP in this study is Lutetium (^{177}Lu) rhPSMA-10.1 injection (the therapeutic IMP or IMP). The diagnostic IMP is rhPSMA-7.3 (^{18}F) (for Phase 1 only).

8.1.1 Lutetium (^{177}Lu) rhPSMA-10.1 Injection

The IMP Handling Manual provides additional information for the IMP.

8.1.2 Diagnostic IMP: rhPSMA-7.3 (^{18}F) Injection (For Phase 1 Only – Completed)

Flotufolastat (^{18}F) injection (rhPSMA-7.3 (^{18}F)) is a radiopharmaceutical, which was approved for use by the US Food and Drug Administration (FDA) on 25 May 2023 under the tradename POSLUMA® (flotufolastat F 18 injection) for PET of PSMA-positive lesions in men with prostate cancer (POSLUMA® Prescribing Information):

- with suspected metastasis who are candidates for initial definitive therapy
- with suspected recurrence based on elevated serum PSA level

The IMP Handling Manual provides additional information for the diagnostic IMP.

8.2 Method of Assigning Subjects to Treatment Groups

8.2.1 Phase 1

Phase 1 of the study is a single-arm study and the treatment phase in Phase 1 is now completed (as of Protocol Version 5.0).

8.2.2 Phase 2

In Phase 2, subjects will be initially randomised to Cohorts 2A and 2B in a 1:1 ratio by block randomisation using an Interactive Response Technology (IRT) system. If Cohort 2C is initiated, the randomisation will be opened to allow the third cohort. This will be a 1:1:1 ratio if Cohorts 2A and 2B are still open, or may be solely Cohort 2C, if recruitment is complete for Cohorts 2A and 2B.

8.3 Blinding

Not applicable. This is an open-label study.

8.4 Formulation of Test and Control Products

8.4.1 Formulation of Lutetium (^{177}Lu) rhPSMA-10.1 Injection

The therapeutic IMP is formulated as a sterile solution for IV administration. The solution contains rhPSMA-10.1 and ^{177}Lu-rhPSMA-10.1, which are respectively, the chemical precursor/non-radioactive carrier and radiopharmaceutical drug substance, in which the beta

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emitting radionuclide Lutetium-177 (^{177}Lu), with a radioactive half-life of 6.7 days, provides a means for systemic radiation therapy.

The molecular structure of the drug substance comprises a PSMA binding motif (glutamate-urea), conjugated via a peptide-like spacer to a SiFA and lutetium-chelator complex (^{177}Lu-DO3A). It is designed to target the extracellular epitope of PSMA which is over-expressed in the majority of prostate tumours (Figure 1). The therapeutic IMP represents an optimised version of ^{177}Lu-rhPSMA-7.3C (pre-Investigational New Drug # 155824), with the only change to the chemical structure being the removal of a propanoate group (on the linker between the ^{177}Lu-chelator complex and the glutamate-urea PSMA binding site).

International Nonproprietary Name:	Not yet assigned
Chemical Name:	Lutetium-^{177}Lu 2,2',2''-(10-((4*R*,7*R*,21*R*,26*S*,30*S*)-7,21,26,30,32-pentacarboxy-4-((4-(di-*tert*-butylfluorosilyl)benzamido)methyl)-2,5,13,16,23,28-hexaoxo-3,6,12,17,22,27,29-heptaazadotriacontyl)-1,4,7,10-tetraazacyclododecane-1,4,7-triyl)triacetate
Molecular Formula:	$C_{60}H_{92}F^{177}LuN_{12}O_{23}Si$

8.4.2 Formulation of rhPSMA-7.3 (^{18}F) Injection (For Phase 1 only – Completed)

The diagnostic IMP is formulated as a sterile, aqueous solution for IV administration. The molecular structure of the rhPSMA-7.3 (^{18}F) drug substance comprises a PSMA binding motif, a peptide spacer, an ^{18}F-radiolabelled silicon-fluoride acceptor moiety and a gallium chelator complex. Please see the rhPSMA-7.3 (^{18}F) IB for more information on the formulation of the diagnostic IMP.

Company Name:	rhPSMA-7.3 (^{18}F)
Chemical Name:	Gallium 2,2',2''-(10-((3S,7S,12R,26R,34S)-1,3,7,12,26,34-hexacarboxy-29-((4-(di-tert-buty[^{18}F]fluorosilyl)benzamido)methyl)-5,10,17,20,28,31-hexaoxo-4,6,11,16,21,27,30-heptaazatetratriacontan-34-yl)-1,4,6,10-tetraazacyclododecane-1,4,7-triyl)triacetate
Molecular Formula:	$C_{63}H_{96}{}^{18}FGaN_{12}O_{25}Si$

8.4.3 Formulation of Control Product

Not applicable. There is no control product in this study.

8.4.4 Packaging and Labelling

8.4.4.1 Phase 1 (Completed)

The therapeutic IMP will be supplied as a multi- or single-dose vial sealed with a synthetic rubber closure and aluminium overseal or in a single unit dose syringe. The product label is region-specific, but as a minimum will include the radioactivity concentration, and/or total radioactivity in GBq and/or mCi and calibration date and time.

The diagnostic IMP will be supplied either in a multi-dose vial sealed with a synthetic rubber closure and aluminium overseal or in a single unit dose syringe and transported in a lead or tungsten radiation-shielding secondary container. The product label is region-specific, but as a minimum will include the radioactive concentration and/or total radioactivity (activity in MBq and/or mCi) and calibration date and time.

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8.4.4.2 Phase 2

The IMP will be supplied as a multi- or single-dose vial closed with a stopper and secured with an overseal or in a single unit dose syringe. The product label is region-specific, but as a minimum will include the total radioactivity in GBq and/or mCi and calibration date and time.

8.5 Supply and Handling of the Investigational Medicinal Product

The quality control (QC) analysis of a sample of the drug product may be performed in parallel with transportation of the drug product to the study site. If this is the case, the investigator (or nominated deputy) will receive release information for the product. Only product for which confirmation of release has been received, or product which is released prior to shipping, shall be used. If at any time in the transport or storage, the integrity of the primary packaging (vial/syringe) is compromised, the product shall not be used.

Waste must be disposed of according to Federal, State, and local regulations for radioactive material. Investigator sites must comply with all applicable regulations. Precautions for the safe handling of radioactive materials should be observed. Sites will be provided with an IMP Handling Manual for each IMP, which will provide guidance on the ordering, supply, receipt, storage, and handling of the product.

8.5.1 Dispensing

Where the product is supplied in a syringe, the dose will be measured in a qualified radionuclide dose calibrator before administration. Where the product is supplied in a vial, the volume of injection for each subject will be calculated and dispensed according to the instructions in the IMP Handling Manual. The unit dose dispensed must be used for a single subject.

Further information is provided in the IMP Handling Manual.

8.5.2 Administration Instructions

The IMP will be administered under direct supervision of study personnel. Local site guidelines for the administration of radiopharmaceuticals should be followed wherever possible. Where local guidelines are not explicit, IV access should be obtained using a large bore cannula placed into a palpable vein. Once established, the IV cannula should be easily flushable with sterile 0.9% sodium chloride solution and the area visually inspected for any sign of swelling. The subject should be asked whether they experienced any discomfort during the flushing. Where peripheral venous access is not achievable or cannot be used due to site policy, central venous access may be considered.

If a cannula is not correctly placed, radiopharmaceutical extravasation can occur, and potentially lead to severe soft tissue damage and unintended extended local radiation exposure. In the event of suspected extravasation, clinical management should follow local site guidelines, and the investigator and Sponsor should be informed of the incident (to be reported as a serious adverse event [SAE] in the eCRF). The appropriate regulatory authorities should be informed per the guidelines in that jurisdiction e.g. radiation safety committees. For further information, please refer to the IMP Handling Manual.

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8.5.2.1 Lutetium (^{177}Lu) rhPSMA-10.1 Injection

8.5.2.1.1 Phase 1 (Completed)

The radioactivity levels of the IMP to be administered per cycle and cohort in Phase 1 of the study are described in Table 2. There will be a permissible tolerance of ±10% on the administered therapeutic IMP radioactivity level.

8.5.2.1.2 Phase 2

Radioactivity levels of the IMP to be administered in the Phase 2 cohorts are presented in Figure 5. There will be a permissible tolerance of ±10% on the administered IMP radioactivity level.

Details of administration of the IMP are provided in the IMP Handling Manual.

*8.5.2.2 rhPSMA-7.3 (^{18}F) Injection **(Phase 1 only – Completed)***

The diagnostic IMP will be administered at an activity of 296 MBq (8 mCi) ±20% diluted with normal saline solution for injection up to 10 mL. The diagnostic IMP will be administered as an IV injection followed by 10 mL flush with normal saline solution.

8.5.3 Storage

The therapeutic IMP (and diagnostic IMP in Phase 1) will be provided in a shielded container and should be stored at controlled temperatures.

The relevant IMP Handling Manuals provide detailed instructions regarding IMP storage.

Precautions for the safe handling of radioactive materials should be observed according to the investigator site's usual operating procedures and processes.

8.6 Investigational Medicinal Product Accountability

Investigational medicinal product accountability will be managed in accordance with principles of International Council for Harmonisation (ICH) Good Clinical Practice (GCP) and regulatory requirements. The investigator or their delegate will account for all IMP received at the institution and dispensed to the subjects, keeping up-to-date and accurate accountability records.

Accountability records must include information on:

- IMP received: lot identifiers, date of receipt, and quantity received,
- IMP dispensed: subject identifiers, dates and times dispensed, and radioactivity dispensed,
- IMP destruction: date of destruction, lot numbers, and quantities destroyed.

Spillages should be managed in accordance with local regulations and IMP Handling Manual instructions.

All shipments from the Sponsor to the participating sites will be trackable and relevant documentation maintained.

The study monitor will verify the IMP Accountability Records throughout the course of the study. Any discrepancies noted on final reconciliation will be explained by the investigator or designee.

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8.7 Measures of Treatment Compliance

Subjects will receive the IMP under direct supervision of study personnel. Each administration volume and total radioactivity injected will be checked. The lot identifier and activity per administration will be recorded in each subject's eCRF and source documents.

8.8 End of Study

The primary endpoint analysis will occur after the last subject in the Phase 2 portion of the study has completed the 6-week follow-up period after their final treatment. At this point in the study, a formal interim analysis will be completed, and an interim study report will be written.

At the end of Phase 1, a top line results memo/analysis report will be written, which will include primary endpoint and dosimetry data.

The end of study is defined as when the last subject has completed long-term passive safety follow-up, which is defined as the point at which they have died, or are lost to follow-up, whichever occurs first. At this time point, the clinical study database will be closed to new data and a final study report will be written.

Due to the complexities and the need for radiation dosimetry, subjects in Phase 1 and Phase 2 cannot receive additional treatment cycles of the IMP other than those specified in the protocol.

9 STUDY PROCEDURES AND GUIDELINES

A Study Schedule representing the required testing procedures to be performed for the duration of the study is summarised in Appendix 1.

Written informed consent will be obtained prior to conducting any study-related activities (see Section 18.3). Subjects will provide written informed consent and be assessed for eligibility for study participation at screening (performed within 42 days before first IMP administration). Subject eligibility will also be checked before IMP administration for all treatment cycles. Written informed re-consent will also be obtained if the subject's next dosing cycle is predicted to exceed the normal organ dose limits as defined in this protocol (see Section 18.3).

Subjects are considered as enrolled in the study when they sign the ICF, are screened, and satisfy the eligibility criteria.

9.1 General Assessments

9.1.1 Medical History

Relevant medical history, including history of prostate cancer disease and comorbidities will be recorded at screening and reviewed by the investigator pre-IMP dosing on Day 1. Evidence of histological confirmation of adenocarcinoma of the prostate must be available in source data and provided at screening. Where subjects have undergone genetic sequencing for the management of their prostate cancer (e.g. germline, circulating tumour DNA, or tumour tissue) this will be collected to evaluate whether particular mutations may influence outcomes.

9.1.2 Performance Status

Eastern Cooperative Oncology Group performance status will be collected at the timepoints indicated in the Study Schedule (Appendix 1) and recorded on the eCRF.

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9.1.3 Physical Examination

Physical examination will be performed at the timepoints indicated in the Study Schedule (Appendix 1). At any point after screening, any clinically significant physical examination changes (as determined by the site investigator) should be reported as an AE. Physical examination findings should be recorded on the eCRF.

9.1.4 Physical Measurements

Height and body weight measurements will be performed at the timepoints indicated in the Study Schedule (Appendix 1). Height and body weight should be measured without shoes.

In Phase 1 (completed), body weight should be measured pre-diagnostic IMP administration on the day of diagnostic PSMA/PET CT imaging and pre-therapeutic IMP administration on each day of dosing.

In Phase 2, body weight should be measured pre-IMP administration on each day of dosing.

9.1.5 Demographics

Demographic information as permitted by local regulation (year of birth, age at the time of screening, gender/sex, ethnicity/race) will be recorded at screening.

9.2 Efficacy Assessments

Efficacy will be determined by both clinical laboratory values and radiographic assessments.

9.2.1 Prostate-specific Antigen Blood Sampling

Prostate-specific antigen blood sampling will be performed at timepoints specified in the Study Schedule (Appendix 1) to assess blood plasma PSA levels as a measure of anti-tumour activity.

In Phase 1 (completed), blood samples for PSA testing will be collected every 2 weeks during the treatment phase, and monthly during the efficacy follow-up phase until radiographic progression, death, or loss to follow-up.

In Phase 2, blood samples for PSA testing will be collected at screening, pre-IMP administration on Cycle 1 Day 1, and then every 6 weeks during the treatment phase, and every 12 weeks during the efficacy follow-up phase until centrally-determined radiographic progression, death, or loss to follow-up, whichever occurs first.

A ≥50% reduction in PSA level from baseline is chosen as a surrogate biomarker of anti-tumour activity in this subject population.

9.2.2 Radiological Assessments

The extent of the prostate cancer disease will be assessed every 12 weeks (from Cycle 1 Day 1) until centrally-determined radiographic progression (by ceCT or MRI of the chest/abdomen and pelvis, and a full body ^{99m}Tc bone scan based on PCWG3-recommended application of the RECIST v1.1 criteria), death, or loss to follow-up, whichever occurs first. The full body ^{99m}Tc bone scan must be performed within 5 days prior to IMP administration during the treatment phase to avoid inference with dosimetry measurements.

Further information on ceCT, MRI, and ^{99m}Tc bone scan imaging is presented in Section 9.4.2.

All radiological imaging will be performed as described in the Image Acquisition Guidelines, and scans will be interpreted by the Central Imaging vendor (see Section 9.4.4).

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9.3 Safety Assessments

In all subjects, safety evaluations will include AE monitoring and reporting from the time of informed consent throughout the study. However, once subjects enter the passive long term safety follow-up, AE collection will be via chart review and will not involve any further study related visits or procedures. Clinical laboratory measurements for safety assessments are described in Section 9.5.

9.3.1 Vital Signs

Vital signs (body temperature, blood pressure, pulse, and respiration rates) will be recorded after resting for at least 5 minutes at screening and pre- and post-IMP administration (reference time is the end time of IMP administration) on each dosing day.

Any clinically significant changes in vital sign measurements (as determined by the site investigator) should be reported as an AE.

9.3.2 Electrocardiogram

Twelve-lead ECGs will be performed at the timepoints indicated in the Study Schedule (Appendix 1). The subject must be in supine position in a rested state for at least 5 minutes before ECG assessment is conducted. If a subject is unable to be in the supine position, the subject should be in the most recumbent position achievable. Electrocardiograms should be performed in a standardised method and should be run consecutively.

In Phase 1 (completed), single 12-lead ECGs will be performed at screening and at the EoT visit. Triplicate 12-lead ECGs will be performed pre-therapeutic IMP administration and immediately post-therapeutic IMP administration (between 30 to 120 minutes post-injection is acceptable) for each treatment cycle.

In Phase 2, single 12-lead ECGs will be performed at screening, pre-IMP administration in each cycle, and at the EoT visit. A single 12-lead ECG will also be performed post-IMP administration in Cycle 1 only. This must be performed within 5 hours of the IMP administration.

At any point after screening, any clinically significant ECG changes (as determined by the site investigator) should be reported as an AE. All ECG results should be reviewed by an appropriately qualified physician.

9.3.3 Adverse Events

Adverse events will be captured for all subjects throughout the study. Adverse events will be monitored from the time of informed consent until the last study visit. Duration (start and stop dates and times), severity/grade, outcome, treatment, and relationship to IMP will be recorded on the eCRF. However, once subjects enter the passive long term safety follow-up, AE collection will be via chart review and will not involve any further study related visits or procedures. Adverse events ongoing at the time of the next visit will be reviewed and updated or closed, as required. Details regarding reporting and capture of SAEs are provided in Section 12.2.

9.3.3.1 Lutetium (^{177}Lu) rhPSMA-10.1 Injection

Any untoward medical occurrence that occurs from the time of signed informed consent to the time immediately prior to IMP administration will be recorded in the eCRF as an AE. Only AEs reported after a subject has received the IMP will be assessed for relatedness to the IMP.

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Subjects who experience any SAE or an AE that was assessed as related to the IMP will be followed until resolution (CTCAE Version 5.0 grade of ≤1) or stabilisation of these events is recorded.

Throughout the study, subjects will be asked to report AEs as they occur to the site team but will also be prompted to report at each visit. However, once subjects enter the passive long term safety follow-up, AE collection will be via chart review and will not involve any further study related visits or procedures. Details regarding reporting and capture of SAEs are provided in Section 12.2.

9.3.3.2 rhPSMA-7.3 (^{18}F) (Phase 1 only – Completed)

The diagnostic IMP is a microdose PET radiopharmaceutical that was approved for use by the US FDA on 25 May 2023 under the tradename POSLUMA® (flotufolastat F 18 injection) for PET of PSMA-positive lesions in men with prostate cancer:

- with suspected metastasis who are candidates for initial definitive therapy
- with suspected recurrence based on elevated serum PSA level.

The IB for rhPSMA-7.3 (^{18}F) will be supplied to all investigator sites. The Reference Safety Information (RSI) will be the IB. If the diagnostic IMP is approved during the conduct of the study, the IB will remain the RSI as the use is expected to be off-label. Adverse events will be monitored and recorded from the time of injection until 48 hours later as this is the period of time in which potentially related AEs may plausibly occur. Where AEs occur, duration (start and stop dates and times), severity/grade, outcome, treatment, and relationship to diagnostic IMP will be recorded on the eCRF. If the AE occurs during the eligibility rhPSMA-7.3 (^{18}F) PET/CT scan (i.e. the subject has not received the therapeutic IMP yet), then the investigator will only be asked to assess relatedness to the diagnostic IMP. If the AE occurs during the 48 hours following the response assessment rhPSMA-7.3 (^{18}F) PET/CT scan, which takes place after 2 cycles of therapy (4-6 weeks after Cycle 2, Day 1), the investigator will be asked to assess relatedness to both IMPs.

Subjects who experience any SAE or an AE that was assessed as related to the diagnostic IMP will be followed until resolution (CTCAE Version 5.0 grade of ≤1) or stabilisation of these events is recorded.

9.4 Tumour Imaging and Dosimetry

9.4.1 PSMA Diagnostic Positron Emission Tomography/Computed Tomography Scan

9.4.1.1 Phase 1 (Completed)

In Phase 1, all subjects will receive an IV bolus injection of the diagnostic IMP and undergo a PET/CT scan at screening for imaging eligibility as described in the Image Acquisition Guidelines and the inclusion/exclusion criteria. An additional PET/CT scan with the diagnostic IMP will be completed within 4-6 weeks after Cycle 2, of the therapeutic IMP administration (4-6 weeks after Cycle 2, Day 1); this scan will be used for exploratory analyses to evaluate response as determined by PSMA PET imaging but will not be used to define progression for the purposes of this study.

The results of the PSMA PET/CT scan at screening will be used to select suitable subjects for the study. The criteria for inclusion require at least 1 PSMA-positive lesion with greater uptake than normal liver using visual assessment. In Phase 1 only, there must be at least 1 lesion

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>1.5 cm in diameter to facilitate tumour dosimetry assessments. In addition, the exclusion criteria require the exclusion of subjects with PSMA-negative disease which is defined as any large PSMA-negative lymph node >1 cm in the short axis and/or a PSMA-negative bone metastasis which has a significant soft tissue component suggesting ongoing disease activity and/or a PSMA-negative solid organ metastasis >1 cm in the long axis. In Phase 1, image assessment will be performed by the local site team, which may include the site investigator, a nuclear medicine physician and/or a radiologist. The PSMA PET/CT scans must be submitted to the Central Imaging vendor but will not be read centrally.

9.4.1.2 Phase 2

In Phase 2, subjects will undergo a PSMA PET/CT scan at screening for imaging eligibility as described in inclusion criterion #7 and exclusion criterion #2 for Phase 2. Imaging eligibility determination will be performed by the Central Imaging vendor. The PSMA PET/CT scans and baseline cross-sectional imaging must be submitted promptly to the Central Imaging vendor. Further details are provided in the Image Review Charter.

The PSMA PET/CT tracer selected by the investigator should be approved under marketing authorisation where available, or already be in use at the study site via an applicable local authorisation or permission for use at the study site. In order to reduce unnecessary interventions and radiation exposure, where PSMA PET/CT imaging has been recently completed as part of standard of care and the images are available for submission to the central reader, these images may be used to determine eligibility, provided that the scan was acquired no more than 28 days prior to informed consent.

Further guidance for PSMA PET/CT imaging for both study phases is provided in the Image Acquisition Guidelines.

9.4.2 Contrast Enhanced Computed Tomography /Magnetic Resonance Imaging, and ^{99m}Tc Bone Scan

To establish baseline imaging, ceCT or MRI of the chest/abdomen and pelvis and a full body ^{99m}Tc bone scan will be completed at screening and read locally. In order to reduce unnecessary interventions and radiation exposure, where cross-sectional imaging has been recently completed as part of standard of care and the images are available for submission to the central reader, these images may be used to determine eligibility and for baseline assessment, provided that the scan was acquired no more than 28 days prior to informed consent.

Throughout the study, this cross-sectional imaging assessment will be performed every 12 weeks ±5 days from Cycle 1 Day 1 for all subjects until centrally-determined radiographic progression, death, or loss to follow-up, whichever occurs first. Brain MRI/CT imaging is not required unless there is clinical suspicion of brain metastases. The ceCT or MRI and ^{99m}Tc bone scans must be submitted to the Central Imaging vendor promptly for efficacy central assessment according to PCWG3 criteria.

Each scan will be centrally assessed per the PCWG3 criteria, including the recommended usage of RECIST v1.1. If a subject shows radiological progression, based on the results of the central assessment before completing all planned cycles of treatment, the subject will not receive further doses of the IMP. If a subject has PSA progression without radiological progression at any time during treatment, they will continue to receive further cycles of the IMP per PCWG3 recommendation (Scher 2016; Appendix 2).

If a subject has renal insufficiency or an intolerance to iodinated IV contrast agent (or if they develop these issues during the course of the study), non-contrast CT of the chest is acceptable

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and should be supplied with an MRI of the abdomen and pelvis (with contrast, if possible). Further guidance for ceCT, MRI, and ^{99m}Tc bone scans is provided in the Image Acquisition Guidelines.

9.4.3 Single-photon Emission Computed Tomography/Computed Tomography for Dosimetry

9.4.3.1 Phase 1 (Completed)

After each therapeutic IMP administration in Phase 1, organ and tumour dosimetric assessment will be performed by serial quantitative SPECT/CT at 3 hours, 24 hours, 48 hours, and 168 hours of each treatment cycle. The SPECT/CT scans must be submitted promptly to the Central Imaging vendor on the day of scan acquisition (and no more than 24 hours after the final timepoint at 168 hours) to allow sufficient time for the central dosimetry analyses to be performed. Dosimetry results will be reported to investigators prior to the next therapeutic IMP administration (within 28 days) to allow for a decision to be taken on the administration of the next dose.

Estimated absorbed radiation dose to the heart will be recorded for each administration to allow assessment of a dose-response relationship in the event of observed cardiac toxicity. All data will be presented to the SRC after the completion of a cohort to guide the decision on cohort escalation (see Section 15).

9.4.3.2 Phase 2

Dosimetry data from Phase 1 shows that cumulative absorbed doses can be predicted using first-cycle dose estimates when scaling time-activity curves from a single SPECT/CT time point for subsequent cycles. As a result, a simplified dosimetry protocol will be used in Phase 2 to assess absorbed doses in organs and tumours. In Phase 2, SPECT/CT imaging will be performed at 3 hours (±1 hour) post-IMP administration, and then at 24 hours (±4 hours), 48 hours (±4 hours), and at 168 hours (±24 hours) post-IMP administration for the first treatment cycle only. For all subsequent cycles, a single time-point SPECT/CT imaging will be performed at 48 hours (±4 hours). SPECT/CT imaging at 168 hours (Day 8) post-IMP administration should be performed within a 24-hour window (168 hours). With the exception of the first 2 treatment cycles in Cohort 2B, available dosimetry data must be reviewed every 6 weeks and prior to any subsequent treatment cycles to assess organ dose limits and confirm eligibility for further treatment.

Further guidance for the SPECT/CT scans is provided in the Study Dosimetry Manual.

9.4.4 Image Interpretation

The assessment of the PSMA PET/CT images and baseline ceCT or MRI and ^{99m}Tc Bone Scan images for the purposes of eligibility in Phase 1 will be performed by the local site team, which may include the site investigator, a nuclear medicine physician, and/or a radiologist. In Phase 2, assessment of the baseline ceCT or MRI and ^{99m}Tc Bone Scan images for the purposes of eligibility will also be performed by the local site team; however, imaging scans should be submitted to the Central Imaging vendor. Other image interpretations in the study will be performed by the Central Imaging vendor (Table 12).

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Table 12: Sites for Image Assessment

Procedure	Phase 1	Phase 2
ceCT or MRI, and ^{99m}Tc Bone Scan eligibility	Local site read	Local site read
PSMA PET/CT eligibility	Local site read	Central read[a]
ceCT or MRI, and ^{99m}Tc Bone Scan efficacy analyses	Central read	Central read
PSMA PET/CT efficacy analyses	Central read	Not applicable
SPECT/CT dosimetry	Central analyses with specialist vendor	Central analyses with specialist vendor

Abbreviations: ^{99m}Tc=technetium-99m; ceCT=contrast enhanced computed tomography; CT=computed tomography; MRI=magnetic resonance imaging; PET=positron emission tomography; PSMA=prostate-specific membrane antigen; SPECT/CT=single-photon emission computed tomography.
[a] This will include a concurrent read of the cross-sectional images to assess PSMA-negative disease as per exclusion criterion #2.

9.5 Clinical Laboratory Measurements

Blood samples will be collected at the timepoints indicated in the Study Schedule (Appendix 1). Laboratory evaluations will be conducted at the local site laboratory following locally approved site guidelines.

A summary of the blood sample collection schedule at each visit in Phase 1 is presented in Table 9 of Protocol Version 4.0 (dated 22 September 2023).

Laboratory assessments on the day (or within 72 hours) of every cycle before Lutetium (^{177}Lu) rhPSMA-10.1 Injection administration will be used to assess if the subject can receive the subsequent treatment cycle. If collected within 72 hours of Day 1, then the Day 1 blood collection for safety assessments may be waived, unless clinically indicated. The results must be reviewed before administering the next treatment dose.

Phase 1

Blood safety laboratory tests, including haematology (full blood count), biochemistry, thyroid function, and coagulation, will be performed during screening, at baseline (pre-therapeutic IMP administration) and repeated at 24 hours (±8 hours) and 7 days post-therapeutic IMP administration. Blood samples will also be collected at weekly intervals during the 6-week treatment cycle, and a final blood sample will be collected at the EoT visit. In addition, blood will be taken at set timepoints for radiation dosimetry assessments as detailed in Section 9.6.1. Wherever possible, sites should attempt to take blood with a single needle puncture in order to minimise discomfort to the subject e.g. the use of a vacutainer system.

Phase 2

Blood safety laboratory tests, including haematology, biochemistry, and coagulation will be performed at screening and baseline (pre-IMP administration) in each treatment cycle. In between cycles of therapy in Cohorts 2A and 2C, and during the 6-week dosing intervals in Cohort 2B, a blood sample will be taken every 2 weeks on Days 15, 29 and 43 (± 24 hours) for safety monitoring. For Cohort 2B, blood safety laboratory tests will also be performed on Day 22 of the first 2 treatment cycles during the 3-weekly dosing intervals. Additional haematology, coagulation, and biochemistry samples may be collected as clinically indicated during the study and must be recorded in the eCRF including the reason for sampling. A final

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blood sample will also be collected at the EoT visit. Wherever possible, sites should attempt to take blood with a single needle puncture in order to minimise discomfort to the subject e.g. the use of a vacutainer system.

In both study phases, ad-hoc blood testing may be performed if required, depending on clinical need as directed by the investigator (e.g. in the event of myelosuppression requiring more frequent monitoring than weekly).

Where greater than 42 days have elapsed between the screening blood tests used to determine eligibility and the planned IMP administration, the eligibility values should be re-confirmed with additional haematology, biochemistry, thyroid function (Phase 1 only), and coagulation testing prior to IMP administration. The repeat values should be reassessed against the inclusion/exclusion criteria.

At any timepoint after screening, abnormal clinically significant changes from the blood values recorded at screening (as determined by the site investigator) in clinical laboratory measurements should be reported as an AE. Ongoing SAEs or related AEs will be followed until resolution or stabilisation.

9.5.1 Haematology

Blood will be obtained and sent to the site's local laboratory for a complete blood count (haemoglobin, haematocrit, red blood cell count, WBC count, WBC differential [absolute or percentage], and platelet count determinations) for assessment.

9.5.2 Biochemistry

Blood will be obtained and sent to the site's local laboratory for determination of serum sodium, potassium, chloride, bicarbonate, random glucose, blood urea nitrogen, creatinine, magnesium, calcium, AST, ALT, ALP, total bilirubin, direct bilirubin, gamma-glutamyl transferase, albumin, lactate dehydrogenase, and C-reactive protein. Estimated glomerular filtration rate should be calculated using Chronic Kidney Disease Epidemiology Collaboration Creatinine Equation (2009) (Levey 2009).

9.5.3 Coagulation

In Phase 1, blood will be obtained and sent to the site's local laboratory for the determination of prothrombin time, activated partial thromboplastin time, and thrombin time.

In Phase 2, blood will be obtained and sent to the site's local laboratory for the determination of prothrombin time, activated partial thromboplastin time, and international normalised ratio.

9.5.4 Thyroid Function Panel (Phase 1 only – Completed)

Blood will be obtained and sent to the site's local laboratory for the determination of thyroid stimulating hormone (TSH), free thyroxine (T4), and free triiodothyronine (T3).

9.5.5 Prostate-specific Antigen Testing

Blood will be obtained and sent to the site's local laboratory for determination of serum PSA levels.

9.5.6 Urinalysis

In both phases of the study, urinalysis will be performed at screening with a particular view to reducing the risk of concurrent urinary tract infection. Colour, specific gravity, pH, protein, glucose, ketones, nitrites, and blood will be measured. The presence of nitrites on dipstick will

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require IMP dosing to be withheld pending urine culture and the treatment of any infection. Urinalysis will also be performed at the EoT visit in Phases 1 and 2 of the study.

Of note, subjects with indwelling urinary catheters will not undergo urinalysis due to the high probability of false positive findings. If there is clinical suspicion of a catheter associated urinary tract infection, then the subject should be assessed and treated as per local site practice. On the day of IMP dosing, any new urinary symptoms reported by the subject should trigger assessment by the investigator. Therapeutic IMP administration should be delayed until the suspected infection has resolved.

For Phase 2 only, a spot urine protein creatinine ratio (UPCR) should be obtained pre-IMP administration on the morning of each dosing day. Wherever possible, this sample should be collected at approximately the same time of the day. The sample should not be the first void of the day. Subjects should eat and drink as normal. The results of the UPCR will be tracked over the treatment phase and during active follow-up. Results of the UPCR will not be used in isolation to stop or alter IMP dosing.

9.6 Pharmacokinetic Measurements (Phase 1 Only)

In Phase 1, the PK of the therapeutic IMP in blood will be determined through multi-time point whole blood sampling post-therapeutic IMP administration.

9.6.1 Blood Sample Collection for Radioactivity Measurements (Phase 1 Only- Completed)

In Phase 1, blood samples for radioactivity measurement will be collected within 30 minutes prior to therapeutic IMP administration, and 30 minutes, 90 minutes (±10 minutes), 4 hours (±30 minutes), 24 hours (±1 hour), and 48 hours (±1 hour) post-therapeutic IMP administration in all treatment cycles (see Table 9 of Protocol Version 4.0 [dated 22 September 2023]).

Blood samples should be collected from the contralateral arm to that of the therapeutic IMP injection, or from another site.

The accurate time of sample collection and the duration for measuring the activity concentration must be recorded. Any issues associated with sample collection or processing should be reported to the study monitor. Blood sample collection is further detailed in the Study Dosimetry Manual.

No blood radioactivity measurements will be completed in Phase 2.

9.7 Patient-reported Outcomes (*Phase 2 Only*)

Patient-reported outcome measures will not be recorded in Phase 1.

All subjects in Phase 2 will complete the assessments below at baseline, and at 6-week intervals (and at 3-week intervals for the first 3 cycles of Cohort 2B) to coincide with IMP dosing in each treatment cycle. The PRO questionnaires will be completed pre-IMP administration. The FACT-P questionnaire should be completed first, followed by the FACT-RNT questionnaire. A final PRO assessment will be completed at the EoT visit. Subject assessments will be captured on paper at the participating institution and questionnaires will be provided in the local language.

9.7.1 Functional Assessment of Cancer Therapy-Prostate (FACT-P)

The FACT-P combines the FACT-General plus a 12-item prostate cancer-specific supplement or subscale, which includes questions related to pain, appetite, sexual function, and urinary and

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bowel functions. This measure is widely recommended for PROs in the prostate cancer clinical trial literature and has strong evidence of validity and feasibility and a high suitability for the study population.

9.7.2 Functional Assessment of Cancer Therapy–Radionuclide Therapy (FACT-RNT)

The FACT-RNT is a standardised measure to monitor relevant symptoms and toxicities among prostate cancer patients in RNT trials and clinical settings. It was developed first by identifying relevant symptoms and toxicities by reviewing published trials and interviewing prostate cancer patients receiving radionuclide therapy (n=529), caregivers (n=514), and clinicians (n=511), then by selecting items for inclusion. The item list was refined with input from experts in RNTs and PRO measures.

10 EVALUATIONS BY VISIT

The study procedures and assessments for Phase 1 and Phase 2 of the study are summarised in the Study Schedule (Appendix 1). A signed and dated ICF will be obtained before any study-specific procedures, including those required for screening, are performed.

10.1 Phase 1 (Completed)

10.1.1 Visit 1 (Screening; Day -42 [Within 42 Days Before Therapeutic IMP Administration])

1. Review the study with the subject and obtain written informed consent and Health Insurance Portability and Accountability Act (HIPAA) authorisation (or equivalent, as required).
2. Confirm eligibility criteria.
3. Record demographics data.
4. Record ongoing and past medical history, including confirmation of a history of adenocarcinoma of the prostate.
5. Record concomitant medications.
6. Record ECOG performance status.
7. Perform a focused physical examination, including height and body weight.
8. Perform and record vital signs, including single 12-lead ECG.
9. Collect urine sample for dipstick urinalysis to exclude urinary tract infection.
10. Collect blood samples for screening haematology, coagulation, biochemistry, thyroid function panel, and PSA laboratory testing.
11. Perform ceCT or MRI of chest/abdomen and pelvis and full body ^{99m}Tc bone scan to establish baseline imaging. In order to reduce unnecessary interventions and radiation exposure, where cross-sectional imaging has been recently completed as part of standard of care and the images are available for submission to the central reader, these images may be used to determine eligibility and for baseline assessment, provided that the scan was acquired no more than 28 days prior to informed consent.

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12. The Patient Eligibility Review Checklist part 1 will be completed and will be reviewed and approved by the medical monitor before administration of diagnostic IMP (rhPSMA-7.3 (^{18}F)) and the PET/CT scan can be performed.
13. Administer diagnostic IMP (rhPSMA-7.3 (^{18}F)) and perform a PET/CT scan for imaging eligibility assessment. Note: body weight should be measured before this assessment.
14. Review all screening visit data and PSMA PET/CT imaging eligibility and confirm subject suitability for enrolment.
15. The Patient Eligibility Review Checklist part 2 will be completed and will be reviewed and approved by the medical monitor before the therapeutic IMP can be ordered.
16. Commence AE monitoring.
17. Order therapeutic IMP and schedule subject for Day 1 visit within 42 days of consent.
18. Book any visit logistics, as required.
19. Give subject contact card.

10.1.2 Day 1 of Treatment Cycle 1 (Therapeutic IMP Administration)

1. Record any changes in concomitant medications since screening.
2. Ask about and record any AEs occurring since screening.
3. Record ECOG performance status pre-therapeutic IMP administration.
4. Perform and record physical examination including body weight pre-therapeutic IMP administration.
5. Perform and record vital signs pre-and post-therapeutic IMP administration.
6. Perform and record triplicate 12-lead ECG pre-therapeutic IMP administration.
7. Review suitability to administer therapeutic IMP.
8. Collect blood sample pre-therapeutic IMP administration for haematology, coagulation, biochemistry, thyroid function panel, and PSA laboratory testing.
9. Collect blood sample within 30 minutes prior to the therapeutic IMP administration for radiation dosimetry.
10. Administer therapeutic IMP (IV Lutetium (^{177}Lu) rhPSMA-10.1 Injection).
11. Collect blood samples at 30 minutes, 90 minutes, and 4 hours post-therapeutic IMP administration for radiation dosimetry.
12. Perform triplicate 12-lead ECG between 30 to 120 minutes post-therapeutic IMP administration.
13. Perform SPECT/CT imaging for dosimetry at 3 hours post-therapeutic IMP administration.
14. Record any injection site reactions and any other AEs that occur during the visit.

10.1.3 Day 1 of Treatment (Cycle 2 and Subsequent Cycles of Therapeutic IMP Administration)

1. Approximately 2 weeks before any subsequent treatment cycles, review dosimetry data, clinical assessments, and laboratory evaluations to confirm subject eligibility for further treatment cycles.

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2. Record any changes in concomitant medications since last visit.
3. Ask about and record any AEs occurring since last visit.
4. Perform and record physical examination including body weight pre-therapeutic IMP administration.
5. Perform and record vital signs pre-and post-therapeutic IMP administration.
6. Perform triplicate 12-lead ECG pre-therapeutic IMP administration.
7. Collect blood sample pre-therapeutic IMP administration for haematology, coagulation, biochemistry, and thyroid function panel laboratory testing.
8. Collect blood sample within 30 minutes prior to the therapeutic IMP administration for radiation dosimetry.
9. Administer therapeutic IMP (IV Lutetium (^{177}Lu) rhPSMA-10.1 Injection).
10. Collect blood samples at 30 minutes, 90 minutes, and 4 hours post-therapeutic IMP administration for radiation dosimetry.
11. Perform triplicate 12-lead ECG between 30 to 120 minutes post-therapeutic IMP administration.
12. Perform SPECT/CT imaging for dosimetry at 3 hours post-therapeutic IMP administration.
13. Record any injection site reactions and any other AEs that occur during the visit.

10.1.4 24 Hours to 168 Hours of Each Treatment Cycle (Post-therapeutic IMP Dose of Each Treatment Cycle)

Evaluations during Cycle 1 and subsequent cycles of therapy are summarised in the Study Schedule (Appendix 1).

Subjects may be hospitalised from Day 1 to Day 3 at the discretion of the investigator and in compliance with local rules and regulations in that jurisdiction. For convenience, subjects may be offered transport to and from the clinic to their home, and/or overnight accommodation in a nearby hotel. Subjects may also be outpatients and return home at the end of the day to return the next day for additional interventions/monitoring.

1. Record any changes in concomitant medications since last visit.
2. Ask about and record any AEs occurring since last visit.
3. Perform and record physical examination at 168 hours.
4. Collect blood samples for haematology, coagulation, and biochemistry laboratory testing 24 hours (± 8 hours) and 7 days post-therapeutic IMP administration.
5. Collect blood samples for radiation dosimetry at 24 hours and 48 hours post-therapeutic IMP administration in all treatment cycles.
6. Perform SPECT/CT imaging for dosimetry at 24 hours, 48 hours, and 168 hours post-therapeutic IMP administration (all treatment cycles).

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10.1.5 Six-week Treatment Interval Period

Evaluations during the 6-week treatment cycles are summarised in the Study Schedule (Appendix 1). The subject will be asked to present at the study site/centre on a weekly basis. The list of procedures at weekly visits (±1 day) are as follows:

Week 2 (Day 14), Week 3 (Day 21), Week 4 (Day 28), Week 5 (Day 35), and Week 6 (Day 42)

1. Record any changes in concomitant medications at weekly intervals.
2. Ask about and record any AEs occurring at weekly intervals.
   Note: Subjects will be asked to report AEs as they occur to the site team but will also be prompted to report at each weekly visit.
3. Collect blood samples for PSA, haematology, coagulation, and biochemistry laboratory testing. Ad-hoc blood testing may be performed if required, depending on clinical need as directed by the investigator (e.g. in the event of myelosuppression requiring more frequent monitoring than weekly).

Note: Week 3 (Day 21) and Week 5 (Day 35) can be conducted as remote visits and therefore laboratory safety tests would not be conducted at these visits and only AEs or changes in concomitant medication will be recorded.

10.1.6 Twelve-weekly Imaging

1. At 4-6 weeks after Cycle 2, Day 1, a second PET/CT scan with the diagnostic IMP (rhPSMA-7.3 (^{18}F)) will be performed. This assessment should be performed within the 14 days prior to the next therapeutic IMP administration. Where subjects are receiving a 3rd cycle of therapeutic IMP, it should not be performed in the days after therapeutic IMP administration.
2. Every 12 weeks (i.e. every other cycle of therapeutic IMP dosing) following the administration of the first therapeutic IMP cycle, a ceCT or MRI of the chest/abdomen and pelvis and full body ^{99m}Tc bone scan will be performed. Whilst subjects are receiving therapeutic IMP cycles, the cross-sectional imaging should be performed ±5 days of the next cycle of therapeutic IMP that is due. However, for the bone scan component, this must take place within 2-5 days prior to therapeutic IMP administration to avoid interference with dosimetry measurements (i.e. at 12 weeks -5 days and 24 weeks -5 days).

Please note, the PSMA PET/CT scan should not be performed on the same day as the ^{99m}Tc bone scan to avoid potential interference of the 2 radionuclides on image quality. If this is unavoidable, then the 2 administrations should be separated by as much time as is practicable and the Sponsor informed.

10.2 Phase 2 (Cohort 2A and Cohort 2C [if Opened])

10.2.1 Visit 1 (Screening; Day -42 [Within 42 Days Before IMP Administration])

1. Review the study with the subject and obtain written informed consent and HIPAA authorisation (or equivalent, as required).
2. Confirm eligibility criteria.
3. Record demographics data.
4. Record ongoing and past medical history, including confirmation of a history of adenocarcinoma of the prostate.

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5. Record concomitant medications.
6. Record ECOG performance status.
7. Perform a physical examination, and measure height and body weight.
8. Perform and record vital signs and single 12-lead ECG.
9. Collect urine sample for dipstick urinalysis to exclude urinary tract infection.
10. Collect blood samples for screening haematology, coagulation, biochemistry, and PSA laboratory testing.

    Note: If the screening period exceeds 42 days, screening blood tests should be repeated prior to Cycle 1 Day 1.
11. Perform ceCT or MRI of chest/abdomen and pelvis and full body ^{99m}Tc bone scan to establish baseline imaging. This scan will be read locally.
12. Perform a PSMA PET/CT scan for imaging screening/eligibility assessment. Note: body weight should be measured before this assessment.

    Note for imaging scans: In order to reduce unnecessary interventions and radiation exposure, where PSMA PET/CT imaging or cross-sectional imaging has been recently completed as part of standard of care and the images are available for submission to the central reader, these images may be used to determine eligibility, provided that the scan was acquired no more than 28 days prior to informed consent.
13. Send ceCT or MRI of chest/abdomen and pelvis and full body ^{99m}Tc bone scan, and PSMA PET/CT imaging scans to the Central Imaging vendor.
14. Confirm subject eligibility for study.
15. Commence AE monitoring.
16. Randomise subject in the IRT system.
17. Order IMP and schedule subject for Day 1 visit within 42 days of consent.
18. Book any visit logistics, as required.
19. Give subject contact card.

10.2.2 Day 1 of Treatment Cycle 1 (IMP Administration)

1. Record any changes in concomitant medications since screening.
2. Ask about and record any AEs occurring since screening.
3. Ask subject to complete PRO questionnaire prior to all other procedures (subject should complete FACT-P first, followed by FACT-RNT).
4. Record ECOG performance status pre-IMP administration.
5. Perform and record physical examination and measure body weight pre-IMP administration.
6. Perform and record single 12-lead ECG pre-IMP administration.
7. Collect urine sample for UPCR test pre-IMP administration.
8. Collect blood sample for haematology, coagulation, biochemistry, and PSA laboratory testing pre-IMP administration.

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Note: Blood samples for safety laboratory assessments may be collected within 72 hours of Day 1 and must be reviewed prior to IMP administration. If collected within 72 hours of Day 1, then the Day 1 blood collection for safety assessments may be waived, unless clinically indicated.

9. Perform and record vital signs pre-IMP administration.
10. Confirm subject eligibility for study.
11. Administer IMP (IV Lutetium (^{177}Lu) rhPSMA-10.1 Injection).
12. Perform and record vital signs post-IMP administration.
13. Perform SPECT/CT imaging for dosimetry at 3 (±1) hours post-IMP administration.
14. Perform single 12-lead ECG within 5 hours post-IMP administration.
15. Record any injection site reactions and any other AEs that occur during the visit.

10.2.3 Day 1 of Treatment (Cycle 2 and Subsequent Cycles of IMP Administration)

1. Before any subsequent treatment cycles, review available dosimetry data and clinical assessments to confirm subject eligibility for further treatment cycles.
2. Record any changes in concomitant medications since last visit.
3. Ask about and record any AEs occurring since last visit.
4. Ask subject to complete PRO questionnaires prior to all other procedures (subject should complete FACT-P first, followed by FACT-RNT).
5. Record ECOG performance status pre-IMP administration.
6. Perform and record physical examination and measure body weight pre-IMP administration.
7. Perform and record single 12-lead ECG pre-IMP administration.
8. Collect urine sample for UPCR test pre-IMP administration.
9. Collect blood samples pre-IMP administration for haematology, coagulation, biochemistry, and PSA laboratory testing.

   Note: Blood samples for safety laboratory assessments may be collected within 72 hours of Day 1 and must be reviewed prior to IMP administration. If collected within 72 hours of Day 1, then the Day 1 blood collection for safety assessments may be waived, unless clinically indicated.
10. Perform and record vital signs pre-IMP administration.
11. Confirm subject eligibility for study.
12. Administer IMP (IV Lutetium (^{177}Lu) rhPSMA-10.1 Injection).
13. Perform and record vital signs post-IMP administration.
14. Record any injection site reactions and any other AEs that occur during the visit.

10.2.4 24 Hours to 168 Hours Post-IMP Dose

10.2.4.1 Treatment Cycle 1 Only

Evaluations during Cycle 1 of therapy are summarised in the Study Schedule (Appendix 1).

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Subjects may be hospitalised at the discretion of the investigator and in compliance with local rules and regulations in that jurisdiction. For convenience, subjects may be offered transport to and from the clinic to their home, and/or overnight stay in suitable accommodation. Subjects may also be outpatients and return home at the end of the day to return the next day for additional interventions/monitoring.

1. Record any changes in concomitant medications since last visit.
2. Ask about and record any AEs occurring since last visit.
3. Perform SPECT/CT imaging for dosimetry at 24 (±4) hours, 48 (±4) hours, and 168 (±24) hours post-IMP administration.

10.2.4.2 Treatment Cycle 2 and Subsequent Treatment Cycles

Evaluations during subsequent treatment cycles of therapy are summarised in the Study Schedule (Appendix 1).

Subjects may be hospitalised at the discretion of the investigator and in compliance with local rules and regulations in that jurisdiction. For convenience, subjects may be offered transport to and from the clinic to their home, and/or overnight stay in suitable accommodation. Subjects may also be outpatients and return home at the end of the day to return the next day for additional interventions/monitoring.

1. Record any changes in concomitant medications since last visit.
2. Ask about and record any AEs occurring since last visit.
3. Perform SPECT/CT imaging for dosimetry at 48 (±4) hours post-IMP administration.

10.2.5 Six-weekly Treatment Interval Period

Evaluations during the 6-week treatment cycles are summarised in the Study Schedule (Appendix 1). The subject will be asked to present at the study site/centre every 2 weeks from Week 2 (Day 15). The list of procedures at the 2-weekly visits are as follows:

Week 2 (Day 15), Week 4 (Day 29)

1. Record any changes in concomitant medications since the last visit.
2. Ask about and record any AEs occurring since the last visit.

   Note: Subjects will be asked to report AEs as they occur to the site team but will also be prompted to report at the 2-weekly visits.
3. Collect blood samples for haematology, coagulation, and biochemistry laboratory testing. Ad-hoc blood testing may be performed if required, depending on clinical need as directed by the investigator (e.g. in the event of myelosuppression requiring more frequent monitoring than 2-weekly).

Week 6 (Day 43)

1. Record any changes in concomitant medications since the last visit.
2. Ask about and record any AEs occurring since the last visit.

   Note: Subjects will be asked to report AEs as they occur to the site team but will also be prompted to report at the 2-weekly visits.

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3. Collect blood samples for haematology, coagulation, biochemistry, and PSA laboratory testing. Blood sample may also be collected pre-IMP administration on Day 1 of the next treatment cycle.

   Ad-hoc blood testing may be performed if required, depending on clinical need as directed by the investigator (e.g. in the event of myelosuppression requiring more frequent monitoring than weekly).

4. Follow-up on the results of the 12-weekly ceCT or MRI, and ^{99m}Tc bone scanning, if applicable.

10.2.6 Twelve-weekly Imaging

Every 12 weeks (i.e. every other cycle of IMP dosing) following the administration of the first IMP cycle, a ceCT or MRI of the chest/abdomen and pelvis, and full body ^{99m}Tc bone scan will be performed until centrally-determined radiographic progression, death, or loss to follow-up, whichever occurs first. Whilst subjects are receiving IMP cycles, the cross-sectional imaging should be performed ±5 days of the next cycle of IMP that is due. However, for the bone scan component, this must take place within 5 days prior to IMP administration to avoid interference with dosimetry measurements.

10.3 Phase 2 (Cohort 2B)

10.3.1 Visit 1 (Screening; Day -42 [Within 42 Days Before IMP Administration])

1. Review the study with the subject and obtain written informed consent and HIPAA authorisation (or equivalent, as required).
2. Confirm eligibility criteria.
3. Record demographics data.
4. Record ongoing and past medical history, including confirmation of a history of adenocarcinoma of the prostate.
5. Record concomitant medications.
6. Record ECOG performance status.
7. Perform a physical examination, and measure height and body weight.
8. Perform and record vital signs and single 12-lead ECG.
9. Collect urine sample for dipstick urinalysis to exclude urinary tract infection.
10. Collect blood samples for screening haematology, coagulation, biochemistry, and PSA laboratory testing.

    Note: If the screening period exceeds 42 days, screening blood tests should be repeated prior to Cycle 1 Day 1.

11. Perform ceCT or MRI of chest/abdomen and pelvis and full body ^{99m}Tc bone scan to establish baseline imaging. Note: this scan will be read locally.
12. Perform a PSMA PET/CT scan for imaging screening/eligibility assessment. Note: body weight should be measured before this assessment.

    Note for imaging scans: In order to reduce unnecessary interventions and radiation exposure, where PSMA PET/CT imaging or cross-sectional imaging has been recently completed as part of standard of care and the images are available for submission to the

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central reader, these images may be used to determine eligibility, provided that the scan was acquired no more than 28 days prior to informed consent.

13. Send ceCT or MRI of chest/abdomen and pelvis and full body ^{99m}Tc bone scan, and PSMA PET/CT imaging scans to the Central Imaging vendor.
14. Confirm subject eligibility for study.
15. Commence AE monitoring.
16. Randomise subject in the IRT system.
17. Order the IMP and schedule subject for Day 1 visit within 42 days of consent.
18. Book any visit logistics, as required.
19. Give subject contact card.

10.3.2 Cycles 1 – 2

10.3.2.1 Day 1 of Treatment Cycle 1 (IMP Administration)

1. Record any changes in concomitant medications since screening.
2. Ask about and record any AEs occurring since screening.
3. Ask subject to complete PRO questionnaires pre-IMP administration (subject should complete FACT-P first, followed by FACT-RNT).
4. Record ECOG performance status pre-IMP administration.
5. Perform and record physical examination and measure body weight pre-IMP administration.
6. Perform and record single 12-lead ECG pre-IMP administration.
7. Collect urine sample for UPCR test pre-IMP administration.
8. Collect blood sample pre-IMP administration for haematology, coagulation, biochemistry, and PSA laboratory testing.

   Note: Blood samples for safety laboratory assessments may be collected within 72 hours of Day 1 and must be reviewed prior to IMP administration. If collected within 72 hours of Day 1, then the Day 1 blood collection for safety assessments may be waived, unless clinically indicated.
9. Perform and record vital signs pre-IMP administration.
10. Confirm subject eligibility for study.
11. Administer IMP (IV Lutetium (^{177}Lu) rhPSMA-10.1 Injection).
12. Perform and record vital signs post-IMP administration.
13. Perform SPECT/CT imaging for dosimetry at 3 hours post-IMP administration.
14. Perform single 12-lead ECG within 5 hours post-IMP administration.
15. Record any injection site reactions and any other AEs that occur during the visit.

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10.3.2.2 Day 1 of Treatment (Cycle 2 and Subsequent Cycles of IMP Administration)

1. Before any subsequent treatment cycles, review dosimetry data and clinical assessments to confirm subject eligibility for further treatment cycles.
2. Record any changes in concomitant medications since last visit.
3. Ask about and record any AEs occurring since last visit.
4. Ask subject to complete PRO questionnaires pre-IMP administration (subject should complete FACT-P first, followed by FACT-RNT).
5. Record ECOG performance status pre-IMP administration.
6. Perform and record physical examination and measure body weight pre-IMP administration.
7. Perform and record single 12-lead ECG pre-IMP administration.
8. Collect urine sample for UPCR test pre-IMP administration.
9. Collect blood samples pre-IMP administration for haematology, coagulation, biochemistry, and PSA laboratory testing.

   Note: Blood samples for safety laboratory assessments may be collected within 72 hours of Day 1 and must be reviewed prior to IMP administration. If collected within 72 hours of Day 1, then the Day 1 blood collection for safety assessments may be waived, unless clinically indicated.
10. Perform and record vital signs pre-IMP administration.
11. Confirm subject eligibility for study.
12. Administer IMP (IV Lutetium (^{177}Lu) rhPSMA-10.1 Injection).
13. Perform and record vital signs post-IMP administration.
14. Record any injection site reactions and any other AEs that occur during the visit.

10.3.2.3 24 Hours to 168 Hours of Each Treatment Cycle (Post-IMP Dose of Each Treatment Cycle)

10.3.2.3.1 Treatment Cycle 1 Only

Evaluations during Cycle 1 of therapy are summarised in the Study Schedule (Appendix 1).

Subjects may be hospitalised at the discretion of the investigator and in compliance with local rules and regulations in that jurisdiction. For convenience, subjects may be offered transport to and from the clinic to their home, and/or overnight stay in suitable accommodation. Subjects may also be outpatients and return home at the end of the day to return the next day for additional interventions/monitoring.

1. Record any changes in concomitant medications since last visit.
2. Ask about and record any AEs occurring since last visit.
3. Perform SPECT/CT imaging for dosimetry at 24 (±4) hours, 48 (±4) hours, and 168 (±24) hours post-IMP administration.

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10.3.2.3.2 Treatment Cycles 1 and 2

Evaluations during treatment Cycles 1 and 2 of therapy are summarised in the Study Schedule (Appendix 1).

Subjects may be hospitalised at the discretion of the investigator and in compliance with local rules and regulations in that jurisdiction. For convenience, subjects may be offered transport to and from the clinic to their home, and/or overnight stay in suitable accommodation. Subjects may also be outpatients and return home at the end of the day to return the next day for additional interventions/monitoring.

1. Record any changes in concomitant medications since last visit.
2. Ask about and record any AEs occurring since last visit.
3. Perform SPECT/CT imaging for dosimetry at 48 (±4) hours post-IMP administration.

10.3.2.4 Three-weekly Treatment Interval Period

Evaluations during the 3-week treatment cycles are summarised in the Study Schedule (Appendix 1). The subject will be asked to present at the study site/centre on Day 15 (Week 2) and Day 22 (Week 3) of each 3-week treatment cycle. After the third dose has been administered, the subjects will enter the 6-week treatment cycles. The list of procedures at the weekly visits are as follows:

Week 2 (Day 15) and Week 3 (Day 22)

1. Record any changes in concomitant medications at weekly intervals.
2. Ask about and record any AEs occurring at weekly intervals.

   Note: Subjects will be asked to report AEs as they occur to the site team but will also be prompted to report on a weekly basis.
3. Collect blood samples for haematology, coagulation, and biochemistry testing. Blood samples will also be collected for PSA laboratory testing every 6 weeks. Ad-hoc blood testing may be performed if required, depending on clinical need as directed by the investigator (e.g. in the event of myelosuppression requiring more frequent monitoring than weekly).

   Note: blood samples will only be collected on Day 22 of the first 2 treatment cycles only. The subject will enter the 6-week treatment cycle after the third dose is administered.

10.3.3 Cycles 3 – 8

10.3.3.1 Day 1 of Treatment Cycles 3 – 8 (IMP Administration)

1. Record any changes in concomitant medications since the last visit.
2. Ask about and record any AEs occurring since screening.
3. Ask subject to complete PRO questionnaire prior to all other procedures (subject should complete FACT-P first, followed by FACT-RNT).
4. Record ECOG performance status pre-IMP administration.
5. Perform and record physical examination and measure body weight pre-IMP administration.
6. Perform and record single 12-lead ECG pre-IMP administration.

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7. Collect urine sample for UPCR test pre-IMP administration.
8. Collect blood sample pre-IMP administration for haematology, coagulation, biochemistry, and PSA laboratory testing.

   Note: Blood samples for safety laboratory assessments may be collected within 72 hours of Day 1 and must be reviewed prior to IMP administration. If collected within 72 hours of Day 1, then the Day 1 blood collection for safety assessments may be waived, unless clinically indicated.
9. Perform and record vital signs pre-IMP administration.
10. Confirm subject eligibility to continue study.
11. Administer IMP (IV Lutetium (^{177}Lu) rhPSMA-10.1 Injection).
12. Perform and record vital signs pre-IMP administration.
13. Record any injection site reactions and any other AEs that occur during the visit.

10.3.3.2 48 Hours Post-IMP Dose

Evaluations during Cycles 3 – 8 of therapy are summarised in the Study Schedule (Appendix 1).

Subjects may be hospitalised at the discretion of the investigator and in compliance with local rules and regulations in that jurisdiction. For convenience, subjects may be offered transport to and from the clinic to their home, and/or overnight stay in suitable accommodation. Subjects may also be outpatients and return home at the end of the day to return the next day for additional interventions/monitoring.

1. Record any changes in concomitant medications since last visit.
2. Ask about and record any AEs occurring since last visit.
3. Perform SPECT/CT imaging for dosimetry at 48 (±4) hours post-IMP administration.

10.3.3.3 Six-weekly Treatment Interval Period

Evaluations during the 6-week treatment cycles are summarised in the Study Schedule (Appendix 1). The subject will be asked to present at the study site/centre every 2 weeks from Week 2 (Day 15). The list of procedures at the 2-weekly visits are as follows:

Week 2 (Day 15), Week 4 (Day 29)

1. Record any changes in concomitant medications since the last visit.
2. Ask about and record any AEs occurring since the last visit.

   Note: Subjects will be asked to report AEs as they occur to the site team but will also be prompted to report at the 2-weekly visits.
3. Collect blood samples for haematology, coagulation, and biochemistry laboratory testing. Ad-hoc blood testing may be performed if required, depending on clinical need as directed by the investigator (e.g. in the event of myelosuppression requiring more frequent monitoring than 2-weekly).

Week 6 (Day 43)

1. Record any changes in concomitant medications since the last visit.

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2. Ask about and record any AEs occurring since the last visit.

   Note: Subjects will be asked to report AEs as they occur to the site team but will also be prompted to report at the 2-weekly visits.

3. Collect blood samples for haematology, coagulation, biochemistry, and PSA laboratory testing. Blood sample may also be collected pre-IMP administration on Day 1 of the next treatment cycle.

   Ad-hoc blood testing may be performed if required, depending on clinical need as directed by the investigator (e.g. in the event of myelosuppression requiring more frequent monitoring than weekly).

4. Follow-up on the results of the 12-weekly ceCT or MRI, and ^{99m}Tc bone scanning, if applicable.

10.3.4 Twelve-weekly Imaging

Every 12 weeks (i.e. every other cycle of IMP dosing) following the administration of the first IMP cycle, a ceCT or MRI of the chest/abdomen and pelvis and full body ^{99m}Tc bone scan will be performed until centrally-determined radiographic progression, death, or loss to follow-up, whichever occurs first. Whilst subjects are receiving IMP cycles, the cross-sectional imaging should be performed ±5 days of the next cycle of IMP that is due. However, for the bone scan component, this must take place within 5 days prior to IMP administration to avoid interference with dosimetry measurements.

10.4 End of Treatment Visit

For subjects who complete either phase of the study, the EoT visit will be the last day of the final 6-week cycle (±48 hours) as per cohort described in this protocol. Vital signs, blood (haematology, coagulation, biochemistry, and thyroid function panel [for Phase 1 only]), AEs, concomitant medication, ECOG performance status, physical examination, body weight, 12-lead ECG, urinalysis records, and PRO assessments should be updated at the EoT visit. Subjects who have not experienced centrally-determined radiographic progression will be advised to return to the study site to monitor PSA levels (monthly for Phase 1 subjects, and 12-weekly for Phase 2 subjects) and for 12-weekly ceCT or MRI of the chest/abdomen and pelvis and full body ^{99m}Tc bone scans until evidence of progression. Subjects will be advised regarding long-term follow-up at this visit. Where subjects have ongoing toxicity at the EoT visit (e.g. evidence of myelosuppression), additional visits should be scheduled to perform necessary investigations (e.g. blood draws for laboratory testing) until the toxicity has resolved or stabilised in the judgement of the investigator.

For subjects who withdraw prematurely for any reason, the EoT visit should be completed within 10 days of withdrawal. Lost to follow-up should be avoided if possible, and the site should be able to demonstrate that they have tried to contact the subject to have them return for regular follow-up. Once a subject is recorded as lost to follow-up, it is assumed that no further visits/data are possible for this subject.

10.5 Follow-up Period

10.5.1 Efficacy Follow-up

In the follow-up period after IMP treatment, PSA will be measured monthly (Phase 1) or 12-weekly (Phase 2) until centrally-determined radiographic progression, death, or loss to follow-up, whichever occurs first. Contrast enhanced CT/MRI and full body ^{99m}Tc bone scans

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will be performed every 12 weeks (±5 days from Cycle 1 Day 1) for all subjects until centrally-determined radiographic progression, death, or loss to follow-up, whichever occurs first.

If a subject initiates treatment with another anti-cancer therapy during efficacy follow-up, efficacy follow-up will be terminated and the subject will continue long-term passive safety follow-up (see Section 10.5.2).

10.5.2 Safety Follow-up

Whilst subjects are in efficacy follow-up (i.e. undergoing PSA monitoring and imaging assessments as described in Section 10.5.1), the investigator should continue recording all AEs, irrespective of causality, in the eCRF.

Once a subject experiences radiographic progression and/or initiates treatment with another anti-cancer therapy, they will continue to be monitored for long-term safety in a passive follow-up. The investigator site will review subject notes at 6-monthly intervals until death. Subjects will not be asked to attend any formal study visits during this period, but any standard of care visit source data will be reviewed. The following AEs, irrespective of causality, will be collected during passive follow-up:

- Change in renal function/evidence of renal impairment
- Presence of new haematological disorders
- Development of new malignancy
- New cardiac events.

Once in the passive long term safety follow-up, AEs which are not in any of these categories of special interest will only be reported if they are serious and related. Any new prostate cancer treatments which the site becomes aware of should be entered into the eCRF. If a subject dies, the date, and cause of death will be collected.

Note: Where an AE is experienced that may be related to the IMP e.g. new renal impairment, the circumstances surrounding the presentation should be considered and where possible, the subject's treating physician will be asked to assess the likelihood of it being related to the IMP. As an example, disease progression leading to a new urinary tract obstruction causing acute kidney injury is unlikely to be related to the IMP.

11 IMAGING AND CALIBRATION PROTOCOLS

Full details on the diagnostic imaging protocols, including PET/CT scanner calibration and QC, and the rhPSMA-7.3 (^{18}F) PET/CT acquisition (applicable to Phase 1 only) are in the Image Acquisition Guidelines.

Full details on the calibration protocols for the SPECT/CT scanner, well/gamma counter, and dose calibrator are in the Study Dosimetry Calibration Manual.

Full details on the dosimetry imaging protocols, including SPECT/CT acquisition and SPECT/CT scanner QC, blood sample collection and assay procedures (applicable to Phase 1 only) are in the Study Dosimetry Manual.

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11.1 Prostate-specific Membrane Antigen Diagnostic Imaging

11.1.1 Phase 1 (Completed)

Prostate-specific membrane antigen PET/CT scans will be acquired to assess eligibility, and a second PSMA PET/CT scan will be completed for evaluation of PSMA-defined disease response. A dedicated hybrid PET scanner (i.e. PET/CT) is mandatory. The selected PET scanner must be calibrated and qualified by a qualified vendor. For each subject, the same qualified scanner should be use for the pre-therapeutic and post-therapeutic PSMA PET/CT imaging.

11.1.1.1 rhPSMA-7.3 (^{18}F) Acquisition

Details of diagnostic IMP administration are presented in Section 8.5.2.2, and briefly described here. A 296 MBq (8 mCi) ±20% activity of -rhPSMA-7.3 (^{18}F) will be administered via a venous cannula with arms down, followed by a 10 mL flush with normal saline solution.

The time from the end of injection of the diagnostic IMP to the start of imaging will be between 50 to 70 minutes. Subjects will be imaged for approximately 20 minutes. All local and central readers will undergo training in interpretation of rhPSMA-7.3 (^{18}F) PET/CT and will have a training set available for reference.

11.1.2 Phase 2

Prostate-specific membrane antigen PET/CT scans will be acquired at screening for imaging eligibility assessment as described in inclusion criterion #7 and exclusion criterion #2 in Phase 2 of the study. The locally approved standard of care PSMA PET/CT tracer may be used for the diagnostic PSMA PET/CT screening and local guidelines/standard of care for image acquisition and reconstruction should be followed. Alternatively, an existing positive PSMA PET/CT scan obtained as part of the subject's standard of care may be used to determine eligibility, provided that the scan was obtained within 4 weeks (28 days) prior to informed consent and is available for use.

All PSMA PET/CT images must be available for submission to the Central Imaging vendor, who will determine subject eligibility for the study. Further details on the central reads, including reader training, are provided in the Image Review Charter.

11.2 Contrast Enhanced CT/MRI, and ^{99m}Tc Bone Scan

Contrast enhanced CT/MRI and full body ^{99m}Tc bone scans will be acquired as per local guidelines/standard of care.

11.3 Single-photon Emission Computed Tomography/Computed Tomography Imaging

Quantitative SPECT/CT scans will be acquired for dosimetry assessment. A dedicated hybrid SPECT/CT scanner is mandatory. The SPECT/CT scanner must be qualified by the Central Imaging vendor. Study sites will receive a Lutetium calibration vial from the Sponsor and will follow instructions in the Study Dosimetry Calibration Manual. For each subject, the same qualified scanner should be used for each SPECT/CT acquisition throughout the study.

The SPECT/CT scan data will be assessed by the Central Imaging vendor for evaluability. All scan data will undergo QC based on parameters detailed in the Study Dosimetry Manual.

Note that as the SPECT/CT acquisition uses the radiation emitted from the IMP, no additional radiopharmaceutical administration is required.

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11.4 Ancillary Equipment Calibration

11.4.1 Dose Calibrator Calibration

Any dose calibrator used to measure the IMP must be qualified by the Central Imaging vendor. Sites will receive a Lutetium calibration vial from the Sponsor and will follow instructions in the Study Dosimetry Calibration Manual.

11.4.2 Well/Gamma Counter Calibration

In Phase 1, the gamma counter used to measure blood radioactivity must be qualified by the Central Imaging vendor. Sites will receive a Lutetium calibration vial from the Sponsor and will follow instructions in the Study Dosimetry Calibration Manual.

No well/gamma counter will be required for Phase 2.

11.5 Image Transfer and Queries

Following the completion of each test scan and diagnostic scan, the scan data (i.e. Digital Imaging and Communications in Medicine [DICOM] files) and any ancillary data will be sent to the Central Imaging vendor using the Central Imaging vendor software. Sites will receive online training on the Central Imaging vendor software transfer and query response process.

The test and diagnostic scan image data must be submitted to the Central Imaging vendor promptly. Queries generated by the vendor must be addressed promptly by the site. Additional information regarding image data submission and query responses are provided in the Image Acquisition Guidelines.

12 ADVERSE EVENT REPORTING AND DOCUMENTATION

12.1 Adverse Events

An AE is any untoward medical occurrence in a clinical investigation of a subject administered a pharmaceutical product and that does not necessarily have a causal relationship with this treatment. An AE is therefore any unfavourable and unintended sign (including an abnormal laboratory finding), symptom or disease temporally associated with the administration of an IMP, whether or not related to that IMP. An unexpected AE is one of which the nature, severity, or outcome is not expected when assessed against the current RSI (Lutetium (^{177}Lu) rhPSMA-10.1 Injection IB for the therapeutic IMP and rhPSMA-7.3 (^{18}F) IB for the diagnostic IMP [applicable to Phase 1 only]).

The investigator will collect AEs through non-leading questions and examination of the subject during each visit and for 48 hours following administration of the diagnostic IMP (applicable to Phase 1 only) and IMP and record the information in the site's source documents. For each AE, the following information should be recorded: description of signs/symptoms, duration (start and stop dates and times), seriousness, severity/grade, outcome, action(s) taken, and relationship to IMP, or if unrelated, the cause.

12.1.1 Severity of Adverse Events

The National Cancer Institute's CTCAE Version 5 (2017) will be used to assess and grade event severity, including laboratory abnormalities judged to be clinically significant. The severity grading is provided below (Table 13).

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Table 13: Common Terminology Criteria for Adverse Events (Version 5) Adverse Event Severity Grading

Severity (Toxicity Grade)	Description
Grade 1	Mild; asymptomatic or mild symptoms; clinical or diagnostic observations only; intervention not indicated.
Grade 2	Moderate; minimal, local or non-invasive intervention indicated; limiting age-appropriate instrumental ADL.[a]
Grade 3	Severe or medically significant but not immediately life-threatening; hospitalisation or prolongation of hospitalisation indicated; disabling; limiting self-care ADL.[b]
Grade 4	Life-threatening consequences; urgent intervention indicated.
Grade 5	Death related to AE.

Abbreviations: ADL=activities of daily living; AE=adverse event.

[a] Instrumental ADL refer to preparing meals, shopping for groceries or clothes, using the telephone, managing money, etc.

[b] Self-care ADL refers to bathing, dressing, and undressing, feeding self, using the toilet, taking medications, and not bedridden.

If the experience is not covered in the CTCAE criteria, the guidelines shown in Table 14 below will be used to grade severity. Note that the term "severe" is a measure of intensity and that a severe event is not necessarily serious.

Table 14: Adverse Event Severity Grading

Severity (Toxicity Grade)	Description
Mild (1)	Transient or mild discomfort; no limitation in activity; no medical intervention or therapy required. The subject may be aware of the sign or symptom but tolerates it reasonably well.
Moderate (2)	Mild to moderate limitation in activity, no or minimal medical intervention/therapy required.
Severe (3)	Marked limitation in activity, medical intervention/therapy required, hospitalisations possible.
Life-threatening (4)	The subject is at risk of death due to the adverse experience as it occurred. This does not refer to an experience that hypothetically might have caused death if it were more severe.
Death (5)	Death related to AE.

Abbreviation: AE=adverse event.

12.1.2 Relationship of Adverse Events to Investigational Medicinal Product

The relationship of an AE to the IMP will be assessed based on the question of whether there is a "reasonable causal relationship" to the specific IMP. Possible answers are "yes" or "no".

An assessment of "no" would include:

1. The existence of a highly likely alternative explanation, or
2. Non-plausibility (e.g. cancer detected a few days after the first administration of the IMP), or
3. Lack of a reasonable temporal sequence from administration of the IMP.

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An assessment of "yes" would indicate that the AE is reasonably associated with the use of the IMP. Important factors to be considered in assessing the relationship of the AE to the IMP include:

- The temporal sequence from IMP administration: The event follows a reasonable temporal sequence from administration of the IMP. The length of time from contrast exposure to event should be evaluated in the clinical context of the event.
- Underlying concomitant diseases or conditions: Each event should be evaluated in the context of the natural history and course of the disease being treated and any other disease the subject may have.
- Concomitant medication or treatment: The other drugs that the subject is taking or the treatment that the subject receives will be examined to determine whether any of them might have caused the event in question.
- Known AEs associated with an IMP. The pharmacology and PK (including absorption, distribution, metabolism, and excretion) of the IMP, coupled with the individual subject's conditions, should be considered.
- There is evidence of partial or complete disappearance of the event after withdrawal of the product (positive dechallenge).
- If the report of the event contains conflicting data and/or dubious or insufficient/poor evidence, the AE will, by default, be considered to be related.

12.2 Serious Adverse Events

An SAE is any untoward medical occurrence that at any dose:

- results in death,
- is life-threatening (i.e. the subject was at risk of death at the time of the event/reaction; it does not refer to an event/reaction which hypothetically might have caused death if it were more severe),
- requires inpatient hospitalisation or prolongation of existing hospitalisation,
- results in persistent or significant disability/incapacity (where disability is defined as a permanent or substantial disruption of ability to carry out normal life functions, either reported or defined as per clinical judgement),
- is a congenital anomaly /birth defect,
- is an important medical event that may not result in death, be life-threatening, or require hospitalisation but may be considered serious when, based upon appropriate medical judgement, it may jeopardise the subject and may require medical or surgical intervention to prevent any of the outcomes listed in the definition above. Examples of such events include allergic bronchospasm requiring intensive treatment in an emergency room or at home, and blood dyscrasias or convulsions that do not result in hospitalisation, or development of drug dependency or drug abuse.

Any suspected transmission via a medicinal product of an infectious agent is also considered an SAE.

A non-serious AE is any AE that does not meet the criteria listed above for a SAE.

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12.2.1 Serious Adverse Event Reporting

Study sites will document all SAEs that occur (whether or not related to IMP) on an SAE Report Form via the eCRF system. The collection period for all SAEs will begin after informed consent is obtained and end once the subject completes the (active) follow-up (defined as centrally-determined radiographic progression, loss to follow-up, or death, whichever occurs first) and enters into the passive long term safety follow-up. Once subjects have entered the long-term follow-up, only SAEs which are assessed as related to the IMP will be reported. However, a small number of AEs of special interest will be assessed at 6-monthly medical chart reviews and these will be reported into the relevant section of the eCRF irrespective of causality.

All SAEs must be reported immediately (as soon as possible after the investigator becomes aware of the event but no later than 24 hours) by completing the relevant part of the eCRF. All sites will be trained how to complete this. Completion of the SAE page within the eCRF will send automatic notification to the Sponsor's Pharmacovigilance department who will act upon receipt.

In case of death, which is suspected to be related to the IMP, if an autopsy is performed, a copy of the autopsy report should be actively sought by the site investigator and sent to the Sponsor as soon as available. Additional and further requested information (follow-up or corrections to the original event) will be completed within the eCRF/electronic data capture (EDC) system.

12.2.2 Suspected Unexpected Serious Adverse Reactions

A suspected unexpected serious adverse reaction (SUSAR) is an adverse reaction, the nature and severity of which is not consistent with the RSI for the IMP.

SUSARs will be collected and reported for the IMP based on the RSI set out in the Lutetium (^{177}Lu) rhPSMA-10.1 Injection IB. The Sponsor will be responsible for reporting all SUSARs for the IMP to the relevant authorities, Institutional Review Board (IRB)/Independent Ethics Committee (IEC), and other parties, as applicable.

Investigators will be informed of all SUSARs for Lutetium (^{177}Lu) rhPSMA-10.1 Injection for all studies sponsored by Blue Earth Therapeutics Limited, whether or not the event occurred in the current study.

12.3 Urgent Safety Measures

An urgent safety measure is an action that the Sponsor and/or investigator may take in order to protect study subjects against an immediate hazard to their health or safety (e.g. it is identified that there is a significant higher incidence of death at 1 investigator site and as a result recruitment is suspended at that site as an urgent safety measure).

An urgent safety measure taken by the investigator must be immediately notified to the Sponsor. For all urgent safety measures, the IRB/IEC and Regulatory Agency/Competent Authority will be notified in accordance with ICH GCP and applicable regulations.

Initial notification may be followed by submission of a protocol amendment (see Section 18.2), notification of early termination of the study within 15 days (if applicable) and notification of AEs and serious adverse reactions (if applicable).

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12.4 Pregnancy Safety Monitoring

If the female partner of a treated male subject becomes pregnant, the male subject should notify the investigator as soon as possible after learning of the pregnancy. The investigator must make every effort to ensure that the pregnant female is aware of the need to notify her healthcare provider regarding her male partner's participation in this clinical study, and his exposure to the IMP.

The investigator should notify the study assigned Medical Monitor within 24 hours of knowledge of the pregnancy (see Section 12.5).

12.5 Medical Monitoring

The study assigned Medical Monitor (Clinical Research Organisation [CRO]) should be contacted directly using the following contact information to report medical concerns or questions regarding safety.

Phone: +49 151 4024 1973

E-mail: BETPSMA121.Medical@psi-cro.com

13 DISCONTINUATION AND REPLACEMENT OF SUBJECTS

13.1 Discontinuation of Therapeutic Investigational Medicinal Product

Discontinuation from IMP treatment does not mean discontinuation from the study, and remaining study procedures should be completed as indicated by the study protocol. In the event that a subject has a complete biochemical response (i.e. undetectable PSA) and has no visible disease on imaging, the investigator may wish to discontinue dosing with the IMP when the dose limits have been exceeded.

Subjects who discontinue IMP treatment for reasons other than disease progression but remain in the study will enter the follow-up period for efficacy and safety assessments as outlined in Section 10.5. If a subject who has discontinued from IMP treatment initiates treatment with another anti-cancer therapy, efficacy follow-up will be terminated and long-term passive safety follow-up will continue as described in Section 10.5.2.

The data to be collected at the time of IMP discontinuation will include AE assessment and concomitant medication at the time of discontinuation.

13.2 Discontinuation/Withdrawal from the Study

Subject participation in the study is entirely voluntary and subjects may decline to participate or withdraw from the study, at any time, without prejudice to their future care.

Although subjects are not obliged to give a reason(s) for premature withdrawal, the investigator should make a reasonable effort to ascertain the reason(s), whilst fully respecting the subject's rights.

All withdrawals and dropouts of enrolled subjects from the study should be reported and explained on the eCRF. Rules for replacement of subjects that self-withdraw or who are withdrawn from study treatment are presented in Section 13.3.

For subjects who withdraw prematurely for any reason or are deemed lost to follow-up, the EoT eCRF page should be completed within 10 days. All subjects who self-withdraw will have

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no further data entered into the clinical database. Wherever possible, subjects who are withdrawn by the investigator will remain in the passive long term safety follow-up, providing they continue to consent.

An investigator may discontinue or withdraw a subject for any of the following reasons:

- Significant non-compliance/protocol violation requiring discontinuation from the study.
- If any AE, laboratory abnormality, or other medical condition or situation occurs such that continued participation in the study would not be in the best interests of the subject. This may include locally-determined clinical progression of disease irrespective of centrally-determined radiographic progression, which means the investigator judges that continued participation is no longer appropriate.
- If the subject meets any of the exclusion criteria (either newly developed or not previously recognised) that precludes further study participation.
- Lost to follow-up (see Section 13.4).
- Sponsor request for early termination of study.

The reason for subject discontinuation or withdrawal from the study will be recorded in the subject's source documents and on the eCRF.

13.3 Replacement of Subjects

Subjects who sign the ICF but do not receive the therapeutic IMP will be replaced. Subjects who sign the ICF and who receive the therapeutic IMP, but subsequently withdraw before any follow-up procedures are obtained will also be replaced as specified below.

In Phase 1 of the study, subjects who only receive 1 cycle of the therapeutic IMP for reasons other than the presence of a DLT (e.g. intercurrent illness that results in subject withdrawal from the study) will be replaced but remain in the study for safety assessments. Subjects who receive a single cycle of the therapeutic IMP but cannot receive subsequent cycles due to normal organ radiation dose limits being exceeded with further radiation will not be replaced. Any subject who is non-evaluable (i.e. has received no cycles or only a single cycle of therapeutic IMP), and experiences a DLT during the DLT observation period, which is deemed to be non-related to the therapeutic IMP by the site investigator will also be replaced but remain in the study for safety assessments.

In Phase 2 of the study, subjects who have been randomised but do not receive IMP administration prior to study withdrawal will be replaced.

13.4 Lost to Follow-up

A subject will be considered lost to follow-up if they fail to return for any study visit and is unable to be contacted by the study site staff.

The following actions must be taken if a subject fails to return to the clinic for a required study assessment:

- Before a subject is deemed lost to follow-up, the investigator or designee will make every effort to regain contact with the subject (where possible, 3 telephone calls and, if necessary, a recorded delivery letter to the subject's last known mailing address or local equivalent methods). These contact attempts should be documented in the subject's medical record or study file.

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- Where contact is made, the subject should be counselled on the importance of maintaining the assigned visit schedule and the investigator/designee should ascertain if the subject wishes to and/or should continue in the study.
- Should the subject continue to be unreachable, they will be considered to have withdrawn from the study with a primary reason of lost to follow-up.

13.5 Study Discontinuation or Termination

Subject enrolment will be temporarily halted or stopped if any of the following occur.

- SAE of severe acute hypersensitivity reaction attributed to the therapeutic IMP.
- SAE of death attributed to the therapeutic IMP.
- Multiple and/or severe AEs, which are unexpected and inconsistent with the known mechanism of action of the therapeutic IMP.
- Data from other clinical trials which greatly and negatively influence the current Lutetium (^{177}Lu) rhPSMA-10.1 Injection benefit/risk assessment.

An urgent meeting of the SRC or TSC will be convened if any of the above events occur during the study conduct.

14 PROTOCOL DEVIATIONS

A protocol deviation occurs when the subject, investigator, or Sponsor fails to adhere to protocol requirements or where there is a non-compliance to ICH GCP affecting subject safety or the scientific value of the study.

Any non-compliance with the protocol or GCP, which is likely to effect to a significant degree the safety or physical or mental integrity of the study subjects or the scientific value of the study should be notified to the Sponsor (or designee) within 1 working day.

Major protocol deviations for this study include, but are not limited to, the following:

- Failure to meet inclusion/exclusion criteria (see Sections 6.2 and 6.3).
- Use of a prohibited concomitant medication (see Section 7.3).
- IMP administration but no subsequent SPECT/CT scan.
- Administration of the therapeutic IMP in a subject who has exceeded normal organ radiation dose limits (applicable to Phase 1 only).
- Administration of the IMP in a subject who has exceeded normal organ radiation dose limits and does not meet the continuation criteria defined in Section 4.1.2.2 and Section 4.1.2.3 (applicable to Phase 2 only).
- Administration of the IMP in a subject who has already experienced centrally-determined radiographic progression.

The Sponsor will determine if a protocol deviation results in withdrawal of a subject. Protocol deviations categorised as major will be defined and documented in the SAP, which will be written and approved before database lock.

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15 DATA SAFETY MONITORING

Two governance committees will be formed to oversee the conduct of this Phase 1 & 2 study: A SRC in Phase 1, and a TSC in Phase 2. Further details are provided in the Safety Review Charter.

The SRC will consist of the Principal Investigators taking part in Phase 1 of the study, plus an independent medical oncologist and a nuclear medicine physician. In addition, the Sponsor's statistics/data management representative, medical representative and drug safety representative will attend. One of the independent physicians will chair the meeting. The SRC's role will be to review the data generated from Phase 1 Cohorts A and B. The SRC will also make recommendations to the Sponsor regarding the planned RP2D to test in the Phase 2 component of the study (see Section 4.1.1.5). In addition, the SRC will contribute to the drafting of rules to govern treatment discontinuation, dose delay, and dose modification in the event that subjects experience toxicity in Phase 2.

All study data will be reviewed by the SRC with a view to recommending the RP2D at the end of Phase 1 dose escalation. The SRC will also assess the benefit/risk for the therapeutic IMP at the end of Cohort A, and at the end of Phase 1 (i.e. end of Cohort B) and decide whether it is in favour of continuing the development of the therapeutic IMP.

The SRC will not permit escalation to a higher dose cohort until all relevant safety data and radiation dosimetry is available for subjects in the prior cohort. This includes the entire DLT observation period from the day of the first injection of IMP until 6 weeks following the final cycle of IMP for each subject in that cohort is administered. For dose escalation and maximum cycle decisions in between Cohorts A and B, the SRC will be provided with at least the following set of data:

- Listings of reported AEs for each subject in the cohort.
- Tables of blood laboratory values at baseline, and during the treatment and follow-up period for each subject. These may also be provided graphically in order to better observe trends.
- Tables of radiation dose estimates for all organs listed in Table 3 and for the estimated tumour absorbed doses for each subject in the cohort.

At each convened SRC meeting, safety data will also be reviewed in a cumulative manner (i.e. AEs and laboratory values will be reviewed for subjects in Cohort A as they have been collected). Thus, late emerging safety issues that may have occurred after the time-point for dose escalation decision can be considered. The SRC will also consider the planned dose in Cohort B and whether it should be adapted, or whether other areas of the protocol should be adapted to ensure continuing safety of study subjects.

It is anticipated that there will be a minimum of 2 weeks between the end of Cohort A and the beginning of Cohort B. However, should a longer review be necessary, then this time will be extended.

Minutes from the dose escalation SRC meetings will be taken and the Chair of the SRC will be asked to sign to confirm they represent an accurate account of the discussion, and that the SRC agrees regarding the recommendations.

At the end of Phase 1, the SRC will review the safety data and radiation dosimetry for every subject taking part in Phase 1, plus available efficacy data. Efficacy data will consist of PSA values, data regarding evidence of centrally-determined radiographic progression, and information regarding which subjects have died of their disease (if any). The SRC will use the safety and efficacy data, in conjunction with radiation dosimetry estimates to recommend the

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injected radioactivity level and the number of cycles to be applied in Phase 2. The SRC will produce a written report at this end of Phase 1 meeting, which summarises their recommendations and this will be signed and approved by the Chair of the SRC.

In the event of an urgent safety issue, the SRC will be convened at short notice and further dosing paused pending recommendations from the SRC.

The Phase 2 TSC will consist of the Coordinating/Chief Investigator plus country level Principal Investigators and the Sponsor's clinical operations, medical and safety representatives. The committee will meet monthly, and its role will be the oversight of trial conduct, with a focus on reviewing study data on an ongoing basis, ensuring ongoing compliance with GCP, and timely communication with sites and relevant authorities, where necessary. Opening of Cohort 2C will be approved by the TSC. The committee will operate under a pre-defined charter. The TSC will serve as the forum for the discussion of any issues as they arise, including safety observations. The TSC will be the body that makes recommendations for the study.

16 STATISTICAL METHODS AND CONSIDERATIONS

Prior to the analysis of the final study data, a detailed SAP will be written and approved describing all analyses that will be performed. The SAP will be finalised prior to the database lock for this study and will contain any modifications to the analysis plan described below.

16.1 Data Analysis Sets

All Screened Set: All subjects who sign the ICF and are screened. This set will be used to summarise the screening data.

Diagnostic Set: All subjects who receive at least one dose of diagnostic IMP (rhPSMA-7.3 (^{18}F)) (in Phase 1). This set will be used to summarise exposure to diagnostic tracers and AEs in Phase 1 only.

Full Analysis Set (FAS): All subjects assigned (in Phase 1) or randomised (in Phase 2) to study intervention and who receive any amount of IMP will be included in the FAS. This set will be used for baseline characteristics and efficacy and safety analyses.

Efficacy Analysis Population (EAP): All subjects in the FAS who receive the full planned dose of IMP and have no major protocol deviations that could affect efficacy will be included in the EAP. The final decision on including or removing subjects from the EAP will be made during the Data Review Meeting prior to the database lock.

PK Population (PKP): All subjects in the FAS who have evaluable PK data will be included in the PKP. This set will be used for the PK analyses.

16.2 Accountability of Subjects and Imaging

The completion status of all subjects will be summarised. Further, availability of results of subject disease assessments, completion of visits, and information on image assessments performed will be reported.

16.3 Demographic and Baseline Characteristics

Demographic variables and baseline characteristics at screening will be summarised descriptively. Descriptive statistics on continuous data will include means, medians, standard

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deviations, and ranges, as appropriate. The set of data that will be collected will be in line with the country requirements in each of the participating countries; only the data that is allowed to be collected will be summarised.

Planned statistical analysis will be presented in the final SAP. The first version of the SAP will be approved before the first subject is enrolled in the study and a final version of the SAP will be approved before database lock.

16.4 Analysis of Primary Endpoints

16.4.1 Phase 1

The primary endpoints in Phase 1 of the study are the incidence of DLTs during the DLT observation period, and the frequency and nature of TEAEs. The definition of DLTs in this study is described in Section 4.1.1.4.

Safety data will be summarised descriptively. For AEs and SAEs, the Medical Dictionary for Regulatory Activities System Organ Class and Preferred Terms will be used to code the reported events. The number of events, number of subjects with 1 or more event, and percentage of subjects with 1 or more events will be provided for all AEs, related AEs, SAEs, and all related SAEs. Subject incidence of DLTs will be summarised by dose level.

16.4.2 Phase 2

The primary endpoint in Phase 2 of the study is the number of subjects with an anti-tumour response defined as ≥50% reduction in PSA level from baseline to the end of the treatment phase. Data will be presented independently for each cohort. Point estimates for the number of subjects with PSA reduction ≥50% will be presented together with 95% CIs. To compare differences between the cohorts, the difference in PSA50 responses will be calculated, along with 95% CIs. This summary will be for descriptive purposes only.

Median duration of response will be estimated using Kaplan Meier methodology and presented together with 95% CIs. Methods will also be used to compare the rates in each cohort to a historical control. Bayesian methods may also be used to estimate probability distributions and subsequent inferences, for example, to estimate the probability that the PSA response rate exceeds a historical rate by a certain amount.

Due to the exploratory nature of this study, there will be no formal hypothesis testing.

16.5 Analysis of Secondary Efficacy Endpoints

Secondary endpoints will be summarised descriptively, with 95% CIs, where applicable. Response endpoints will be analysed similarly to the PSA response endpoint, with time-to-event endpoints (e.g. duration of response) analysed using Kaplan-Meier methods.

Further details of the summary and presentation of efficacy endpoints will be provided in the SAP.

16.6 Analysis of Safety Endpoint(s)

Safety data will be summarised descriptively. For AEs and SAEs, the Medical Dictionary for Regulatory Activities System Organ Class and Preferred Terms will be used to code the reported events. The number of events, number of subjects with 1 or more event, and percentage of subjects with 1 or more events will be provided for all AEs, related AEs, SAEs,

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and all related SAEs. For vital signs results, the change from baseline results will also be summarised.

16.7 Analysis of Pharmacokinetic Endpoint

Pharmacokinetic analysis will be reported in the form of radiation dosimetry data acquired during the study.

16.8 Analysis of Patient-reported Outcome Assessments

The analysis of PRO will include all subjects in the FAS who had at least one PRO assessment throughout the study.

16.9 Laboratory Test Results

Blood biochemistry, haematology, and coagulation data will be reviewed for each subject. Summary statistics over time and/or changes from baseline over time may be provided. Shifts in grades of safety laboratory values from baseline for selected laboratory values may also be provided.

16.10 Vital Signs

Vital signs will be reviewed for each subject. Summary statistics over time and/or changes from baseline over time may be provided.

16.11 Physical Measurements

Physical measurements will be reviewed for each subject.

16.12 Electrocardiogram

Summaries over time and/or changes from baseline over time will be provided for all ECG parameters. All on-study ECG data will be listed and select parameters of interest may be plotted.

16.13 Exposure to Therapeutic Investigational Medicinal Product

Details of therapeutic IMP administration will be listed for every subject.

16.14 Exposure to Concomitant Medication

All medication will be coded using the World Health Organisation dictionary. A subject listing of all prior and concomitant medications will be presented.

16.15 Planned Interim Analysis

During the conduct of Phase 2, a safety and preliminary efficacy review will be performed to decide on whether or not to open Cohort 2C. Following the safety review, Cohort 2A or 2B may be closed for futility (i.e. estimated PSA50 response rate ≤20% within a cohort).

The Sponsor may also perform additional interim reviews of the emerging data to assess preliminary efficacy and safety. Full details of such reviews will be described in the SAP.

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Due to the exploratory nature of this study, no adjustment for multiplicity (i.e. alpha) will be made.

16.16 Sample Size and Randomisation

16.16.1 Assumptions

16.16.1.1 Phase 1 (Completed)

At least 12 subjects are proposed for the Phase 1 dose escalation part of the study, which is consistent with other early phase studies and enables collation of high-quality radiation dosimetry data to inform later clinical development as well as first assessment of safety.

16.16.1.2 Phase 2

For the Phase 2 component of the study, historical comparator PSA response rates (≥50% reduction) were estimated from literature with adjustments to reflect the target population in this study.

16.16.2 Sample Size

Overall, up to approximately 83 subjects will be enrolled in this study.

16.16.2.1 Phase 1 (Completed)

Approximately 30 subjects will be screened (including rhPSMA-7.3 (^{18}F) PET imaging) in Phase 1 of the study, with at least 12 subjects (Cohort A: at least 3 subjects; Cohort B: at least 9 subjects) enrolled to receive the therapeutic IMP.

16.16.2.2 Phase 2

Approximately 70 subjects will be enrolled and treated in Phase 2. Subjects will be initially randomised to one of 2 dosing regimens, with approximately 30 subjects randomised to Cohort 2A and Cohort 2B. If Cohort 2C is opened, approximately 10 subjects will be randomised to this cohort. If Cohort 2C is not opened, additional subjects will be randomised to Cohort 2A and Cohort 2B, resulting in a total of approximately 35 subjects in each cohort.

The pooled cohort size of approximately 70 subjects was selected based on simulations under a range of scenarios for PSA50 response rate when performing interim analyses after 20 evaluable subjects, then subsequently every 10 evaluable subjects using Bayesian posterior probabilities (Lee 2020). In particular, under a range of potential scenarios, the simulations showed that the average sample size will be between 51 and 61 (hence approximately 70 subjects) to enable reliable estimation of PSA50 response when compared with relevant historical controls (PSMAfore, Morris 2024).

The individual cohort sample size of approximately 30 subjects per cohort was selected to provide sufficient exploratory data on dose optimisation without exposing an excessive number of subjects to the IMP at this stage of development. The Cohort 2C sample size of approximately 10 subjects is based on assessing gross differences in efficacy or safety. If data from this cohort are compelling, a protocol amendment may be submitted to increase its sample size.

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17 DATA COLLECTION, RETENTION AND MONITORING

17.1 Data Collection Instruments

The investigator will prepare and maintain adequate and accurate study records and source documents designed to record all observations and other pertinent data on each of the site's study subjects. Particular care should be taken to ensure all data points are recorded in source documentation, especially those which are not part of standard practice. Source data should be attributable, legible, contemporaneous, original, accurate and complete. Changes to source data should be traceable, should not obscure the original entry, and should be explained if necessary. The eCRF will not be used as a source document for data collection.

Study personnel at each site will enter data from source documents corresponding to a subject's visit into the protocol-specific eCRF within 3 business days of subject visit for key data points (dosimetry and imaging).

Subjects will not be identified by name in the study database or on any study documents to be collected by the Sponsor (or designee) but will be identified by a site number combined with subject number. Data are considered pseudo-anonymised as the subject can be identified by the study site and as such data remains personal data and within the scope of US, EU, and UK law (US HIPAA 1996; EU General Data Protection Regulation [GDPR] 2018 [EU] 2016/679; UK Data Protection Act (DPA) 2018, and UK GDPR 2021).

If a correction is required for an eCRF, an electronic audit trail will be created to track the person entering or updating eCRF data, the changes made, and the date/time the change was made.

Patient-reported outcome measures will be completed by study subjects on paper and will be transcribed into the EDC. These will be recorded at pre-IMP injection on each day of dosing, and at the EoT visit.

The investigator is responsible for all information collected on subjects screened and enrolled in this study. All data collected during the course of this study must be reviewed and verified for completeness and accuracy by the investigator. A copy of the eCRF will remain at the investigator's site at the completion of the study.

17.2 Data Management Procedures

The data will be entered into validated databases. The Data Management group at the CRO will be responsible for data processing, in accordance with procedural documentation. Database lock will occur once QC procedures have been completed.

All procedures for the handling and analysis of data will be conducted to meet the FDA guidelines for the handling and analysis of data for clinical trials.

17.3 Data Quality Control and Reporting

After data have been entered into the study database, a system of computerised data validation checks will be implemented and applied to the database on a regular basis. Queries are entered, tracked, and resolved through the EDC system directly. The study database will be updated in accordance with the resolved queries. All changes to the study database will be documented in the audit trail.

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17.4 Archival of Data

The database is safeguarded against unauthorised access by established security procedures; appropriate backup copies of the database and related software files will be maintained. Databases are backed up by the database administrator in conjunction with any updates or changes to the database.

At critical junctures of the protocol (e.g. production of the final report), data for analysis is locked and cleaned per established procedures.

At the end of the study, the database will be archived by the Sponsor in accordance with applicable regulatory requirements.

17.5 Availability and Retention of Investigational Records

The investigator/institution should maintain adequate and accurate source documents and study records that include all pertinent observations on each of the site's study subjects. Source data should be attributable, legible, contemporaneous, original, accurate, and complete. Changes to source data should be traceable, should not obscure the original entry, and should be explained, if necessary.

The investigator/institution should maintain the study documents as specified in Section 8 of ICH GCP E6 (R2) and as required by the applicable regulatory requirement(s). The investigator/institution should take measures to prevent accidental or premature destruction of these documents.

Upon request of the monitor, auditor, IRB/IEC, or regulatory authority, the investigator/institution must make available for direct access all requested study-related records.

Study records should be retained for at least 25 years after the end of the clinical study. These documents should be retained for a longer period, however, if required by applicable local regulations. No records will be destroyed without the written consent of the Sponsor, if applicable. It is the responsibility of the Sponsor to inform the investigator when these documents no longer need to be retained.

Subject medical files should be archived in accordance with applicable local regulations.

17.6 Monitoring and Auditing

Monitoring visits will be conducted by representatives of the Sponsor according to ICH GCP and relevant regulations. By signing this protocol, the investigator grants permission to the Sponsor's and designee's monitors and auditors, as well as the IRB/IEC and regulatory authorities to conduct on-site monitoring and/or auditing and provide direct access to all requested study-related records. If on-site monitoring or auditing is not possible, then the investigator will work remotely with the Sponsor and designee's monitor or auditor to facilitate management or audit of the study including review of study-related records.

17.7 Subject Confidentiality

In order to maintain subject confidentiality, only a site number combined with subject number will identify all study subjects on eCRFs and other documentation submitted to the Sponsor. Additional subject confidentiality issues (if applicable) are covered in the Clinical Study Agreement.

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To maintain confidentiality, all laboratory specimens, evaluation forms, reports and other records will be identified by a coded number and initials only. All study records will be retained for 25 years.

Clinical information will not be released without written permission of the subject, except as necessary for monitoring by regulatory authorities. The investigator must also comply with all applicable privacy regulations (e.g. US HIPAA 1996; EU GDPR 2018; UK DPA 2018; UK GDPR 2021).

18 REGULATORY, ETHICAL AND STUDY OVERSIGHT CONSIDERATIONS

The study will be conducted in accordance with ICH GCP guidelines, EU Clinical Trials Regulation No 536/2014, applicable national laws and regulations (including the Medicinal Products Act and GCP Ordinance), and the Declaration of Helsinki. The investigator will assure that no deviation from, or changes to the protocol will take place without prior agreement from the Sponsor and documented approval from the relevant regulatory authority (if applicable) and IRB/IEC, except where necessary to eliminate an immediate hazard(s) to the study subjects. All personnel involved in the conduct of this study must have completed ICH GCP training, relevant to their role.

The governance committees that will oversee the conduct of the Phase 1 and Phase 2 components of the study are described in Section 15.

18.1 Institutional Review Boards and Independent Ethics Committees

Any documents that the IRB/IEC may need to fulfil its responsibilities (such as protocol, protocol amendments, IB, consent forms, information concerning subject recruitment, payment or compensation procedures, or other pertinent information) will be submitted to the IRB/IEC.

The IRB/IECs written unconditional approval/favourable opinion of the study, and any additional local approvals (e.g. hospital management, Administration of Radioactive Substances Advisory Committee etc.), must be obtained prior to shipment of IMP to the site and prior to any subjects undergoing study-specific procedures. The investigator will obtain assurance of IRB/IEC compliance with regulations.

Note: Regulatory authority approvals may also be required.

The IRB/IEC's Standard Operating Procedures and policies will be followed for the submission of SAEs and progress reports during the conduct of the study.

An end of study notification will be submitted as determined by each country's regulatory requirements.

18.2 Amendments

Any decision to amend the clinical trial application and/or associated documents (e.g. protocol, ICF, eCRF, IB etc.) will be made by the Sponsor.

The relevant regulations will be followed to determine what approvals from regulatory, IRB/IEC or local bodies are required. All required approvals will be obtained prior to implementation of the amendment, except as necessary to eliminate immediate safety hazards to subjects (see Section 12.3, Urgent Safety Measures). The Sponsor (or designee) will notify each participating investigator site when the amendment can be implemented.

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All changes to the consent form will be IRB/IEC approved; including making a determination regarding whether a new consent needs to be obtained from subjects who provided consent using a previously approved consent form.

18.3 Subject Information and Consent

In obtaining and documenting subject informed consent, the investigator must comply with the applicable regulatory requirement(s), ICH GCP and the ethical principles that have their origin in the Declaration of Helsinki.

Subject information and consent forms, and any other written material provided to the subject, must be approved by the relevant IRB/IEC (and by any other body as required by national regulations) prior to the start of the study (defined as site activation) at each study site.

The investigator (or an appropriately qualified designee) will explain the study to the subject and answer any questions that arise. A verbal explanation will be provided in terms suited to the subject's comprehension, of the purposes, procedures, and potential risks of the study and the rights of research subjects. Subjects will have the opportunity to carefully review the written information and consent form, to discuss the study with their family or surrogates, and be given ample time to think about the study and ask questions before agreeing to participate.

Subjects must be informed that participation is voluntary and that they may withdraw from the study at any time, without prejudice. The rights and welfare of the subjects will be protected by emphasising to them that the quality of their medical care will not be adversely affected if they decline to participate in this study.

Prior to the subject undergoing any study-specific procedures, the written ICF must be signed and personally dated by the subject and by the person who conducted the informed consent discussion. The informed consent process will also be documented in the source document (including the date/time consent was obtained).

If a subject is unable to read, an impartial witness should be present during the entire informed consent discussion. The subject may orally consent to their participation, if they are not capable of signing and personally dating the consent form. Once the subject has provided consent, the witness should also sign and personally date the consent form. By signing the consent form the witness attests that the information sheet/consent form/any other written information was accurately explained to, and apparently understood by, the subject and that informed consent was freely given by the subject.

The distribution of the signed information sheet/consent form will be as required by any applicable local regulations. Otherwise, a copy of the signed ICF will be given to the subject and the original maintained with the subject's records.

If the subject's next dosing cycle is predicted to exceed the normal organ dose limits as defined in Section 4.1.2.3, the subject must be re-informed of the potential risks of acute, delayed, and cumulative toxicities that may result from the IMP, and the uncertainty regarding appropriate radiation dose limits. The subject must then be re-consented prior to continuing treatment. If the subject is willing to continue participation in the study, re-consent will be documented in an ICF addendum, which both the subject and the Principal Investigator must sign and date.

The subject will be informed in a timely manner if new information becomes available that may be relevant to the subject's willingness to continue participation in the study. The communication of this information will be documented in the source documentation. The written subject information/consent form and any other written information provided to the subjects should be revised whenever important new information becomes available that may

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be relevant to the subject's consent. Any revised written subject information and consent form should receive IRB/IEC approval/favourable opinion prior to use. The subject should sign and personally date any revised consent form and receive a copy (or original, if required by applicable regulations). All subjects will receive all the required information for collecting and processing their personal data for the purposes of the study and for potential reuse of data in other studies or scientific research. A copy of the Privacy Information Notice and Consent will be given to subjects, as applicable.

18.4 Publications

The preparation and submittal for publication of abstracts and manuscripts containing the study results shall be in accordance with a process determined by mutual written agreement among the study Sponsor and participating institutions. The publication or presentation of any study results shall comply with all applicable privacy laws, including, but not limited to, the US HIPAA 1996 and the EU GDPR 2018.

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APPENDIX 1: STUDY SCHEDULE

Study Schedule Phase 1: Cohort A

Recruitment into Phase 1 Cohort A is now completed. Please refer to Protocol Version 3.0 (dated 11 April 2022) for the Study Schedule applicable to this cohort.

Study Schedule Phase 1: Cohort B

Recruitment into Phase 1 Cohort B and the Phase 1 part of the study is now completed. Please refer to Protocol Version 4.0 (dated 22 September 2023) for the Study Schedule applicable to this cohort.

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Study Schedule Phase 2: Cohort A (and Cohort C, if Opened)

Day	Screening	Cycle 1-7 (6-week dosing interval)[a]								
		Week 1				W2	W4	W6		
	Visit 1	D1 (dosing)	D2 (24 hr)	D3 (48 hr)	D8 (168 hr)	D15	D29	D43[b]	EoT Visit[c]	Efficacy & Safety Follow-up[d]
Window (days)	**D-42 to D-1[e]**	**±1[f]**			**±1**	**±1**	**±1**	**±1**	**±2**	
ICF	X									
Inclusion/exclusion criteria[g]	X									
Demographics	X									
Medical history	X									
Concomitant medication	X	X	X	X	X	X	X	X	X	
AE assessment	X	X	X	X	X	X	X	X	X	X
ECOG PS	X	X							X	
Physical examination	X	X							X	
Height	X									
Body weight[h]	X	X							X	
Vital signs[i]	X	X							X	
12-lead ECG[j]	X	X[j]							X	
Urinalysis	X[k]	X[l]							X	
Haematology, coagulation, and biochemistry (safety labs)[m]	X	X				X	X	X	X	
PSA testing[n]	X	X						X		X[o]

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Day	Screening	Cycle 1-7 (6-week dosing interval)[a]								
		Week 1				W2	W4	W6		
	Visit 1	D1 (dosing)	D2 (24 hr)	D3 (48 hr)	D8 (168 hr)	D15	D29	D43[b]	EoT Visit[c]	Efficacy & Safety Follow-up[d]
Window (days)	**D-42 to D-1[e]**	**±1[f]**			**±1**	**±1**	**±1**	**±1**	**±2**	
Imaging										
PSMA PET/CT for imaging screening/eligibility[p]	X									
ceCT or MRI and ^{99m}Tc bone scanning for imaging screening/eligibility[q]	X									
Confirm eligibility to continue study	X	X[a]								
^{177}Lu-rhPSMA-10.1 injection administration[r]		X								
SPECT/CT for dosimetry[s]		X	X	X	X					
ceCT or MRI and ^{99m}Tc bone scanning follow-up[t]								X[t]		X[t]
Patient-reported outcomes[u]										
FACT-P		X							X	
FACT-RNT		X							X	

Abbreviations: ^{177}Lu=Lutetium 177; ^{99m}Tc=technetium-99m; AE=adverse event; ce=contrast enhanced; CT=computed tomography; D=day; ECG=electrocardiogram; ECOG PS=Eastern Cooperative Oncology Group Performance Status; eCRF=electronic case report form; Eff=efficacy; EoT=End of Treatment; FACT-P=Functional Assessment of Cancer Therapy-Prostate; FACT-RNT=Functional Assessment of Cancer Therapy- Radionuclide Therapy; GBq=Gigabecquerel; ICF=informed consent form; IMP=investigational medicinal product; MRI=magnetic resonance imaging; PET=positron emission tomography; PSA=prostate-specific antigen; PSMA=prostate-specific membrane antigen; rh=radiohybrid; SFU=safety follow-up; SPECT=single-photon emission computed tomography; UPCR=urine protein creatinine ratio; W=week.

[a] Subjects will receive up to 6 (Cohort 2C) or 7 (Cohort 2A) cycles of therapy. All doses administered at 6-week intervals. Dosimetry data, clinical assessments, and laboratory evaluations to confirm eligibility for further treatment will be completed before each IMP administration.

[b] If there is no dosing delay, Day 43 assessments may be combined with Day 1 of the subsequent cycle. See Section 4.1.2.2 for dose delay rules.

[c] EoT visit will occur on the last day of the final 6-week cycle (±48 hours) in the study. For subjects who withdraw from the study prematurely, the EoT visit should be completed within 10 days of withdrawal.

[d] If a subject experiences radiographic progression and/or initiates treatment with another anti-cancer therapy, efficacy follow-up will be terminated, and the subject will continue to be monitored for long-term safety in passive follow-up only.

[e] Screening and imaging eligibility should take place within 42 days of first IMP administration. However, in the event of logistical issues e.g. PET scan availability, it would be permissible to extend screening for an additional 28 days. If the screening period exceeds 42 days, screening blood tests should be repeated prior to Cycle 1 Day 1.

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[f] With the exception of Cycle 1 Day 1, a dosing window of ±1 day is allowed for Day 1 IMP dosing in subsequent treatment cycles.

[g] As per the inclusion criteria, serum testosterone levels should be <50 ng/dL (1.73 nmol/L) after surgical or continued chemical castration, at screening. If a sample has been taken within the last 3 months before the screening date and there has been no change to medication, it can be used to confirm eligibility.

[h] Body weight should be measured pre-IMP administration.

[i] Vital signs, (body temperature, pulse, blood pressure and respiration rates) will be recorded after resting for at least 5 minutes at screening, pre-, and post-IMP administration. Reference time is the end time of IMP administration on each dosing day.

[j] Single 12-lead ECG will be performed at screening and pre-IMP administration in each cycle. A single 12-lead ECG will also be performed within 5 hours post-IMP administration in Cycle 1 only.

[k] Urinalysis using a dipstick will be performed at screening to exclude interval renal pathology, particularly urinary tract infection. Subjects with indwelling urinary catheters will be excluded from this assessment as urinalysis is persistently abnormal in this group.

[l] Spot UPCR will be performed pre-IMP administration on the morning of each dosing day. Wherever possible, this sample should be collected at approximately the same time of the day and should not be the first void of the day.

[m] Haematology, coagulation, and biochemistry blood samples will be collected at screening and pre-IMP administration in each treatment cycle. Laboratory assessments taken on the day of IMP dosing (or within 72 hours of dosing) must be reviewed prior to IMP administration. In between cycles of therapy, a blood sample will be taken every 2 weeks D15, D29 and D43 (±24 hours) for safety monitoring. Additional haematology, coagulation, and biochemistry samples may be collected as clinically indicated during the study and must be recorded in the eCRF including the reason for sampling. Wherever possible, a single needle should be used to draw blood for laboratory testing; timepoints should be aligned at the site to aid this.

[n] PSA samples will be collected at screening, pre-IMP administration on Cycle 1 Day 1, and then every 6 weeks during the treatment phase.

[o] During efficacy follow-up, PSA samples will be collected every 12 weeks until centrally-determined radiographic progression, death, or loss to follow-up, whichever occurs first.

[p] The locally approved standard of care PSMA PET/CT tracer may be used for the diagnostic PSMA PET/CT screening. Alternatively, if available, an existing positive PSMA PET/CT scan obtained as part of the subject's standard of care within 4 weeks (28 days) prior to informed consent may be used for subject selection.

[q] Contrast enhanced CT/MRI of chest/abdomen and pelvis. Full body ^{99m}Tc bone scan. Brain MRI/CT imaging not required unless there is clinical suspicion of brain metastases. Imaging scans obtained no more than 28 days prior to informed consent may be used to determine imaging eligibility, if available for submission to the central reader.

[r] First 2 doses at 10.00 GBq (in Cohort 2A) or 14.80 GBq (in Cohort 2C) followed by up to 5 (or 4 in Cohort 2C) additional doses at 7.40 GBq.

[s] SPECT/CT imaging will be performed at 3 (±1) hours post-therapeutic IMP administration, and then at 24 (±4) hours, 48 (±4) hours, and at 168 (±24) hours post-IMP administration for the first treatment cycle only. For all subsequent cycles, a single time-point SPECT/CT imaging will be performed at 48 (±4) hours.

[t] Cross-sectional imaging assessment (ceCT or MRI/full body ^{99m}Tc bone scan) will take place every 12 weeks ±5 days from Cycle 1 Day 1 until centrally-determined radiographic progression, death, or loss to follow-up, whichever occurs first. The full body ^{99m}Tc bone scan must be performed within 5 days prior to IMP administration during the treatment phase to avoid inference with dosimetry measurements.

[u] Patient-reported outcomes must be completed prior to all other procedures on the visit day; FACT-P should be completed before FACT-RNT.

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Study Schedule Phase 2: Cohort B

Day	Screening	Cycle 1-2[a]						Cycle 3-8[b]						
		Week 1				W2	W3	Week 1		W2	W4	W6		
	Visit 1	D1 (dosing)	D2 (24 hr)	D3 (48 hr)	D8 (168 hr)	D15	D22[c]	D1 (dosing)	D3 (48 hr)	D15	D29	D43[d]	EoT Visit[e]	Efficacy & Safety Follow-up[f]
Window (days)	D-42 to D-1[g]	±1[h]			±1	±1	±1			±1	±1	±1	±2	
ICF	X													
Inclusion/exclusion criteria[i]	X													
Demographics	X													
Medical history	X													
Concomitant medication	X	X	X	X	X	X	X	X	X	X	X	X	X	
AE assessment	X	X	X	X	X	X	X	X	X	X	X	X	X	X
ECOG PS	X	X						X					X	
Physical examination	X	X						X					X	
Height	X													
Body weight[j]	X	X						X					X	
Vital signs[k]	X	X						X					X	
12-lead ECG[l]	X	X						X					X	
Urinalysis	X[m]	X[n]						X[n]					X	
Haematology, coagulation, and biochemistry (safety labs)[o]	X	X				X	X	X		X	X	X	X	
PSA testing[p]	X	X					X[q]	X				X		X[q]

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Day	Screening	Cycle 1-2[a]						Cycle 3-8[b]						
		Week 1				W2	W3	Week 1		W2	W4	W6		
	Visit 1	D1 (dosing)	D2 (24 hr)	D3 (48 hr)	D8 (168 hr)	D15	D22[c]	D1 (dosing)	D3 (48 hr)	D15	D29	D43[d]	EoT Visit[e]	Efficacy & Safety Follow-up[f]
Window (days)	D-42 to D-1[g]	±1[h]			±1	±1	±1			±1	±1	±1	±2	
Imaging														
PSMA PET/CT for imaging screening/eligibility[s]	X													
ceCT or MRI and ^{99m}Tc bone scanning for imaging screening/eligibility[t]	X													
Confirm eligibility to continue study	X	X[b]						X						
^{177}Lu-rhPSMA-10.1 injection therapeutic administration[u]		X						X						
SPECT/CT for dosimetry[v]		X	X	X	X				X					
ceCT or MRI and ^{99m}Tc bone scanning follow-up[w]												X[w]		X[v]
Patient-reported outcomes[x]														
FACT-P		X						X					X	
FACT-RNT		X						X					X	

Abbreviations: ^{177}Lu=Lutetium 177; ^{99m}Tc=technetium-99m; AE=adverse event; ce=contrast enhanced; CT=computed tomography; D=day; ECG=electrocardiogram; ECOG PS=Eastern Cooperative Oncology Group Performance Status; eCRF=electronic case report form; Eff=efficacy; EoT=End of Treatment; FACT-P=Functional Assessment of Cancer Therapy-Prostate; FACT-RNT=Functional Assessment of Cancer Therapy- Radionuclide Therapy; GBq=Gigabecquerel; ICF=informed consent form; IMP=investigational medicinal product; MRI=magnetic resonance imaging; PET=positron emission tomography; PSA=prostate-specific antigen; PSMA=prostate-specific membrane antigen; rh=radiohybrid; SFU=safety follow-up; SPECT=single-photon emission computed tomography; UPCR=urine protein creatinine ratio; W=week.

[a] The first 3 doses will be administered once every 3 weeks. After the third dose has been administered, the subject will enter the 6-week treatment cycle. Dosimetry data, clinical assessments, and laboratory evaluations to confirm eligibility for further treatment will be completed approximately 1 week (respectively 2 weeks from cycle 3 onwards) before IMP administration.

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[b] Subjects will receive up to 8 cycles of therapy. Doses for Cycles 3-8 will be administered at 6-week intervals. Dosimetry data, clinical assessments, and laboratory evaluations to confirm eligibility for further treatment will be completed before each IMP administration.

[c] If there is no dosing delay, Day 22 assessments may be combined with Day 1 of the subsequent cycle. See Section 4.1.2.2 for dose delay rules.

[d] If there is no dosing delay, Day 43 assessments may be combined with Day 1 of the subsequent cycle. See Section 4.1.2.2 for dose delay rules.

[e] EoT visit will occur on the last day of the final 6-week cycle (±48 hours) in the study. For subjects who withdraw from the study prematurely, the EoT visit should be completed within 10 days of withdrawal.

[f] If a subject experiences radiographic progression and/or initiates treatment with another anti-cancer therapy, efficacy follow-up will be terminated, and the subject will continue to be monitored for long-term safety in passive follow-up only.

[g] Screening and imaging eligibility should take place within 42 days of first IMP administration. However, in the event of logistical issues e.g. PET scan availability, it would be permissible to extend screening for an additional 28 days. If the screening period exceeds 42 days, screening blood tests should be repeated prior to Cycle 1 Day 1.

[h] With the exception of Cycle 1 Day 1, a dosing window of ±1 day is allowed for Day 1 IMP dosing in subsequent treatment cycles.

[i] As per the inclusion criteria, serum testosterone levels should be <50 ng/dL (1.73 nmol/L) after surgical or continued chemical castration at screening. If a sample has been taken within the last 3 months before the screening date and there has been no change to medication, it can be used to confirm eligibility.

[j] Body weight should be measured pre-IMP administration.

[k] Vital signs, (body temperature, pulse, blood pressure and respiration rates) will be recorded after resting for at least 5 minutes at screening, pre-, and post-IMP administration. Reference time is the end time of IMP administration on each dosing day.

[l] Single 12-lead ECG will be performed at screening and pre-IMP administration in each cycle. A single 12-lead ECG will also be performed within 5 hours post-IMP administration in Cycle 1 only.

[m] Urinalysis using a dipstick will be performed at screening to exclude interval renal pathology, particularly urinary tract infection. Subjects with indwelling urinary catheters will be excluded from this assessment as urinalysis is persistently abnormal in this group.

[n] Spot UPCR will be performed pre-IMP administration on the morning of each dosing day. Wherever possible, this sample should be collected at approximately the same time of the day and should not be the first void of the day.

[o] Haematology, coagulation, and biochemistry blood samples will be collected at screening and pre-IMP administration in each treatment cycle. Laboratory assessments taken on the day of IMP dosing (or within 72 hours of dosing) must be reviewed prior to IMP administration. In between the 6-weekly cycles of therapy, a blood sample will be taken every 2 weeks on D15, D29 and D43 (±24 hours) for safety monitoring. Blood safety laboratory tests will also be collected only on Day 22 of the first 2 treatment cycles during the 3-week dosing intervals. Additional haematology, coagulation, and biochemistry samples may be collected as clinically indicated during the study and must be recorded in the eCRF including the reason for sampling. Wherever possible, a single needle should be used to draw blood for laboratory testing; timepoints should be aligned at the site to aid this.

[p] PSA samples will be collected at screening, pre-IMP administration on Cycle 1 Day 1, and then every 6 weeks during the treatment phase.

[q] PSA sample will be collected in Cycle 2 only (6 weeks after the first dose of the IMP).

[r] During efficacy follow-up, PSA samples will be collected every 12 weeks until centrally-determined radiographic progression, death, or loss to follow-up, whichever occurs first.

[s] The locally approved standard of care PSMA PET/CT tracer may be used for the diagnostic PSMA PET/CT screening. Alternatively, if available, an existing positive PSMA PET/CT scan obtained as part of the subject's standard of care within 4 weeks (28 days) prior to informed consent may be used for subject selection.

[t] Contrast enhanced CT/MRI of chest/abdomen and pelvis. Full body ^{99m}Tc bone scan. Brain MRI/CT imaging not required unless there is clinical suspicion of brain metastases. Imaging scans obtained no more than 28 days prior to informed consent may be used to determine imaging eligibility, if available for submission to the central reader.

[u] Up to 8 doses at 7.40 GBq. The first 3 doses will be administered once every 3 weeks, with the remaining doses being administered once every 6 weeks.

[v] SPECT/CT imaging will be performed at 3 hours (±1 hour) post-IMP administration, and then at 24 hours (±4 hours), 48 hours (±4 hours), and at 168 hours (±24 hours) post-IMP administration for the first treatment cycle only. For all subsequent cycles, a single time-point SPECT/CT imaging will be performed at 48 hours (±4 hours).

[w] Cross-sectional imaging assessment (ceCT or MRI and full body ^{99m}Tc bone scan) will take place every 12 weeks ±5 days from Cycle 1 Day 1 until centrally-determined radiographic progression, death, or loss to follow-up, whichever occurs first. The full body ^{99m}Tc bone scan must be performed within 5 days prior to IMP administration during the treatment phase to avoid inference with dosimetry measurements.

[x] Patient-reported outcomes must be completed prior to all other procedures on the visit day; FACT-P should be completed before FACT-RNT.

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APPENDIX 2: PROSTATE CANCER CLINICAL TRIALS WORKING GROUP 3 CRITERIA

Variable	PCWG2 (2008)	PCWG3 (2015)
Blood-based		
• PSA	Obtain sequence of rising values at a minimum of 1-week intervals	Retained
	2.0 ng/mL minimal starting value	1.0 ng/mL is the minimal starting value if confirmed rise is only indication of progression unless pure small-cell carcinoma
	Estimate pretherapy PSADT if at least three values available ≥ 28 days apart	Retained
Imaging		
• Nodes	Nodal progression sufficient for trial entry independent of PSA	Retained
	Measurable lesions not required for entry	Retained
	Use RECIST to record nodal lesions as target or nontarget	Modified RECIST 1.1 criteria, separate pelvic and extra-pelvic disease, up to five nodal lesions total recorded
	Only lymph nodes ≥ 2 cm in diameter (long axis) were actionable as progressive disease	Previously normal (< 1.0-cm) lymph nodes must have grown by ≥ 5 mm in the short axis from baseline or nadir and be ≥ 1.0 cm in the short axis to be considered to have progressed
		If the node progresses to ≥ 1.5 cm in the short axis, it is measurable; nodes that have progressed to 1.0 to less than 1.5 cm are pathologic, subject to clinical discretion, and non-measurable
		For existing pathologic adenopathy, progression is defined per RECIST 1.1
	Record presence of nodal and/or visceral disease separately	Retained with modification
		Nodal sites:
		• Locoregional: pelvic only
		• Extra-pelvic: retroperitoneal, mediastinal, thoracic, or other

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Variable	PCWG2 (2008)	PCWG3 (2015)
• Viscera	Visceral progression sufficient for trial entry independent of PSA	Retained but recorded separately by site of spread (lung, liver, adrenal, CNS); up to five lesions per site of spread
	Measurable lesions not required for entry	Retained
	Use RECIST to record visceral lesions as target or nontarget	Retained
	Record presence of nodal and/or visceral disease separately	Retained with modification
		Visceral sites: lung, liver, adrenal, CNS
• Prostate/prostate bed (primary site)	Record prior treatment of primary tumour	Retained
	Perform directed pelvic imaging (CT, MRI, PET/CT, endorectal MRI, transrectal ultrasound) to document presence or absence of disease	Retained
• Bone	Two new lesions	Retained
	Confirm ambiguous results by other imaging modalities (e.g., CT or MRI)	Retained, but only positivity on the bone scan defines metastatic disease to bone
• Other sites of disease	Patients with treated epidural lesions and no other epidural progression are eligible	Retained
Type of progression at trial entry	Not addressed	Report separately:
		• PSA only
		• Bone only ± nodal disease
		• Nodal disease only (no bone disease present)
		• Visceral (lung, liver, adrenal, CNS) disease (± other sites)
		Record new lesions and site of new lesions *v* growth of pre-existing lesions, or both

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Variable	PCWG2 (2008)	PCWG3 (2015)
Other markers		
• Patient-reported outcomes	Not addressed	For pain palliation analyses, presence of clinically meaningful pain at baseline (eg, ≥4 on a 10-point pain intensity scale) is a prerequisite; for pain progression analyses, patients may have any level of pain at baseline, including no pain

Abbreviations: CNS=central nervous system; CT=computed tomography; MRI=magnetic resonance imaging; PCWG2=Prostate Cancer Clinical Trials Working Group 2; PCWG3=Prostate Cancer Clinical Trials Working Group 3; PET=positron emission tomography; PSA=prostate-specific antigen; PSADT=PSA doubling time; RECIST=Response Evaluation Criteria in Solid Tumours.